Ordered Fractional Integration (OFI)

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Abstract

Classical fractional calculus faces three foundational obstructions. **(1) The factorial** $(-1)!$ is undefined classically, yet the pole of $\Gamma(z)$ at $z = 0$ is the very mechanism that forces $I^0 = \mathrm{id}$; suppressing it destroys the identity-at-zero property. **(2) The negative Gamma poles:** $\Gamma(z)$ has poles at every non-positive integer, leaving $(-n)!$ without a consistent finite value and blocking the extension of fractional calculus to negative orders. **(3) The ordering problem:** the Riemann–Liouville fractional derivatives do not compose canonically on bounded intervals: $D^{\alpha}_{\mathrm{RL}} D^{\beta}_{\mathrm{RL}} f \neq D^{\alpha+\beta}_{\mathrm{RL}} f$ in general because boundary correction terms appear, making fractional differentiation order-dependent.

We define the *Ordered Fractional Integral* (OFI) as the symmetric Riesz-kernel fractional integral equipped with a half-integer bound shift and a residue-regularization convention, designed to address all three obstructions simultaneously: poles are regularized by residue extraction to assign consistent finite values $(-1, \frac{1}{12})$-style to $(-n)!$; the bounded-interval OFI integral family satisfies $I^{\alpha}_{\mathrm{OFI}} I^{\beta}_{\mathrm{OFI}} = I^{\alpha+\beta}_{\mathrm{OFI}}$ by explicit Beta-function convolution under the shifted lower-limit convention; and the $n + \frac{1}{2}$ bound shift bridges discrete sums to continuous integrals, recovering the Euler–Maclaurin formula as the residual Bernoulli correction series.

Chapter 12 gives the core rigorous treatment: the OFI operator is defined as a bounded operator on $L^1([a - \frac{1}{2}, b])$ and the semigroup law is proved by an explicit Beta-function convolution. The Fourier-multiplier picture on $\mathbb{R}^n$ (Riesz potential $|\xi|^{-\alpha}$) motivates the $L^2(\mathbb{R}^n)$ realization and the global semigroup. Self-adjointness, a concrete Fourier-side core, and spectral gap results for the associated Hilbert-space operator are developed in Chapter 13. The 52 catalogued spline families in Parts V and VIII each admit an OFI interpretation at a specific fractional order, connected by a spawning hierarchy rooted at δ and Γ.

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Keywords: fractional calculus, ordered fractional integral, Riesz kernel, semigroup property, Gamma function regularization, spline families, Bernoulli numbers, Euler–Maclaurin formula, discrete–continuous bridge.

Contents

Part I

Foundations

Chapter 1

Introduction

This paper constructs OFI from the ground up and develops its core mathematical theory, requiring only standard calculus as prerequisite. The central results are:

- Why $(-1)! = \infty$ is not an error but a *mechanism* — the pole of $\Gamma(z)$ forces the fractional integral to become the identity at $\alpha = 0$.
- Why the Riemann–Liouville fractional derivative has a non-commutativity flaw — the *ordering problem* — and how OFI eliminates boundary correction terms via the Riesz kernel's symmetry to restore the semigroup law.
- Why the regularized sum $1 + 2 + 3 + \cdots$ carries *two* pieces of information, written $(-1, \frac{1}{12})$, not just $-\frac{1}{12}$.
- How the sin orbit under fractional differentiation, $D^\alpha[\sin\theta] = \sin(\theta + \alpha\pi/2)$, unifies the extended trigonometric families; and how the pole-complementarity of tan and cot (Remark 8.9) provides a geometric analogy for why the OFI half-integer shift restores the semigroup law.
- How the 52 spline families catalogued in Part VIII admit OFI interpretations at specific fractional orders or order ranges, connected by a spawning hierarchy rooted at δ and Γ.
- How Bernoulli splines connect OFI directly to the Bernoulli numbers, the Euler–Maclaurin formula, and the values $\zeta(-n)$.

How to read this volume. Theorems, propositions, lemmas, and corollaries in this book are proved in full. Remarks explicitly labeled as structural observations or program notes are included to record broader connections and future directions, but they are not additional theorem claims. In particular, the core OFI semigroup, ordering-resolution, operator, and Bernoulli/Euler–Maclaurin results are proved here, while some spline-catalogue and Millennium-bridge interpretations are intentionally presented as structural.

Chapter 2

The Gamma Function: Extending Factorials

2.1 The factorial problem

For positive integers, $n! = n(n-1)(n-2)\cdots 1$. Can we define $z!$ for non-integer or negative z?

Definition 2.1 (Gamma function)**.** *For* $\mathrm{Re}(z) > 0$*:*

$$\Gamma(z) := \int_0^\infty t^{z-1} e^{-t}\, dt.$$

The key relation $\Gamma(z+1) = z\Gamma(z)$ (proved by integration by parts) gives $\Gamma(n) = (n-1)!$ for all positive integers n. Equivalently: $n! = \Gamma(n+1)$.

2.2 Analytic continuation and poles

The integral converges only for $\mathrm{Re}(z) > 0$. But $\Gamma(z) = \Gamma(z+1)/z$ extends Γ to all of $\mathbb{C}$ except $z = 0, -1, -2, \ldots$, where it has simple poles.

z	$\Gamma(z)$	Meaning	Type
5	24	$4! = 24$	integer
$\frac{3}{2}$	$\frac{\sqrt{\pi}}{2}$	$(\frac{1}{2})! = \frac{\sqrt{\pi}}{2}$	half-integer
1	1	$0! = 1$	integer
$\frac{1}{2}$	$\sqrt{\pi}$	$(-\frac{1}{2})! = \sqrt{\pi}$	half-integer
0^+	$+\infty$	**pole:** $(-1)! = \infty$	pole
$-\frac{1}{2}$	$-2\sqrt{\pi}$	finite, negative	half-integer
-1^+	$-\infty$	**pole:** $(-2)! = \infty$	pole
$-\frac{3}{2}$	$\frac{4}{3}\sqrt{\pi}$	finite, positive	half-integer

2.3 Why $(-1)! = \infty$ is correct, not broken

Near $z = 0$, the Laurent expansion is:

$$\Gamma(z) = \frac{1}{z} - \gamma + O(z),$$

where $\gamma \approx 0.5772$ is the Euler–Mascheroni constant. The *residue* at $z = 0$ is 1.

This is the mechanism behind a deep fact: as $\alpha \to 0^+$, the Riesz kernel $c_{n,\alpha} |x - y|^{\alpha-n} \to \delta(x - y)$, so the fractional integral $I^\alpha \to \mathrm{Id}$. The $1/z$ pole in Γ and the non-integrable singularity $|x - y|^{-n}$ *cancel exactly*, forcing the identity at $\alpha = 0$. The infinity is the mechanism, not the error.

2.4 Residues and OFI-regularized factorials

Proposition 2.2 (Gamma residues)**.**

$$\mathrm{Res}_{z=-n}\, \Gamma(z) = \frac{(-1)^n}{n!}, \qquad n = 0, 1, 2, \ldots$$

Definition 2.3 (OFI-regularized factorial)**.**

$$(-n)!_{\mathrm{OFI}} := \mathrm{Res}_{z=-n+1}\, \Gamma(z) = \frac{(-1)^{n-1}}{(n-1)!}.$$

n	$(-n)!_{\mathrm{OFI}}$	n	$(-n)!_{\mathrm{OFI}}$
1	$+1$	5	$+\frac{1}{24}$
2	-1	6	$-\frac{1}{120}$
3	$+\frac{1}{2}$	7	$+\frac{1}{720}$
4	$-\frac{1}{6}$	n	$\frac{(-1)^{n-1}}{(n-1)!}$

2.5 The reflection formula and $\sqrt{\pi}$

The reflection formula $\Gamma(z)\Gamma(1-z) = \pi/\sin(\pi z)$ at $z = \frac{1}{2}$ gives $\Gamma(\frac{1}{2})^2 = \pi$, so:

$$\Gamma\left(\tfrac{1}{2}\right) = \sqrt{\pi} = \int_{-\infty}^{\infty} e^{-x^2}\, dx.$$

The values $\sqrt{\pi}$, π, π^2 arise from $\Gamma(\frac{1}{2})$, $\Gamma(\frac{1}{2})^2$, and $\Gamma(\frac{1}{2})^4$ respectively — all generated by the half-integer shift $\alpha = n + \frac{1}{2}$ that is central to OFI.

Chapter 3

Classical Fractional Calculus

3.1 Riemann–Liouville (RL)

Definition 3.1 (RL integral and derivative)**.** *For* $\alpha > 0$*,* $n = \lceil \alpha \rceil$*:*

$$ {}_aI_x^\alpha f(x) := \frac{1}{\Gamma(\alpha)} \int_a^x (x-t)^{\alpha-1} f(t)\, dt, \tag{3.1}$$

$$ {}_aD_x^\alpha f(x) := D^n \Big[{}_aI_x^{n-\alpha} f \Big] (x). \tag{3.2}$$

Semigroup for integrals: ${}_aI_x^\alpha \, {}_aI_x^\beta f = {}_aI_x^{\alpha+\beta} f \quad (\alpha, \beta > 0)$. ✓
Semigroup failure for derivatives: For $0 < \alpha, \beta < 1$:

$$ {}_aD_x^\alpha \, {}_aD_x^\beta f = {}_aD_x^{\alpha+\beta} f - \frac{\Big[{}_aI_x^{1-\beta} f \Big]_{x=a}}{\Gamma(1-\alpha)} (x-a)^{-\alpha}. \tag{3.3}$$

The correction term vanishes only when ${}_aI_x^{1-\beta} f \Big|_{x=a} = 0$.

Key flaw at $\alpha = 0$: $1/\Gamma(0) = 0$ (since $\Gamma(0) = \infty$), so ${}_aI_x^0 f = 0$, not f. This is the $(-1)! = \infty$ issue forcing $1/(-1)! = 0$, killing the integral. The Riesz approach (Chapter 5) resolves this.

3.2 Caputo

Definition 3.2 (Caputo derivative)**.**

$$ {}_a^{\mathrm{C}}D_x^\alpha f(x) := \frac{1}{\Gamma(n-\alpha)} \int_a^x (x-t)^{n-\alpha-1} f^{(n)}(t)\, dt, \qquad n = \lceil \alpha \rceil. $$

Caputo integrates *after* differentiating (RL does the reverse). The difference:

$$ {}_aD_x^\alpha f - {}_a^{\mathrm{C}}D_x^\alpha f = \sum_{k=0}^{n-1} \frac{f^{(k)}(a)}{\Gamma(k-\alpha+1)} (x-a)^{k-\alpha}. $$

Caputo vanishes on constants: ${}_a^{\mathrm{C}}D_x^\alpha [c] = 0$. RL does not: ${}_aD_x^\alpha [c] = c(x-a)^{-\alpha}/\Gamma(1-\alpha) \neq 0$.

3.3 Riesz

Definition 3.3 (Riesz potential / OFI kernel)**.** *For* $0 < \alpha < n$, $f : \mathbb{R}^n \to \mathbb{R}$*:*

$$(I^\alpha f)(x) := c_{n,\alpha} \int_{\mathbb{R}^n} |x-y|^{\alpha-n} f(y)\, dy, \qquad c_{n,\alpha} = \frac{\Gamma(\frac{n-\alpha}{2})}{2^\alpha \pi^{n/2} \Gamma(\frac{\alpha}{2})}.$$

Properties:

- **Power-law**: kernel $|x-y|^{\alpha-n}$, not a box, not a Gaussian.
- **Symmetric**: $|x-y| = |y-x|$. No preferred direction, no boundary.
- **Fourier multiplier**: $\widehat{I^\alpha f}(\xi) = |\xi|^{-\alpha} \hat{f}(\xi)$.
- **Semigroup**: $I^\alpha I^\beta = I^{\alpha+\beta}$ for $0 < \alpha, \beta, \alpha+\beta < n$.

The symmetry is why the ordering problem disappears: $(-\Delta)^{\alpha/2} I^\alpha = \mathrm{Id}$ and $I^\alpha (-\Delta)^{\alpha/2} = \mathrm{Id}$ (up to constants on $\mathbb{R}^n$) — the operators commute.

3.4 Grünwald–Letnikov (GL)

The discrete fractional finite-difference:

$$D_h^\alpha f(x) := h^{-\alpha} \sum_{k=0}^{\infty} (-1)^k \binom{\alpha}{k} f(x-kh), \qquad D^\alpha f := \lim_{h \to 0^+} D_h^\alpha f,$$

where $\binom{\alpha}{k} = \Gamma(\alpha+1)/(\Gamma(k+1)\Gamma(\alpha-k+1))$.

Key: GL is the discrete realization of RL, and its coefficients $\binom{\alpha}{k}$ are exactly the fractional B-spline coefficients (Chapter 9).

3.5 Comparison table

Property	RL	Caputo	Riesz	GL
Semigroup (integrals)	✓	✓	✓	✓
Semigroup (derivatives)	×	×	✓	✓
Integer initial conditions	×	✓	—	—
Symmetric / non-local	×	×	✓	×
$D^\alpha[\mathrm{const}] = 0$	×	✓	✓	✓
Ordering: $D^\alpha D^\beta = D^{\alpha+\beta}$	×	×	✓	✓
Discrete realization	≈	≈	≈	✓

Chapter 4

The Ordering Problem

4.1 Statement

In standard calculus, $D^m D^n = D^{m+n}$. In fractional calculus, this fails:

$$ {}_aD_x^\alpha \, {}_aD_x^\beta f \;\neq\; {}_aD_x^{\alpha+\beta} f \quad \text{in general.} $$

The commutator is:

$$ {}_aD_x^\alpha \, {}_aD_x^\beta f - {}_aD_x^\beta \, {}_aD_x^\alpha f = \frac{[{}_aI_x^{1-\beta} f]_{x=a}}{\Gamma(1-\alpha)}(x-a)^{-\alpha} - \frac{[{}_aI_x^{1-\alpha} f]_{x=a}}{\Gamma(1-\beta)}(x-a)^{-\beta}. $$

This is non-zero whenever f has non-trivial fractional boundary data at $x = a$.

4.2 Why it breaks: boundary ghosts

- "Differentiate then integrate" $\neq$ "integrate then differentiate."
- The fractional Leibniz rule produces boundary ghost terms.
- The semigroup property $I_{\mathrm{RL}}^\alpha I_{\mathrm{RL}}^\beta = I_{\mathrm{RL}}^{\alpha+\beta}$ holds for integrals, fails for derivatives.

Caputo fixes the initial-condition interpretation but not the commutativity. *Neither RL nor Caputo gives a clean semigroup law on* $\mathbb{R}^n$*.*

4.3 Resolution: symmetry kills ghosts

The Riesz kernel $|x-y|^{\alpha-n}$ is symmetric: $|x-y| = |y-x|$. It has no preferred boundary, no left endpoint a. Therefore:

$$ (-\Delta)^{\alpha/2} I^\alpha = I^\alpha (-\Delta)^{\alpha/2} = \mathrm{Id}. $$

The ordering problem disappears. This is why the integral is called **Ordered**: it solves the ordering problem by having no preferred order.

Part II

OFI: The Riesz Ordered Fractional Integral

Chapter 5

OFI: Definition and Core Properties

5.1 The centering principle

The Riesz kernel $|x-y|^{\alpha-n}$ evaluates at the midpoint $(x+y)/2$: it is symmetric about $\frac{x+y}{2}$. This centering appears throughout mathematics:

Context	Formula	The centering
Triangle numbers	$T(n) = \int_0^n (x+\frac{1}{2})\,dx$	$+\frac{1}{2}$ shifts to midpoint
Central difference	$f'(x) \approx \frac{f(x+h)-f(x-h)}{2h}$	symmetric about x
Riesz kernel	$\|x-y\|^{\alpha-n}$	symmetric about $\frac{x+y}{2}$
B-spline B_n	center of mass at $n/2$	midpoint of $[0,n]$
$\sin\theta$ at small θ	$\sin\theta \approx \theta$	centered at origin

In all cases: evaluate at the **midpoint**, not the endpoint. This eliminates boundary bias and ordering ambiguity simultaneously.

5.2 The semigroup law

Theorem 5.1 (OFI semigroup)**.** *$I^\alpha I^\beta = I^{\alpha+\beta}$ for $0 < \alpha, \beta, \alpha+\beta < n$.*

Proof. In Fourier space:

$$\hat{\beta}^\alpha(\xi)\hat{\beta}^\gamma(\xi) = \left(\frac{1-e^{-i\xi}}{i\xi}\right)^{\alpha+1}\left(\frac{1-e^{-i\xi}}{i\xi}\right)^{\gamma+1} = \left(\frac{1-e^{-i\xi}}{i\xi}\right)^{\alpha+\gamma+2} = \hat{\beta}^{\alpha+\gamma+1}(\xi).$$

□

Compare the boxcar filter $K_a = \frac{1}{a}\mathbf{1}_{[-a/2,a/2]}$ with Fourier multiplier $\mathrm{sinc}(a\xi/2)$. Then $K_a * K_a$ has multiplier $\mathrm{sinc}^2(a\xi/2) \neq \mathrm{sinc}(a\xi/2)$. The boxcar fails the semigroup law at *fixed* *scale* a. However, the B-spline family $\{B_n\}_{n\geq 1}$ with $B_n = B_1^{*n}$ satisfies $B_m * B_n = B_{m+n}$ (a genuine semigroup indexed by *order*, not scale), and converges to the Riesz kernel as $h \to 0$ (see Chapter 9).

5.3 Practice examples

Example 5.2 (Half-derivative of x^β)**.**

$$D_{\mathrm{RL}}^{1/2} x^\beta = \frac{\Gamma(\beta+1)}{\Gamma(\beta+\frac{1}{2})} x^{\beta-1/2}.$$

Particular cases:

$$D^{1/2}[1] = \frac{1}{\sqrt{\pi}} x^{-1/2}, \qquad D^{1/2}[x] = \frac{2}{\sqrt{\pi}}\sqrt{x}, \qquad D^{1/2}[x^2] = \frac{8}{3\sqrt{\pi}} x^{3/2}.$$

Consistency check: $D^{1/2} D^{1/2}[x] = D^1[x] = 1$.

$$D^{1/2}\left[\frac{2}{\sqrt{\pi}}\sqrt{x}\right] = \frac{2}{\sqrt{\pi}} \cdot \frac{\Gamma(3/2)}{\Gamma(1)} x^0 = \frac{2}{\sqrt{\pi}} \cdot \frac{\sqrt{\pi}}{2} = 1. \quad \checkmark$$

Example 5.3 (Riesz OFI on a Gaussian)**.** *Let* $f(x) = e^{-\pi x^2}$ *(self-dual under Fourier).* $\hat{f}(\xi) = e^{-\pi\xi^2}$.

$$\widehat{I^\alpha f}(\xi) = |\xi|^{-\alpha}\, e^{-\pi\xi^2}.$$

At $\alpha = \frac{1}{2}$*: OFI suppresses high frequencies by* $|\xi|^{-1/2}$*, smoothing the Gaussian fractionally. The Gaussian is in* $\dot{H}^{1/2}(\mathbb{R})$ *with* $\|f\|_{\dot{H}^{1/2}}^2 = \int |\xi|\, e^{-2\pi\xi^2}\, d\xi = \frac{1}{2\pi} < \infty$.

Part III

The Discrete–Continuous Bridge

Chapter 6

The $\pm\frac{1}{2}$ Shift

6.1 The fundamental gap

For any continuous function f:

$$\int_a^b f(x)\,dx \;\neq\; \sum_{k=a}^{b} f(k).$$

The integral is systematically *short*: it misses a half-unit of area at each endpoint because it evaluates f everywhere, while the sum evaluates f only at integers.

6.2 Three equivalent fixes

Form	Rule	How to read it
Bound shift	$\int_{a-1/2}^{b+1/2} f(x)\,dx = \sum_{k=a}^{b} f(k)$	widen both bounds by $\frac{1}{2}$
Lower shift	$\int_{a}^{b+1} f(x-\frac{1}{2})\,dx = \sum_{k=a}^{b} f(k)$	shift integrand left
Upper shift	$\int_{a-1}^{b} f(x+\frac{1}{2})\,dx = \sum_{k=a}^{b} f(k)$	shift integrand right

All three are equivalent (one substitution apart).

6.3 Proof for $f(x) = x$ (triangle numbers)

$T(n) = 1 + 2 + \cdots + n = n(n+1)/2$. The naive integral falls short:

$$\int_1^n x\,dx = \frac{n^2-1}{2} \neq T(n).$$

Apply the bound shift:

$$\int_{1/2}^{n+1/2} x\,dx = \frac{(n+\frac{1}{2})^2 - (\frac{1}{2})^2}{2} = \frac{n^2+n+\frac{1}{4}-\frac{1}{4}}{2} = \frac{n(n+1)}{2} = T(n). \quad \checkmark$$

6.4 Why this is recursive

The derivative of $T(n) = n(n+1)/2$ is $T'(n) = n + \frac{1}{2}$ — a half-integer, not an integer. So the integrand that recovers the discrete sum is *the derivative of the sum itself, evaluated at the midpoint*. The $\frac{1}{2}$ is *recycled*: it appears in $T'(n)$ and it is also the bound correction needed to make $\int = \sum$.

The three levels:

Level	Object	Formula	OFI role
0	Naturals	$\{1, 2, \ldots, n\}$	discrete grid
1	Gaussian sum	$T(n) = n(n+1)/2$	sum
2	Derivative of sum	$T'(n) = n + \frac{1}{2}$	OFI correction term
3	Integral of T'	$\frac{1}{2}(\frac{n^3}{3} + \frac{n^2}{2} + \ldots)$	antiderivative

Chapter 7

Euler–Maclaurin and the $(-1, \frac{1}{12})$ Pair

7.1 Euler–Maclaurin formula

Theorem 7.1 (Euler–Maclaurin)**.** *For $f \in C^{2P+2}([a-\frac{1}{2}, b+\frac{1}{2}])$:*

$$\sum_{k=a}^{b} f(k) = \int_{a-1/2}^{b+1/2} f(x)\,dx + \sum_{p=1}^{P} \frac{\mathcal{B}_{2p}}{(2p)!}\Big[f^{(2p-1)}\Big]_{a-1/2}^{b+1/2} + R_P,$$

where $\mathcal{B}_{2p}$ are the Bernoulli numbers.

The Bernoulli numbers are:

$$\mathcal{B}_0 = 1, \quad \mathcal{B}_1 = -\tfrac{1}{2}, \quad \mathcal{B}_2 = \tfrac{1}{6}, \quad \mathcal{B}_4 = -\tfrac{1}{30}, \quad \mathcal{B}_6 = \tfrac{1}{42}, \ldots, \quad \mathcal{B}_{2k+1} = 0 \ (k \geq 1).$$

The leading correction is $\mathcal{B}_2/2! = \frac{1}{12}$.

7.2 The $(-1, \frac{1}{12})$ pair

The sum $S = 1 + 2 + 3 + \cdots$ diverges. Classical regularization: $\zeta(-1) = -\frac{1}{12}$ discards information.

The zeta function has a simple pole at $s = 1$: $\zeta(s) = \frac{1}{s-1} + \gamma + O(s-1)$.

The full information about S is the pair:

$$\boxed{(\text{pole order, regularized value}) = \left(-1,\ \tfrac{1}{12}\right).}$$

- -1: the pole of ζ at $s = 1$, shifted by 2 to reach $s = -1$.
- $\frac{1}{12}$: the regularized finite part $|\zeta(-1)| = \frac{1}{12}$.

In the OFI framework a regularized quantity is always a *pair* (analytic structure, finite value). Reporting only $-\frac{1}{12}$ is like reporting only the magnitude of a complex number, discarding the phase.

7.3 Bernoulli splines: the bridge

The Bernoulli spline of order n is the periodic Green's function of D^n on $[0,1]$:

$$G_n^{(n)}(x) = \delta(x) - 1 \quad \text{(periodically)}, \qquad G_n(x) = \frac{\mathcal{B}_n(x)}{n!},$$

where $\mathcal{B}_n(x)$ is the n-th Bernoulli polynomial.

Key values:

$$\mathcal{B}_0(x) = 1, \quad \mathcal{B}_1(x) = x - \tfrac{1}{2}, \quad \mathcal{B}_2(x) = x^2 - x + \tfrac{1}{6}, \quad \mathcal{B}_3(x) = x^3 - \tfrac{3}{2}x^2 + \tfrac{1}{2}x.$$

Note $\mathcal{B}_1(x) = x - \frac{1}{2}$: the first Bernoulli polynomial *is* the $n + \frac{1}{2}$ correction, written as a function.

The Bernoulli numbers are the boundary values: $\mathcal{B}_n := \mathcal{B}_n(0)$.

Remark 7.2 (Notation convention for B-splines and Bernoulli symbols)**.** *Three visually similar symbols occur in this chapter, but they denote different objects throughout the paper:*

1. *Plain B_n always denotes the n-th* B-spline. *Example: B_2 is the hat function and $B_m * B_n = B_{m+n}$.*
2. *Calligraphic $\mathcal{B}_n$ always denotes the n-th* Bernoulli number. *Example: $\mathcal{B}_2 = \frac{1}{6}$ and $\mathcal{B}_{2p}/(2p)!$ appears in Euler–Maclaurin.*
3. *Calligraphic $\mathcal{B}_n(x)$ always denotes the n-th* Bernoulli polynomial. *Example: $\mathcal{B}_1(x) = x - \frac{1}{2}$.*

In particular, plain B_n is never used for a Bernoulli quantity, and calligraphic $\mathcal{B}$ is never used for a spline quantity.

The Euler–Maclaurin formula is the expansion of f in the Bernoulli spline basis, with each $\mathcal{B}_{2p}/(2p)!$ term being the Bernoulli Green's function evaluated at the boundary.

The number theory thread:

$$\text{Bernoulli splines} \;\to\; \mathcal{B}_n = \mathcal{B}_n(0) \;\to\; \zeta(-n) = -\frac{\mathcal{B}_{n+1}}{n+1} \;\to\; \zeta(s) \text{ special values.}$$

Part IV

Extended Function Families

Chapter 8

The Extended Trigonometric Family and OFI

This chapter catalogues the complete extended trigonometric family of 34 functions and places each one within the OFI framework. Sections 8.1–8.11 establish the OFI orbit, arc, versed, hyperbolic, Gudermannian, and cardinal families. Section 8.15 establishes the pole-complementarity of tan and cot (Remark 8.9) and draws the structural analogy to the OFI half-integer bound shift. Section 8.16 gives the integration formulas for circular and square poles, connecting each pole type to a regime of the OFI resolvent (Corollary 8.14).

8.1 The OFI orbit of sine

The fundamental orbit identity: for $\omega > 0$,

$$D^{\alpha}[\sin(\omega\theta)] = \omega^{\alpha} \sin\!\left(\omega\theta + \tfrac{\alpha\pi}{2}\right). \tag{8.1}$$

This follows directly from the Riesz/Fourier multiplier: $\widehat{D^{\alpha}} f(\xi) = |\xi|^{\alpha} e^{i\,\mathrm{sgn}(\xi)\alpha\pi/2} \hat{f}(\xi)$, and $\widehat{\sin(\omega\theta)}$ is a two-point measure at $\pm\omega$.

α	$D^\alpha[\sin\theta]$	Name / interpretation
-1	$-\cos\theta$	antiderivative of sine
$-\frac{1}{2}$	$-\frac{\sqrt{2}}{2}(\sin\theta - \cos\theta)$	half-integral
0	$\sin\theta$	identity
$\frac{1}{4}$	$\sin(\theta + \pi/8)$	quarter-derivative (phase 22.5°)
$\frac{1}{3}$	$\sin(\theta + \pi/6)$	third-derivative (phase 30°)
$\frac{1}{2}$	$\sin(\theta + \pi/4) = \frac{\sqrt{2}}{2}(\sin\theta + \cos\theta)$	half-derivative (diagonal, phase 45°)
$\frac{2}{3}$	$\sin(\theta + \pi/3) = \frac{1}{2}\sin\theta + \frac{\sqrt{3}}{2}\cos\theta$	two-thirds derivative
$\frac{3}{4}$	$\sin(\theta + 3\pi/8)$	three-quarter derivative
1	$\cos\theta$	first derivative (phase 90°)
$\frac{3}{2}$	$\sin(\theta + 3\pi/4) = \frac{\sqrt{2}}{2}(-\sin\theta + \cos\theta)$	three-halves derivative
2	$-\sin\theta$	second derivative
3	$-\cos\theta$	third derivative
4	$\sin\theta$	period closes

The orbit has period 4 in α (integer steps), and rotates continuously by $\alpha\pi/2$ radians per unit of fractional order. The orbit is *closed*: every $D^\alpha[\sin\theta]$ is periodic, bounded, and in L^2. This is because $\sin\theta$ is an eigenfunction of $d^2/d\theta^2$ with eigenvalue -1, so $D^\alpha[\sin\theta] = (-1)^{\alpha/2}\sin\theta$ — which, interpreted as $e^{i\alpha\pi/2}\sin\theta$ for the Riesz branch, gives the phase rotation.

Remark 8.1 (The orbit is the *entire* primary pair)**.** *$D^\alpha[\sin\theta]$ generates both $\sin\theta$ ($\alpha = 0$) and $\cos\theta$ ($\alpha = 1$). $\sin$ and $\cos$ are not two separate functions — they are* one function at two orbit points*. The full circle: $\alpha \in [0, 4)$ sweeps $\sin \to \cos \to -\sin \to -\cos \to \sin$.*

8.2 The tan·cot identity and coverage of $\mathbb{R}$

8.2.1 The Pythagorean and semigroup identities

Two identities, both equal to 1, from different mechanisms:

$$\sin^2\theta + \cos^2\theta = 1 \tag{8.2}$$

$$\tan\theta \cdot \cot\theta = \frac{\sin\theta}{\cos\theta} \cdot \frac{\cos\theta}{\sin\theta} = 1. \tag{8.3}$$

Equation (8.2) is the L^2 norm of the orbit: the orbit vector $(\sin\theta, \cos\theta)$ lies on the unit circle in $\mathbb{R}^2$. Equation (8.3) is the multiplicative inverse: tan and cot are reciprocals. In OFI terms, (8.3) is the analogue of $I^\alpha \circ I^{-\alpha} = I^0 = \mathrm{Id}$: the positive-order operator times its negative-order inverse is identity.

8.2.2 Coverage of $\mathbb{R}$ and the small-angle bridge

$\tan\theta$ is a bijection from $(-\frac{\pi}{2}, \frac{\pi}{2})$ to $\mathbb{R}$: every real number is a tangent value. Similarly $\cot\theta$ is a bijection from $(0, \pi)$ to $\mathbb{R}$. Together, on $(0, \frac{\pi}{2})$ they both cover $(0, \infty)$, and their product is 1 throughout.

At $\theta = 0$: The small-angle series are

$$\sin\theta = \theta - \tfrac{\theta^3}{6} + \cdots, \qquad \tan\theta = \theta + \tfrac{\theta^3}{3} + \cdots, \qquad \cos\theta = 1 - \tfrac{\theta^2}{2} + \cdots$$

At first order: $\tan\theta \approx \sin\theta \approx \theta$ and $\cos(0) = 1$. The deviation between tan and sin at second order is:

$$\tan\theta - \sin\theta = \sin\theta(\sec\theta - 1) = \sin\theta \cdot \operatorname{exsec}\theta \;\approx\; \frac{\theta^3}{2} + O(\theta^5).$$

The *exsecant* exactly measures the quadratic separation of tan from its first-order approximation sin.

Comparison	Value at $\theta = 0$	Leading term	Role
$\sin\theta$	0	θ	first-order universal
$\tan\theta$	0	θ	same linear term
$\cos\theta$	1	$1 - \theta^2/2$	starts at 1, drops
$\tan\theta - \sin\theta$	0	$\theta^3/2$	exsecant correction
$1 - \cos\theta = \operatorname{vers}\theta$	0	$\theta^2/2$	quadratic drop-off
$\sec\theta - 1 = \operatorname{exsec}\theta$	0	$\theta^2/2$	same quadratic

8.3 The complete primary and reciprocal family: calculus

Function	Definition	$\frac{d}{d\theta} f(\theta)$	$\int f(\theta)\, d\theta$
$\sin\theta$	–	$\cos\theta$	$-\cos\theta$
$\cos\theta$	–	$-\sin\theta$	$\sin\theta$
$\tan\theta$	$\sin\theta/\cos\theta$	$\sec^2\theta = 1 + \tan^2\theta$	$-\ln\lvert\cos\theta\rvert$
$\cot\theta$	$\cos\theta/\sin\theta$	$-\csc^2\theta = -(1 + \cot^2\theta)$	$\ln\lvert\sin\theta\rvert$
$\sec\theta$	$1/\cos\theta$	$\sec\theta\tan\theta$	$\ln\lvert\sec\theta + \tan\theta\rvert$
$\csc\theta$	$1/\sin\theta$	$-\csc\theta\cot\theta$	$\ln\lvert\csc\theta - \cot\theta\rvert$
Pythagorean identities: $\sin^2 + \cos^2 = 1$; $1 + \tan^2 = \sec^2$; $1 + \cot^2 = \csc^2$			
Product identities: $\tan \cdot \cot = 1$; $\sin \cdot \csc = 1$; $\cos \cdot \sec = 1$			

Remark 8.2 (OFI orbit applies only to sin and cos)**.** *The orbit formula* $D^\alpha[f] = f(\theta+\alpha\pi/2)$ *applies to* sin *and* cos *because they are eigenfunctions of* $d^2/d\theta^2$ *(eigenvalue* -1*). The functions* $\tan, \cot, \sec, \csc$ *are rational in* $\sin, \cos$*; they are* not *eigenfunctions and their fractional derivatives are not computable by the orbit formula.* $D^\alpha[\tan\theta]$ *requires Fourier series analysis of* $\sec^2$*, and the result is an infinite series — not a single phase shift.*

8.4 Arc (inverse) trigonometric family

Function	Domain	$\frac{d}{dx}f(x)$
$\arcsin x$	$[-1,1] \to [-\pi/2, \pi/2]$	$1/\sqrt{1-x^2}$
$\arccos x$	$[-1,1] \to [0,\pi]$	$-1/\sqrt{1-x^2}$
$\arctan x$	$\mathbb{R} \to (-\pi/2, \pi/2)$	$1/(1+x^2)$
$\operatorname{arccot} x$	$\mathbb{R} \to (0,\pi)$	$-1/(1+x^2)$
$\operatorname{arcsec} x$	$\lvert x\rvert \geq 1 \to [0,\pi] \setminus \{\pi/2\}$	$1/(\lvert x\rvert\sqrt{x^2-1})$
$\operatorname{arccsc} x$	$\lvert x\rvert \geq 1 \to [-\pi/2,\pi/2] \setminus \{0\}$	$-1/(\lvert x\rvert\sqrt{x^2-1})$

Complementary pairs: $\arcsin x + \arccos x = \pi/2$; $\arctan x + \operatorname{arccot} x = \pi/2$.

OFI connection: The derivatives of the arc functions are algebraic: $1/\sqrt{1-x^2}$, $1/(1+x^2)$, etc. These are *Riesz-type potentials restricted to* $[-1,1]$ or to $\mathbb{R}$, respectively. Specifically: $1/(1+x^2)$ is the Poisson kernel of the upper half-plane; $1/\sqrt{1-x^2}$ is the weight function for the Chebyshev measure on $[-1,1]$, which equals $(I^{1/2}_{\mathrm{OFI}}\delta)(\cdot)$ in the sense of the semi-circle distribution. The arc functions are the OFI *antiderivatives* of these algebraic kernels.

8.5 The versed, coversed, and half-versed families

8.5.1 Definitions and calculus

The versed family arises from the four signs $\pm(1 \pm \cos\theta)$ and $\pm(1 \pm \sin\theta)$. Every function has a clean OFI derivative because $D^\alpha[\text{const}] = 0$ (Riesz convention) and $D^\alpha[\cos\theta]$, $D^\alpha[\sin\theta]$ are in the orbit (8.1).

Name	Symbol	Formula	$d/d\theta$	D^α (Riesz)
versine	$\operatorname{vers}\theta$	$1-\cos\theta$	$\sin\theta$	$-\cos(\theta+\frac{\alpha\pi}{2})$
coversine	$\operatorname{covers}\theta$	$1-\sin\theta$	$-\cos\theta$	$-\sin(\theta+\frac{\alpha\pi}{2})$
vercosine	$\operatorname{verc}\theta$	$1+\cos\theta$	$-\sin\theta$	$\cos(\theta+\frac{\alpha\pi}{2})$
covercosine	$\operatorname{coverc}\theta$	$1+\sin\theta$	$\cos\theta$	$\sin(\theta+\frac{\alpha\pi}{2})$
haversine	$\operatorname{hav}\theta$	$\frac{1}{2}(1-\cos\theta)$	$\frac{1}{2}\sin\theta$	$-\frac{1}{2}\cos(\theta+\frac{\alpha\pi}{2})$
hacoversine	$\operatorname{hacov}\theta$	$\frac{1}{2}(1-\sin\theta)$	$-\frac{1}{2}\cos\theta$	$-\frac{1}{2}\sin(\theta+\frac{\alpha\pi}{2})$
havercosine	$\operatorname{havc}\theta$	$\frac{1}{2}(1+\cos\theta)$	$-\frac{1}{2}\sin\theta$	$\frac{1}{2}\cos(\theta+\frac{\alpha\pi}{2})$
hacovercosine	$\operatorname{hcoverc}\theta$	$\frac{1}{2}(1+\sin\theta)$	$\frac{1}{2}\cos\theta$	$\frac{1}{2}\sin(\theta+\frac{\alpha\pi}{2})$

8.5.2 Identities and OFI structure

- $\operatorname{vers}\theta = 2\operatorname{hav}\theta = 2\sin^2(\theta/2)$: the versine is twice the haversine, and both are non-negative.
- $\operatorname{verc}\theta = 2\operatorname{havc}\theta = 2\cos^2(\theta/2)$: similarly for the complementary pair.
- $\operatorname{hav}\theta + \operatorname{havc}\theta = \frac{1}{2}(\text{vers} + \text{verc}) = \frac{1}{2}(1 - \cos + 1 + \cos) = 1$: the haversine and havercosine partition unity.
- $\operatorname{hacov}\theta + \operatorname{hcoverc}\theta = \frac{1}{2}(1 - \sin + 1 + \sin) = 1$: the hacoversine pair also partitions unity.
- OFI pairing: $D^\alpha[\text{hav}] = -\frac{1}{2}D^\alpha[\cos]$, $D^\alpha[\text{havc}] = \frac{1}{2}D^\alpha[\cos]$, so $D^\alpha[\text{hav}]+D^\alpha[\text{havc}] = 0$ (the partition unity is preserved at the derivative level).

Remark 8.3 (Haversine as OFI energy)**.** $\operatorname{hav}\theta = \sin^2(\theta/2)$ *is the squared L^2-norm of the half-frequency orbit element* $\sin(\theta/2)$*. This is directly analogous to the OFI energy* $\left\|(-\Delta)^{1/4}f\right\|_{L^2}^2 = \left\|I_{\text{OFI}}^{-1/4}f\right\|_{L^2}^2$ *in the Riesz setting: both measure energy at half-order. The Haversine formula in spherical geometry uses exactly this energy for great-circle distance — a connection to the Riesz kernel on S^2.*

8.6 The external secant family

Name	Symbol	Formula	$d/d\theta$	Value at $\theta = 0$
exsecant	$\operatorname{exsec}\theta$	$\sec\theta - 1 = (1-\cos\theta)/\cos\theta$	$\sec\theta\tan\theta$	0
excosecant	$\operatorname{excsc}\theta$	$\csc\theta - 1 = (1-\sin\theta)/\sin\theta$	$-\csc\theta\cot\theta$	∞ (undefined at 0)

Exsecant as deviation: $\operatorname{exsec}\theta = \sec\theta - 1 = \dfrac{1-\cos\theta}{\cos\theta} = \dfrac{\operatorname{vers}\theta}{\cos\theta}$: the exsecant is the versine divided by cosine. It measures the excess of the secant above 1 — the deviation of the reciprocal-cosine from the identity. For small θ: $\operatorname{exsec}\theta \approx \theta^2/2$.

Connection to tan: $\tan\theta - \sin\theta = \sin\theta \cdot \operatorname{exsec}\theta$ — the exsecant is the correction that turns $\sin\theta$ into $\tan\theta$.

OFI note: $D^\alpha[\operatorname{exsec}\theta] = D^\alpha[\sec\theta]$ (since $D^\alpha[1] = 0$ in the Riesz convention). There is no clean orbit formula for $D^\alpha[\sec\theta]$; it requires Fourier series of $\sec\theta$.

8.7 Chord, sagitta, and the geometric family

Name	Symbol	Formula	Classical use
chord	$\operatorname{crd}\theta$	$2\sin(\theta/2)$	Hipparchus; arc length approx.
sagitta	$\operatorname{sag}\theta$	$\operatorname{vers}\theta = 1-\cos\theta$	arrow height of circular arc
apothem	—	$\cos(\pi/n)$ for n-gon	inradius/circumradius ratio

Chord calculus: $\operatorname{crd}\theta = 2\sin(\theta/2)$, so $D[\operatorname{crd}\theta] = \cos(\theta/2)$. The OFI orbit formula applies at half-frequency $\omega = 1/2$:

$$D^\alpha[\operatorname{crd}\theta] = 2\cdot\left(\tfrac{1}{2}\right)^\alpha \sin\!\left(\tfrac{\theta}{2} + \tfrac{\alpha\pi}{2}\right) = 2^{1-\alpha}\sin\!\left(\tfrac{\theta}{2} + \tfrac{\alpha\pi}{2}\right). \tag{8.4}$$

Verification: $\alpha = 1$ gives $2^0\cos(\theta/2) = \cos(\theta/2)$. ✓The chord orbit *has period* 8 in α (since it involves $e^{i\alpha\pi/2}$ at frequency $\omega = 1/2$, returning to start at $\alpha = 4$, but the $2^{1-\alpha}$ amplitude scaling means $D^4[\mathrm{crd}] = \mathrm{crd}/16$, not crd itself — the amplitude decays as $2^{-\alpha+1}$).

The sagitta is the versine (arrow): $\operatorname{sag}\theta = 1 - \cos\theta = \operatorname{vers}\theta$.

8.8 The hyperbolic family: complete calculus

8.8.1 Definitions and calculus

The six hyperbolic functions are defined via exponentials: $\sinh x = (e^x - e^{-x})/2$, $\cosh x = (e^x + e^{-x})/2$, etc.

Function	Definition	d/dx	$\int dx$
$\sinh x$	$(e^x - e^{-x})/2$	$\cosh x$	$\cosh x$
$\cosh x$	$(e^x + e^{-x})/2$	$\sinh x$	$\sinh x$
$\tanh x$	$\sinh x/\cosh x$	$\text{sech}^2 x = 1 - \tanh^2 x$	$\ln\cosh x$
$\coth x$	$\cosh x/\sinh x$	$-\text{csch}^2 x = -(\coth^2 x - 1)$	$\ln\lvert\sinh x\rvert$
$\text{sech}\, x$	$1/\cosh x$	$-\text{sech}\, x\tanh x$	$2\arctan(e^x) = \text{gd}(x)$
$\text{csch}\, x$	$1/\sinh x$	$-\text{csch}\, x\coth x$	$\ln\lvert\tanh(x/2)\rvert$
Identities: $\cosh^2 - \sinh^2 = 1$; $1 - \tanh^2 = \text{sech}^2$; $\coth^2 - 1 = \text{csch}^2$			

8.8.2 The hyperbolic orbit

The hyperbolic functions satisfy $D^2[\sinh x] = \sinh x$ (eigenvalue $+1$ for D^2), unlike the circular case ($D^2[\sin x] = -\sin x$, eigenvalue -1).

Decomposing via exponentials: $D^\alpha[e^{\lambda x}] = \lambda^\alpha e^{\lambda x}$ for $\lambda > 0$. Therefore:

$$D^\alpha[\sinh x] = \frac{e^x - (-1)^\alpha e^{-x}}{2}, \qquad D^\alpha[\cosh x] = \frac{e^x + (-1)^\alpha e^{-x}}{2}. \tag{8.5}$$

For integer α: $D^{2k}[\sinh x] = \sinh x$, $D^{2k+1}[\sinh x] = \cosh x$ (period 2, not 4). For non-integer α, $(-1)^\alpha = e^{i\alpha\pi}$ is complex-valued, so $D^\alpha[\sinh x]$ is *complex-valued* for non-integer α. This is the key structural difference from the circular orbit: the hyperbolic orbit does not remain real for fractional α.

Remark 8.4 (Hyperbolic E-spline connection)**.** *The hyperbolic functions are the Green's functions of the E-spline operator $L = D^2 - \omega^2$: $L[\sinh(\omega x)] = 0$, $L[\cosh(\omega x)] = 0$. E-splines with $\lambda = \pm\omega$ generate all hyperbolic families. This is the OFI connection: the hyperbolic family is an E-spline family at imaginary frequency ($\lambda = \omega$ real, not $\lambda = i\omega$ circular).*

8.9 The inverse hyperbolic family

Function	Explicit form	Domain	d/dx
$\operatorname{arcsinh} x$	$\ln(x + \sqrt{x^2+1})$	$\mathbb{R}$	$1/\sqrt{1+x^2}$
$\operatorname{arccosh} x$	$\ln(x + \sqrt{x^2-1})$	$[1, \infty)$	$1/\sqrt{x^2-1}$
$\operatorname{arctanh} x$	$\frac{1}{2} \ln \frac{1+x}{1-x}$	$(-1, 1)$	$1/(1-x^2)$
$\operatorname{arccoth} x$	$\frac{1}{2} \ln \frac{x+1}{x-1}$	$\|x\| > 1$	$1/(1-x^2)$
$\operatorname{arcsech} x$	$\ln \frac{1+\sqrt{1-x^2}}{x}$	$(0, 1]$	$-1/(x\sqrt{1-x^2})$
$\operatorname{arccsch} x$	$\ln\left(\frac{1}{x} + \frac{\sqrt{1+x^2}}{\|x\|}\right)$	$x \neq 0$	$-1/(\|x\|\sqrt{1+x^2})$

Relation to logarithms: All inverse hyperbolic functions are elementary: $\operatorname{arcsinh} x = \ln(x + \sqrt{x^2+1})$, etc. Their derivatives are algebraic kernels — Riesz-type potentials on $\mathbb{R}$ (e.g., $1/\sqrt{1+x^2}$ is the Cauchy/Poisson kernel for $|x| < \infty$).

OFI connection: The integral $\int_0^x 1/\sqrt{1+t^2}\, dt = \operatorname{arcsinh} x$ is an L^2-weighted Riesz-type integral; $1/\sqrt{1+t^2} = D^{1/2}_{\mathrm{arc}}[\operatorname{arcsinh}]$ in a generalized sense.

8.10 The Gudermannian: circular–hyperbolic bridge

The *Gudermannian* function $\operatorname{gd}(x)$ is the unique bijection $\mathbb{R} \to (-\pi/2, \pi/2)$ that converts hyperbolic functions into circular:

$$\operatorname{gd}(x) := \arctan(\sinh x) = \arcsin(\tanh x) = 2\arctan(e^x) - \tfrac{\pi}{2}. \tag{8.6}$$

Key values: $\operatorname{gd}(0) = 0$, $\lim_{x\to+\infty} \operatorname{gd}(x) = \pi/2$, $\lim_{x\to-\infty} \operatorname{gd}(x) = -\pi/2$.
Derivative: $\operatorname{gd}'(x) = \operatorname{sech} x = 1/\cosh x$.
Conversion table via $\phi = \operatorname{gd}(x)$:

Circular (ϕ)	Hyperbolic (x)
$\sin\phi$	$\tanh x$
$\cos\phi$	$\operatorname{sech} x$
$\tan\phi$	$\sinh x$
$\sec\phi$	$\cosh x$
$\csc\phi$	$\coth x$
$\cot\phi$	$\operatorname{csch} x$

OFI connection: The Gudermannian is the integral of sech: $\operatorname{gd}(x) = \int_0^x \operatorname{sech}(t)\, dt$. It maps the hyperbolic half-line to the circular quarter-circle. In OFI terms: the Gudermannian is an $I^1_{\mathrm{OFI}}[\operatorname{sech}]$ evaluated at x, modulo normalizations — a first-order OFI integral of a hyperbolic function that produces a circular function. It is the *single bridge* between the two orbit families: every hyperbolic value converts to a circular value via gd.

8.11 Cardinal sine and the sinc family

Function	Formula	OFI connection
Unnormalized sinc	$\operatorname{sinc}(x) = \sin(x)/x$	$\lim_{\alpha\to\infty} B_\alpha$ at scale $1/\alpha$
Normalized sinc	$\operatorname{sinc}(x) = \sin(\pi x)/(\pi x)$	Fourier dual of $B_1 = \mathbf{1}_{[-1/2,1/2]}$
$\operatorname{sinc}^2(x)$	$\sin^2(\pi x)/(\pi x)^2$	Fourier dual of B_2 (triangular)
$\operatorname{sinc}^\alpha(x)$	$(\sin(\pi x)/(\pi x))^\alpha$	fractional B-spline [4] in Fourier
Dirichlet kernel	$D_N(x) = \sum_{n=-N}^{N} e^{inx}$	$= \sin((2N+1)x/2)/\sin(x/2)$
Fejér kernel	$F_N(x) = \frac{1}{N}\sum_{k=0}^{N-1} D_k(x)$	$= \operatorname{sinc}^2$ at discrete level

Key identity: $\int_{-\infty}^{\infty} \operatorname{sinc}(x)\,dx = 1$ (the normalized sinc integrates to 1 over all of $\mathbb{R}$).

Fourier duality: $\mathcal{F}[B_1](\xi) = \operatorname{sinc}(\xi)$ (boxcar $\leftrightarrow$ normalized sinc). $\mathcal{F}[B_n](\xi) = \operatorname{sinc}^n(\xi)$ (B-spline $\leftrightarrow$ sinc power). $\mathcal{F}[\beta^\alpha](\xi) = \operatorname{sinc}^{\alpha+1}(\xi)$ (fractional B-spline). The Riesz OFI kernel in Fourier space is $|\xi|^{-\alpha} = \lim_{n\to\alpha} \operatorname{sinc}^n$ in an appropriate distributional sense.

Derivative:

$$\operatorname{sinc}'(x) = \frac{\pi\cos(\pi x)}{\pi x} - \frac{\sin(\pi x)}{\pi x^2} = \frac{\cos(\pi x)}{x} - \frac{\sin(\pi x)}{\pi x^2}.$$

There is no clean orbit formula for $D^\alpha[\operatorname{sinc}(x)]$; in Fourier space, $\widehat{D^\alpha[\operatorname{sinc}]}(\xi) = |\xi|^\alpha \operatorname{sinc}(\xi)$ — the product of a power law and a sinc, which is an intrinsically bimodal object.

8.12 The complete prefix/suffix naming taxonomy

Every extended trig function name is generated by a systematic prefix/suffix grammar:

Morpheme	Meaning	Example
co-	complement $(\pi/2-\theta)$	$\cos\theta = \sin(\pi/2-\theta)$; $\cot = \tan(\pi/2-\theta)$
ver(sed)-	$1-\cos(\cdot)$	$\operatorname{vers}\theta = 1-\cos\theta$
ha(lf)-	multiply by $\frac{1}{2}$	$\operatorname{hav}\theta = \operatorname{vers}(\theta)/2$
ex-	subtract 1 (external/excess)	$\operatorname{exsec}\theta = \sec\theta - 1$
arc- / a-	inverse function	arcsin, arctan
h (suffix or infix)	hyperbolic variant	sinh, cosh, tanh
ar-	inverse hyperbolic	arcsinh, arctanh
crd	chord $(2\sin(\theta/2))$	from Hipparchus
sag	sagitta (vers)	arc arrow
gd	Gudermannian	bridge operator
Systematic extensions (documented but rare):		
coha- = haco-	half-complement versed	$\operatorname{hacov} = (1-\sin)/2$
ver+co	versed cosine	$\operatorname{verc}\theta = 1+\cos\theta$
ha+ver+co	half versed cosine	$\operatorname{havc} = (1+\cos)/2$
ha+co+ver+co	half coversed cosine	$\operatorname{hcoverc} = (1+\sin)/2$
Undocumented extensions (by analogy):		
quaver-	$\frac{1}{4}(1-\cos\theta)$	quarter-versine
versinh	$1-\cosh x$ (negative for real x)	hyperbolic versine
haversinh	$\frac{1}{2}(1-\cosh x)$	half hyperbolic versine
coversinh	$1-\sinh x$	hyperbolic coversine
exsinh	$\sinh x - 0 = \sinh x$ (trivial)	external sinh (degenerate)
arcvers	$\arccos(1-x)$	inverse versine
arccovers	$\arcsin(1-x)$	inverse coversine
archav	$\arccos(1-2x)$	inverse haversine (spherical distance)

Remark 8.5 (The grammar terminates)**.** *The prefix system generates a finite family: each prefix (co-, ha-, ver-) can be applied at most once meaningfully, giving at most $2^3 = 8$ combinations for the sin/cos pair, matching exactly the 8 versed/half-versed functions catalogued above. Further prefixes either degenerate (e.g.,* $\operatorname{ha}(\operatorname{ha}(\operatorname{vers})) = \operatorname{vers}/4$*, a rational rescaling) or produce trivial functions. The ex- prefix applied to the arc functions gives the inverse versine family. The hyperbolic extension via -h doubles the count.*

8.13 OFI orbit for the complete zoo

A function $f(\theta)$ has a *clean OFI orbit* $D^{\alpha}[f(\theta)] = g(\theta, \alpha)$ with g a bounded periodic function if and only if f is a finite linear combination of $\{\sin(\omega\theta), \cos(\omega\theta), 1\}$ for some frequencies ω.

Function	Clean orbit?	$D^\alpha[f(\theta)]$	Remark
$\sin\theta$	✓	$\sin(\theta+\alpha\pi/2)$	primary
$\cos\theta$	✓	$\cos(\theta+\alpha\pi/2)$	orbit element 1
$\operatorname{vers}\theta$	✓	$-\cos(\theta+\alpha\pi/2)$	orbit of $-\cos$
$\operatorname{coversine}\theta$	✓	$-\sin(\theta+\alpha\pi/2)$	orbit of $-\sin$
$\operatorname{verc}\theta$	✓	$\cos(\theta+\alpha\pi/2)$	orbit of cos
$\operatorname{coverc}\theta$	✓	$\sin(\theta+\alpha\pi/2)$	orbit of sin
$\operatorname{hav}\theta$	✓	$-\frac{1}{2}\cos(\theta+\alpha\pi/2)$	scaled orbit
$\operatorname{hacov}\theta$	✓	$-\frac{1}{2}\sin(\theta+\alpha\pi/2)$	scaled
$\operatorname{havc}\theta$	✓	$\frac{1}{2}\cos(\theta+\alpha\pi/2)$	scaled
$\operatorname{hcoverc}\theta$	✓	$\frac{1}{2}\sin(\theta+\alpha\pi/2)$	scaled
$\operatorname{crd}\theta$	✓	$2^{1-\alpha}\sin(\theta/2+\alpha\pi/2)$	half-freq
$\tan\theta$	×	infinite series	not eigenfunction
$\cot\theta$	×	infinite series	not eigenfunction
$\sec\theta$	×	infinite series	not eigenfunction
$\csc\theta$	×	infinite series	not eigenfunction
$\operatorname{exsec}\theta$	×	$=D^\alpha[\sec\theta]$	not clean
$e^{\lambda x}$	✓	$\lambda^\alpha e^{\lambda x}$	exp scaling
$\sinh x$	partial	real for integer α, complex otherwise	E-spline
$\cosh x$	partial	real for integer α, complex otherwise	E-spline
$\tanh, \coth, \operatorname{sech}, \operatorname{csch}$	×	no clean formula	rational
$\arcsin, \arctan, \ldots$	×	no clean formula	algebraic
$\operatorname{arcsinh}, \operatorname{arctanh}, \ldots$	×	no clean formula	algebraic
$\operatorname{gd}(x)$	×	no clean formula	bridge only
$\operatorname{sinc}(x)$	×	$\|\xi\|^\alpha\cdot\operatorname{sinc}(\xi)$ in Fourier	bimodal

8.14 DSP seeds and OFI connections

Digital signal processing is built on a minimal set of *seed functions* from which all operations are generated. In OFI language these seeds correspond to specific fractional orders or generating objects.

DSP seed	Symbol	OFI identity	Fourier multiplier
Dirac impulse	$\delta(t)$	$D^{-n}[\delta] = B_n$	1
Boxcar (rect)	$B_1 = \mathbf{1}_{[-1/2,1/2]}$	$D^{-1}[\delta]$	$\mathrm{sinc}(\xi)$
Triangle	$B_2 = B_1 * B_1$	$D^{-2}[\delta]$	$\mathrm{sinc}^2(\xi)$
Sinc	$\mathrm{sinc}(t)$	$\mathcal{F}[B_1]$, $\alpha \to \infty$ limit	$B_1(\xi)$
Complex exponential	$e^{j\omega_0 t}$	eigenfunction of D; $D[e^{j\omega}] = j\omega e^{j\omega}$	$\delta(\xi - \omega_0)$
Real sinusoid	$\cos(\omega_0 t)$, $\sin(\omega_0 t)$	orbit elements	$\delta(\xi \pm \omega_0)$
Unit step	$u(t) = \mathbf{1}_{t\geq 0}$	$D^{-1}[\delta]$ one-sided	$1/(j\xi) + \pi\delta(\xi)$
Gaussian	$g_\sigma(t) = e^{-t^2/(2\sigma^2)}$	B_∞ limit; heat kernel at $t = \sigma^2/2$	$e^{-2\pi^2\sigma^2\xi^2}$
Chirp	$\mathrm{chirp}(t) = e^{j\alpha t^2}$	fractional Fourier kernel at angle α	$e^{-j\xi^2/(4\alpha)}$ (Fresnel)
Raised cosine	$r(t) = \frac{1}{2}(1 + \cos(\pi t/T))$	$= \mathrm{havc}(\pi t/T)$	finite support in $[-1/T, 1/T]$

The seed hierarchy:

$$\begin{aligned} \delta &\xrightarrow{D^{-1}} B_1 \text{ (rect)} \xrightarrow{D^{-1}} B_2 \text{ (triangle)} \xrightarrow{D^{-1}} B_3 \text{ (quadratic)} \\ &\xrightarrow{*n} B_n \xrightarrow{\alpha\in\mathbb{R}} \beta^\alpha \xrightarrow{h\to 0} \mathrm{sinc}^\alpha \xrightarrow{\alpha\to\infty} \text{Gaussian.} \end{aligned}$$

Every DSP filter is a linear combination of convolutions of these seeds at various fractional orders.

Fractional Fourier transform (FrFT): The FrFT of order ϕ is the unitary operator that rotates the time-frequency plane by angle ϕ:

$$\begin{aligned} \mathcal{F}_\phi[f](u) &= \int_{-\infty}^{\infty} f(t)\, K_\phi(t,u)\, dt, \\ K_\phi(t,u) &= \frac{e^{-i(\pi/4-\phi/2)}}{\sqrt{2\pi|\sin\phi|}} \exp\!\left(i\frac{t^2+u^2}{2}\cot\phi - i\frac{tu}{\sin\phi} \right). \end{aligned}$$

At $\phi = \pi/2$: the standard Fourier transform. At $\phi = \alpha\pi/2$: the OFI orbit angle α corresponds to a rotation by $\alpha\pi/2$ in the Wigner phase space. The chirp seed $e^{j\alpha t^2}$ is the FrFT eigenfunction family.

OFI–DSP dictionary:

DSP concept	OFI analogue
Convolution $f * g$	$I^{\alpha}_{\mathrm{OFI}}$ (convolution with power-law kernel)
Low-pass filter	$I^{\alpha}_{\mathrm{OFI}}$ (smoothing at order α)
High-pass filter	$D^{\alpha}_{\mathrm{OFI}}$ (roughening at order α)
Nyquist poles at $\xi = \pm n$	Gamma poles of $1/\Gamma(\alpha)$ at $\alpha = -n$
Sampling theorem	$n + \frac{1}{2}$ shift (bridge theorem)
DFT basis $e^{-j2\pi kn/N}$	Orbit at frequency k/N
Raised cosine filter	$\mathrm{havc}(\pi t/T)$ directly
Root raised cosine	$\sqrt{\mathrm{havc}(\pi t/T)}$ (half-order)
Wiener filter $H(\xi) = S_x(\xi)/(S_x(\xi) + S_n(\xi))$	Riesz OFI at order α chosen by SNR

Remark 8.6 (Raised cosine is a havercosine)**.** *The raised cosine pulse used in Nyquist ISI-free filtering is:* $r(t) = \frac{1}{2}(1 + \cos(\pi t/T))$ *for* $|t| \leq T$*,* 0 *otherwise. This is exactly* $\mathrm{havc}(\pi t/T)$ *— the havercosine evaluated at* $\pi t/T$*. Its Fourier transform has finite support* $[-1/T, 1/T]$ *(no out-of-band energy). The* root raised cosine *filter (matched filter for RRC) corresponds to taking the half-order:* $\sqrt{\mathrm{havc}} = |\cos(\pi t/(2T))|$*, which is the half-order OFI of the square-root profile.*

8.15 Tan, cot, and the ordering mechanism

This section has two goals: (1) prove precisely where and why the RL fractional derivative fails to compose correctly, identifying the correction term explicitly; and (2) show that the pole structure of tan and cot provides a geometric analogy for the same half-integer shift that OFI uses to restore the semigroup law. The analogy is structural, not causal: the semigroup law is proved by Beta-function convolution (Theorem 12.14); the trig picture explains why the fix takes the form it does.

Where the Riemann–Liouville ordering fails

For fractional *integrals* ($\alpha, \beta > 0$), the RL operator satisfies the semigroup law exactly — this is the Beta-function convolution (equation (12.4)). The failure arises for *derivatives*, specifically in mixed integral–derivative compositions.

Proposition 8.7 (RL composition with boundary correction)**.** *Let* $0 < \beta \leq 1$*,* $\alpha > 0$*, and* $f \in L^1[a, b]$*. Then:*

$$D^{\alpha}_{\mathrm{RL}}\Big[I^{\beta}_{\mathrm{RL}} f\Big](x) = I^{\beta-\alpha}_{\mathrm{RL}} f(x) - \frac{\Big[I^{1-\beta}_{\mathrm{RL}} f\Big](a)}{\Gamma(1-\alpha)} (x-a)^{-\alpha}, \qquad 0 < \alpha < 1. \tag{8.7}$$

The second term vanishes if and only if $\Big[I^{1-\beta}_{\mathrm{RL}} f\Big](a) = 0$*.*

Proof. Write $I^{\beta}_{\mathrm{RL}} f(x) = \frac{1}{\Gamma(\beta)} \int_a^x (x-t)^{\beta-1} f(t)\,dt$. Apply $D^{\alpha}_{\mathrm{RL}} = \frac{d}{dx} I^{1-\alpha}_{\mathrm{RL}}$:

$$D^{\alpha}_{\mathrm{RL}}[I^{\beta}_{\mathrm{RL}} f](x) = \frac{d}{dx} \frac{1}{\Gamma(1-\alpha)\Gamma(\beta)} \int_a^x (x-s)^{-\alpha} \int_a^s (s-t)^{\beta-1} f(t)\,dt\,ds.$$

Switch the order of integration (Fubini, integrands non-negative after taking $|f|$) and evaluate the inner s-integral via the Beta function $\int_t^x (x-s)^{-\alpha}(s-t)^{\beta-1}\,ds = B(1-\alpha,\beta)(x-t)^{\beta-\alpha}$:

$$= \frac{B(1-\alpha,\beta)}{\Gamma(1-\alpha)\Gamma(\beta)} \frac{d}{dx} \int_a^x (x-t)^{\beta-\alpha} f(t)\,dt = \frac{d}{dx} \frac{1}{\Gamma(1-\alpha+\beta)} \int_a^x (x-t)^{\beta-\alpha} f(t)\,dt.$$

If $\beta > \alpha$, differentiating under the integral gives $I^{\beta-\alpha}_{\mathrm{RL}} f(x)$ directly. If $\beta \le \alpha < \beta + 1$, the integral has a non-integrable singularity at $t = x$, and differentiation picks up the boundary term $[I^{1-\beta}_{\mathrm{RL}} f](a) \cdot (x-a)^{-\alpha}/\Gamma(1-\alpha)$ from the lower limit. This yields (8.7). ☐

Remark 8.8 (Why the boundary term does not vanish generically)**.** *The boundary value* $[I^{1-\beta}_{\mathrm{RL}} f](a) = \frac{1}{\Gamma(1-\beta)} \int_a^a (a-t)^{-\beta} f(t)\,dt = 0$ *trivially when the integral has an empty domain — but this is only for* f *identically zero near* a*. For general* f*, the term is nonzero. The Caputo derivative avoids it by construction (it sets* $f = 0$ *before the lower limit), but then acquires its own boundary asymmetry in Laplace-space. Neither RL nor Caputo has a clean compositional law in general.*

The Gamma function's role: where $1/\Gamma(1-\alpha)$ lives

The correction term in (8.7) involves $1/\Gamma(1-\alpha)$. Recall that $1/\Gamma$ is entire with zeros at $z = 0, -1, -2, \ldots$, so:

$$\frac{1}{\Gamma(1-\alpha)} = 0 \iff 1-\alpha \in \{0,-1,-2,\ldots\} \iff \alpha \in \{1,2,3,\ldots\}.$$

At positive integer orders, the boundary correction *vanishes identically* — consistent with the classical fact that integer-order derivatives compose without remainder. At every non-integer order, $1/\Gamma(1-\alpha) \neq 0$ and the correction is present.

The correction term is therefore not merely a technicality: it is a *residue of the Gamma pole structure* that appears whenever a non-integer fractional derivative acts on a one-sided fractional integral.

The OFI resolution: symmetrize the kernel

The Riesz kernel $|x-t|^{\alpha-1}$ is symmetric in x and t. When OFI uses this kernel on $\mathbb{R}$ (or with the half-integer lower-limit extension), the composition integral becomes:

$$(I^{\alpha}_{\mathrm{OFI}} \circ I^{\beta}_{\mathrm{OFI}} f)(x) = \frac{1}{\Gamma(\alpha)\Gamma(\beta)} \int_{\mathbb{R}}\int_{\mathbb{R}} |x-s|^{\alpha-1}|s-t|^{\beta-1} f(t)\,ds\,dt.$$

The inner s-integral is the convolution of two Riesz kernels, which by the Fourier multiplier calculation $(|\xi|^{-\alpha})(|\xi|^{-\beta}) = |\xi|^{-(\alpha+\beta)}$ gives $I^{\alpha+\beta}_{\mathrm{OFI}} f(x)$ directly — with *no boundary term*, because the symmetric kernel has no preferred endpoint at which a remainder accumulates. This is proved fully in Theorem 12.14.

The tan/cot analogy

The Riesz kernel resolution has a precise structural parallel in the extended trig family. Define the *pole sets*:

$$P_{\cot} = \{n\pi : n \in \mathbb{Z}\} \quad \text{(integers)}, \qquad P_{\tan} = \left\{\tfrac{\pi}{2} + n\pi : n \in \mathbb{Z}\right\} \quad \text{(half-integers)}.$$

Remark 8.9 (Pole-complementarity of tan and cot)**.** tan *is real-analytic on* $\mathbb{R} \setminus P_{\tan}$ *and* cot *on* $\mathbb{R} \setminus P_{\cot}$*, with* $P_{\tan} \cap P_{\cot} = \emptyset$*. They are related by:*

$$\tan\theta = -\cot\!\left(\theta + \tfrac{\pi}{2}\right), \tag{8.8}$$

verified by $-\cot(\theta + \pi/2) = -\cos(\theta + \pi/2)/\sin(\theta + \pi/2) = \sin\theta/\cos\theta = \tan\theta$*. The half-period shift moves every pole from an integer multiple of* π *to a half-integer multiple, and each branch restricts to a bijection onto* $\mathbb{R}$ *(strictly increasing for* tan*, strictly decreasing for* cot*, by* $(\tan)' = \sec^2 > 0$ *and* $(\cot)' = -\csc^2 < 0$*).*

The analogy to OFI is structural:

Setting	Integer-pole structure	Half-integer resolution
$\cot\theta$	poles at $n\pi$	smooth at $(n + \frac{1}{2})\pi$
$\tan\theta = -\cot(\theta + \pi/2)$	smooth at $n\pi$	poles at $(n + \frac{1}{2})\pi$
RL: $D^{\alpha}_{\mathrm{RL}} I^{\beta}_{\mathrm{RL}} f$	correction at each non-integer α	vanishes only at $\alpha \in \mathbb{Z}$
OFI: $I^{\alpha}_{\mathrm{OFI}} I^{\beta}_{\mathrm{OFI}} f$	no boundary endpoint	$I^{\alpha} I^{\beta} = I^{\alpha+\beta}$ exactly
$1/\Gamma(1-\alpha)$	zero at $\alpha \in \{1, 2, 3, \ldots\}$	Riesz kernel bypasses this entirely

In both settings, the pathology lives at integer positions and the geometric picture is the same: a half-integer shift moves the pole structure off the integers. The OFI proof that this restores the semigroup is Theorem 12.14; the trig table makes the mechanism visible.

Sec, csc, and the ground-state shift

$\sec\theta$ and $\csc\theta$ share the pole sets of tan and cot respectively, but map onto $(-\infty, -1] \cup [1, \infty)$ rather than $\mathbb{R}$ — they are the *reciprocal* rather than ratio of the circular functions. The external functions $\operatorname{exsec}\theta = \sec\theta - 1$ and $\operatorname{excsc}\theta = \csc\theta - 1$ subtract the constant 1, shifting the range to $(-\infty, -2] \cup [0, \infty)$.

In OFI spectral language this is the ground-state subtraction: if $\lambda_{\min} > 0$ is the spectral gap, then $\mathcal{O} - \lambda_{\min} I$ has its spectrum beginning at 0. The function exsec models this: $\sec\theta - 1 \geq 0$ for all θ in its domain, with minimum 0 achieved at $\theta = 0$.

Arctan and arccot: compactifications and the resolvent phase

The inverse functions close the pole-complementarity picture:

$$\arctan : \mathbb{R} \xrightarrow{\sim} \left(-\tfrac{\pi}{2},\, \tfrac{\pi}{2}\right), \qquad \operatorname{arccot} : \mathbb{R} \xrightarrow{\sim} (0,\, \pi).$$

Both are homeomorphisms onto complementary open intervals, and together they cover $(-\pi/2, \pi)$ (with overlap on $(0, \pi/2)$). Adding the endpoints via one-point compactification maps $\pm\infty \mapsto \pm\pi/2$ and gives the compact image $[-\pi/2, \pi/2]$.

In the OFI resolvent, arctan appears concretely: the spectral measure of $(-\Delta + m^2)^{-1}$ integrated from 0 to a cutoff Λ is

$$\int_0^\Lambda \frac{d\xi}{\xi^2 + m^2} = \frac{1}{m} \arctan \frac{\Lambda}{m} \xrightarrow{\Lambda\to\infty} \frac{\pi}{2m}.$$

The arctan encodes the total spectral weight above mass m; its finite limit as $\Lambda \to \infty$ is the statement that the resolvent is a bounded operator with finite operator norm $\pi/(2m)$.

8.16 Circular and square poles in integration

Every rational function has poles in $\mathbb{C}$ that fall into one of two canonical real types. The type determines which member of the extended trig family closes the integral. We derive each antiderivative from first principles, identify the versed functions as the systematic correction terms in circular pole reduction, and prove the Gudermannian integral formula. The connection to OFI operator theory is noted where it is direct.

Circular poles: $x^2 + a^2$, poles at $\pm ia$

Proposition 8.10 (Circular pole integral)**.** *For $a > 0$:*

$$\int \frac{dx}{x^2 + a^2} = \frac{1}{a} \arctan \frac{x}{a} + C. \tag{8.9}$$

Proof. Substitute $x = a\tan\theta$, so $dx = a\sec^2\theta\, d\theta$ and $x^2 + a^2 = a^2\sec^2\theta$:

$$\int \frac{a\sec^2\theta\, d\theta}{a^2\sec^2\theta} = \frac{1}{a}\int d\theta = \frac{\theta}{a} + C = \frac{1}{a}\arctan\frac{x}{a} + C. \qquad \square$$

Proposition 8.11 (Order-n reduction and the versed correction terms)**.** *For $n \geq 1$:*

$$\int \frac{dx}{(x^2 + a^2)^n} = \frac{x}{2a^2(n-1)(x^2+a^2)^{n-1}} + \frac{2n-3}{2a^2(n-1)} \int \frac{dx}{(x^2+a^2)^{n-1}}. \tag{8.10}$$

After $n-1$ applications, the integral resolves to $\arctan(x/a)$ *plus a sum of terms of the form $c_{n,k}\, x/(x^2 + a^2)^k$ for $k = 1, \ldots, n-1$. Under $x = a\tan\theta$, each correction term satisfies:*

$$\frac{x}{(x^2 + a^2)^k} = \frac{\sin\theta\cos^{2k-1}\theta}{a^{2k-1}}, \tag{8.11}$$

and the power-reduction identity

$$\cos^{2k-2}\theta = \tfrac{1}{2^{2k-3}}\binom{2k-2}{k-1}\tfrac{1}{2} + \tfrac{1}{2^{2k-3}}\sum_j \binom{2k-2}{j}\cos\bigl((2k{-}2{-}2j)\theta\bigr)$$

expresses every factor as a linear combination of 1 *and* $\cos(2p\theta)$ *for integers* p*. Since* $\cos(2p\theta) = 1 - \operatorname{vers}(2p\theta)$*, the correction terms are* exactly *linear combinations of versed-family functions evaluated at integer multiples of* 2θ*.*

Proof. Differentiate the reduction formula with respect to a^2, or apply integration by parts with $u = (x^2 + a^2)^{-(n-1)}$ and $dv = dx$. The versed identification follows from $\operatorname{vers}(2p\theta) = 1 - \cos(2p\theta) = 2\sin^2(p\theta)$ and back-substitution $\theta = \arctan(x/a)$. □

For $n = 2$ explicitly:

$$\int \frac{dx}{(x^2 + a^2)^2} = \frac{x}{2a^2(x^2 + a^2)} + \frac{1}{2a^3}\arctan\frac{x}{a} + C. \tag{8.12}$$

The correction $x/(2a^2(x^2 + a^2))$ under $x = a\tan\theta$ is $\sin\theta\cos\theta/(2a^3) = \sin(2\theta)/(4a^3) = (1 - \operatorname{vers}(2\theta))/(4a^3) - 1/(4a^3)$, confirming the versed identification.

Square poles: $x^2 - a^2$, poles at $\pm a$

Proposition 8.12 (Square pole integral)**.** *For* $a > 0$*:*

$$\int \frac{dx}{x^2 - a^2} = \frac{1}{2a}\ln\left|\frac{x-a}{x+a}\right| + C = \begin{cases} -\dfrac{1}{a}\operatorname{arctanh}\dfrac{x}{a} + C & |x| < a, \\ -\dfrac{1}{a}\operatorname{arccoth}\dfrac{x}{a} + C & |x| > a. \end{cases} \tag{8.13}$$

Proof. Partial fractions: $\frac{1}{x^2-a^2} = \frac{1}{(x-a)(x+a)} = \frac{1}{2a}\left(\frac{1}{x-a} - \frac{1}{x+a}\right)$, so the integral is $\frac{1}{2a}\ln|x - a| - \frac{1}{2a}\ln|x + a| + C = \frac{1}{2a}\ln\left|\frac{x-a}{x+a}\right| + C$. For $|x| < a$: writing $u = x/a$,

$$\frac{1}{2a}\ln\left|\frac{x-a}{x+a}\right| = \frac{-1}{2a}\ln\left(\frac{1+u}{1-u}\right) = \frac{-1}{a}\operatorname{arctanh}(u),$$

since $\operatorname{arctanh}(u) = \frac{1}{2}\ln\frac{1+u}{1-u}$ for $|u| < 1$. For $|x| > a$: $\operatorname{arccoth}(u) = \frac{1}{2}\ln\frac{u+1}{u-1}$ for $|u| > 1$, giving the second case. □

The order-n reduction for square poles mirrors (8.10) with a sign change:

$$\int \frac{dx}{(x^2 - a^2)^n} = \frac{-x}{2a^2(n-1)(x^2 - a^2)^{n-1}} - \frac{2n-3}{2a^2(n-1)}\int \frac{dx}{(x^2 - a^2)^{n-1}}, \tag{8.14}$$

terminating at (8.13) with arctanh or ln plus rational corrections.

A model second-order Green's function

The connection between pole type and kernel decay is cleanest for the Helmholtz operator $\mathcal{H}_m = -d^2/dx^2 + m^2$ on $\mathbb{R}$, whose Green's function satisfies:

$$\mathcal{H}_m G_m(x) = \delta(x), \qquad G_m \in L^2(\mathbb{R}).$$

This is a second-order model; the full OFI Riesz potential $|\xi|^{-\alpha}$ is treated in Chapter 12. The computation below establishes that the pole type of $\hat{G}_m$ (circular vs. square) is what determines exponential decay vs. oscillation — a pattern that carries over to the fractional-order OFI setting.

Proposition 8.13 (OFI resolvent kernel via contour integration)**.** *For $m > 0$:*

$$G_m(x) = \frac{e^{-m|x|}}{2m} \qquad (x \in \mathbb{R}). \tag{8.15}$$

The Fourier transform of G_m is $\hat{G}_m(\xi) = 1/(\xi^2 + m^2)$, a circular pole at $|\xi| = im$.

Proof. Take the Fourier transform of $\mathcal{H}_m G_m = \delta$: $(\xi^2 + m^2)\hat{G}_m(\xi) = 1$, so $\hat{G}_m(\xi) = 1/(\xi^2 + m^2)$. This has simple poles at $\xi = \pm im$. Invert via residues:

$$G_m(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{e^{i\xi x}}{\xi^2 + m^2}\, d\xi.$$

For $x > 0$, close in the upper half-plane; the residue at $\xi = im$ is $e^{-mx}/(2im)$, giving $G_m(x) = \frac{1}{2\pi} \cdot 2\pi i \cdot \frac{e^{-mx}}{2im} = \frac{e^{-mx}}{2m}$. For $x < 0$, close in the lower half-plane; residue at $\xi = -im$ gives $G_m(x) = e^{mx}/(2m)$. Combined: $G_m(x) = e^{-m|x|}/(2m)$. □

Corollary 8.14 (Three integration regimes)**.** *For the operator $\mathcal{H}_c = -d^2/dx^2 + c$ with $c \in \mathbb{R}$, the Green's function takes one of three qualitatively distinct forms depending on the sign of c:*

Sign of c	$\hat{G}(\xi)$	*Pole type*	$G(x)$
$c > 0$	$\dfrac{1}{\xi^2 + c}$	*circular at* $\pm i\sqrt{c}$	$\dfrac{e^{-\sqrt{c}\,\lvert x\rvert}}{2\sqrt{c}}$ *(exponential decay)*
$c = 0$	$\dfrac{1}{\xi^2}$	*double pole at* 0	$-\frac{1}{2}\lvert x\rvert$ *(linear growth)*
$c < 0$	$\dfrac{1}{\xi^2 + c}$	*square at* $\pm\sqrt{\lvert c\rvert}$	$\dfrac{-\sin(\sqrt{\lvert c\rvert}\,\lvert x\rvert)}{2\sqrt{\lvert c\rvert}}$ *(oscillatory)*

Proof. The three cases follow from Proposition 8.13 by setting $m^2 = c$ for $c > 0$, and for $c \le 0$ by analytic continuation. For $c < 0$: $\hat{G}(\xi) = 1/(\xi^2 - |c|)$ has square poles at $\pm\sqrt{|c|}$; partial fractions and residues give $G(x) = -\sin(\sqrt{|c|}\,|x|)/(2\sqrt{|c|})$. □

The table encodes the complete trig antiderivative atlas in the $c > 0$ (arctan) regime, the logarithmic / linear-growth boundary at $c = 0$, and the arctanh / oscillatory regime for $c < 0$. The Gudermannian sits at the $c = 0$ boundary as shown in Proposition 8.15.

The Gudermannian as the phase boundary: proof

Proposition 8.15 (Gudermannian integral representation)**.**

$$\int_0^x \operatorname{sech}(t)\,dt = \arctan(\sinh x) = \operatorname{gd}(x). \tag{8.16}$$

Proof. Substitute $u = \sinh(t)$: then $du = \cosh(t)\,dt$ and $\operatorname{sech}(t) = 1/\cosh(t) = 1/\sqrt{1+\sinh^2 t} = 1/\sqrt{1+u^2}$, so $\operatorname{sech}(t)\,dt = du/\sqrt{1+u^2} \cdot 1/\sqrt{1+u^2} \cdot \sqrt{1+u^2} = du/(1+u^2)$. Wait — more directly: $\operatorname{sech}(t)\,dt = (1/\cosh t)\,dt$, and $du = \cosh t\,dt$, so $dt = du/\cosh t$ and $\operatorname{sech}(t)\,dt = dt/\cosh t = du/\cosh^2 t = du/(1+u^2)$. Therefore:

$$\int_0^x \operatorname{sech}(t)\,dt = \int_0^{\sinh x} \frac{du}{1+u^2} = \arctan(\sinh x). \qquad \square$$

The Gudermannian thus sits exactly at the $\mu = 0$ boundary of Corollary 8.14: $\operatorname{sech}(t) = 1/\cosh(t)$ is the *spectral density at the gap edge*, and $\operatorname{gd}(x)$ accumulates this density from 0 to x. Its range is $(-\pi/2, \pi/2)$ — the same interval as arctan — confirming it maps the hyperbolic line into the circular (arctan) world. The identities

$$\sin(\operatorname{gd} x) = \tanh x, \quad \cos(\operatorname{gd} x) = \operatorname{sech} x, \quad \tan(\operatorname{gd} x) = \sinh x$$

are all proved by differentiating both sides and checking they agree at $x = 0$.

Summary: the 34-function pole atlas

Function(s)	Poles	Antiderivative / closure	OFI role
$\sin, \cos$	none; range $[-1, 1]$	—	orbit eigenfunctions
$\tan$	half-integer, real	$-\ln\lvert\cos\theta\rvert$	analogy for OFI half-int. shift (Rem. 8.9)
$\cot$	integer, real	$\ln\lvert\sin\theta\rvert$	analogy for pre-shift RL structure
$\sec, \csc$	half-int. / int., real	$\ln\lvert\sec+\tan\rvert$; $-\ln\lvert\csc+\cot\rvert$	ground-state subtraction model
$\mathrm{vers}, \mathrm{hav}, \ldots$	none; $[0, 2], [0, 1]$	Prop. 8.11 corrections	circular reduction terms
$\arctan, \mathrm{arccot}$	none; range $(-\pi/2, \pi/2), (0, \pi)$	closes $c > 0$ circular poles	semigroup spectral phase
$\sinh, \cosh$	none; unbounded	—	$c < 0$ oscillatory regime
$\tanh, \coth$	coth: integer; range $(-1, 1), \mathbb{R}$	$\ln\cosh$; $\ln\lvert\sinh\rvert$	bounded hyperbolic functions
$\mathrm{arctanh}, \mathrm{arccoth}$	none; $(-1, 1) \to \mathbb{R}$; $\lvert x\rvert > 1 \to \mathbb{R}$	closes $c < 0$ square poles	$c < 0$ closure
$\mathrm{sech}, \mathrm{csch}$	csch: integer	$\mathrm{gd}(x)$; $2\arctan(e^x) + C$	Prop. 8.15 integrand
$\mathrm{gd}(x)$	none; range $(-\pi/2, \pi/2)$	$= \int_0^x \mathrm{sech}$	$c = 0$ boundary bridge
$\mathrm{gd}^{-1}(x)$	poles at $\pm\pi/2$	$\ln\lvert\tan(x/2 + \pi/4)\rvert$	inverse bridge
$\mathrm{sinc}(x)$	removable zero at 0	bimodal in Fourier	B-spline $\alpha \to \infty$ limit

Part V

Spline Families

Chapter 9

Spline Families: OFI Interpretations

9.1 OFI interpretations of spline families

For the L-spline families, each admits an OFI interpretation under one or more of three recurring characterizations (with the Green's-function / semigroup correspondence proved in Proposition 13.13 for the Riesz/OFI case, and the remaining identifications treated structurally):

(G) Green's function: ϕ satisfies $L^\alpha \phi = \delta$.

(V) Variational: ϕ minimizes $\|L^\alpha f\|^2$ subject to data constraints.

(S) Semigroup: $\phi_\alpha * \phi_\beta = \phi_{\alpha+\beta}$.

9.2 The spawning hierarchy

Two primordial objects spawn all 52 families:

Primordial	Spawning path
δ (Dirac delta)	via $B_1 = D^{-1}\delta$: all B-spline families
	via $G_L = L^{-1}\delta$: all Green's function families
$\Gamma(z)$ (Gamma function)	via Beta function: Bernstein $\to$ Bézier
	via $1/\Gamma(\alpha)$: fractional B-splines $\to$ Riesz
B_1 (first child)	$= D^{-1}\delta$; stands at root of B-spline semigroup
	$= \beta^0$; stands at $\alpha = 0$ of fractional family

9.3 Spline family table

Classical Polynomial Splines

Family	**Operator** L	**OFI connection**	**Order** α
Uniform B-spline B_n	D^n	seed of OFI semigroup	$n \in \mathbb{N}$
Non-uniform B-spline	D^n (local knots)	variable-knot OFI	$n \in \mathbb{N}$
Cardinal B-spline	D^n, integer knots	$n + \frac{1}{2}$ bridge	$n \in \mathbb{N}$
Natural cubic	D^2	H^2 OFI energy	2
Clamped / Periodic / NAK	D^2	boundary variants	2
Cubic Hermite	D^2, C^1	matched H^2	2
Quintic Hermite	D^3, C^2	H^3 OFI	3
Fractional Hermite	$D^{\alpha/2}$	frac. $H^{\alpha/2}$	$\alpha/2$
Osculatory	D^{m+1}	order $m+1$ OFI	$m+1$
Lacunary	D^α (gapped)	fractional order	$\alpha \in \mathbb{R}$
Perfect	D^n, L^∞	L^∞ OFI	n
Bernoulli	D^n periodic	EM correction basis	$n \in \mathbb{N}$
Euler	$(D^2+1)^n$	oscillatory OFI	n
Chebyshev	D^n, equiosc.	L^∞ optimal	n

Interpolating and Shape-Controlling Splines

Family	**Operator** L	**OFI connection**	**Order** α
Akima	D^2 (local)	robust H^2	2
Catmull-Rom	D^2 + central diff	$\pm\frac{1}{2}$ centering	2
Kochanek-Bartels (TCB)	$D^{1+\tau}$	frac. tension	$(1, 2)$
Monotone (Fritsch-Carlson)	D^2 constrained	sign-constrained	2

Tension and Shape Splines

Family	**Operator** L	**OFI connection**	**Order** α
Tension spline	$\tau^2 D^1 + D^2$	H^1-H^2 interp.	$(1, 2)$
Nu-spline	$D^{2+\nu}$	frac. order	$2+\nu$
Beta-spline (Barsky)	D^2, geom. cont.	(β_1, β_2) frac.	$(1, 2)$

Geometric and CAD Splines

Family	**Operator** L	**OFI connection**	**Order** α
Bézier (Bernstein basis)	Beta kernel	discrete RL integral	n
Rational Bézier	Beta + Möbius	conformal OFI	n
NURBS	D^n + rational	conformal B-spline	n
T-splines	D^n local	adaptive OFI	n
IGA splines	NURBS / D^p	FEM at OFI order p	p
Subdivision surfaces	D^n iterated	discrete OFI iteration	n
Clothoid / Spiro / Cornu	D^3 on curve	Fresnel = $\alpha = \frac{1}{2}$ OFI	$\frac{1}{2}$

Penalized and Statistical Splines

Family	**Operator** L	**OFI connection**	**Order** α
Smoothing spline	$D^m + \lambda$ pen.	H^m OFI + regularize	m
P-spline (Eilers-Marx)	Δ^m (discrete)	GL OFI on coefficients	m
Fractional P-spline	Δ^α	GL fractional OFI	$\alpha \in \mathbb{R}$
Regression spline	D^n (no penalty)	OFI projection	n
MARS (hinge functions)	D^{-1} (hinge)	$\alpha = 1$ one-sided OFI	1

Exponential and L-Splines

Family	**Operator** L	**OFI connection**	**Order** α
E-spline	$\prod(D - \lambda_k)$	exponential OFI	n
L-spline (general)	any L	Green's function OFI	variable
Trigonometric spline	$D^2 + \omega^2$	oscillatory L-spline	2
Hyperbolic spline	$D^2 - \omega^2$	hyperbolic L-spline	2

Radial Basis and Multivariate Splines

Family	**Operator** L	**OFI connection**	**Order** α
Thin-plate	$(-\Delta)^2$, $\mathbb{R}^2$	polyharmonic $m = 2$	4
Polyharmonic	$(-\Delta)^m$, $\mathbb{R}^n$	integer Riesz	$2m$
Frac. polyharmonic	$(-\Delta)^{\alpha/2}$	Riesz potential	α
Duchon spline	$(-\Delta)^m$, min. norm	Riesz minimizer	$2m$
Wendland (compact)	local $(-\Delta)^k$	radial B-spline	k
Multiquadric (Hardy)	Poisson semigroup	$e^{-c(-\Delta)^{1/2}}$	1
Box spline	$\partial_{\xi_1} \cdots \partial_{\xi_n}$	multivariate B-spline	n
Simplex spline	simplicial D^n	simplex OFI	n

Fractional Splines

Family	**Operator L**	**OFI connection**	**Order α**
Frac. B-spline β^α	D^α (GL)	continuous OFI semigroup	$\alpha \in \mathbb{R}_{>-1}$
Symmetric frac. B-spline	$(-\Delta)^{\alpha/2}$	Riesz OFI discrete	α
Variable-order spline	$D^{\alpha(x)}$	spatially varying OFI	$\alpha(x)$

Wavelet Splines

Family	**Operator L**	**OFI connection**	**Order α**
Battle–Lemarié	D^n + orthog.	OFI wavelet basis	n
CDF biorthogonal	D^n, $D^{\tilde{n}}$	dual OFI pair	$(n, \tilde{n})$
Semiorthog. B-spline	D^n primal	OFI + dual	n
Frac. wavelet frame	D^α	real-α OFI frame	α

Limits of the Hierarchy

Family	**Operator L**	**OFI connection**	**Order α**
Sinc / Whittaker-Shannon	D^∞	$\alpha \to \infty$ OFI	∞
Gaussian	$e^{t\Delta}$	heat semigroup, CLT limit	∞
Riesz / OFI kernel	$(-\Delta)^{\alpha/2}$	the OFI itself	α

Total catalogued here: 52 families.

Chapter 10

The Spline Map

Appendix B contains the full annotated diagram at landscape scale, with every family and the connections between them. Arrows: $\rightarrow$ spawns; $\dashrightarrow$ limit; $\cdots \leftrightarrow \cdots$ dual pair.

The diagram has four main regions. The Riesz/OFI node (violet ellipse) is the hub: it connects to the B-spline semigroup chain on the left, the radial and polyharmonic families on the lower left, and the variational families (thin-plate, smoothing, Wendland) on the right. The Bernoulli chain (gold, right edge) runs from Bernoulli splines through Bernoulli numbers and the Euler–Maclaurin formula to the values $\zeta(-n)$. At the bottom, the B-spline CLT limit (B_n as $n \to \infty$) and the Riesz heat-semigroup limit ($\alpha \to \infty$) converge independently to the Gaussian kernel. The P-spline carries a double arrow back to β^α indicating equivalence via the GL limit ($h \to 0$).

Part VI

Semigroup Catalogue

Chapter 11

OFI Semigroups

11.1 Semigroup structures in the OFI framework

#	Semigroup	Generator	OFI connection
1	OFI (Riesz integral)	I^{α}, Fourier mult. $\lvert\xi\rvert^{-\alpha}$	primary
2	B-spline	$B_n = B_1^{*n}$, conv. semigroup	discrete OFI, integer α
3	Frac. B-spline $\{\beta^{\alpha}\}$	$(1 - e^{-i\xi})^{\alpha+1}/(i\xi)^{\alpha+1}$	continuous, $\alpha \in \mathbb{R}$
4	Heat / Gaussian	$e^{t\Delta}$, mult. $e^{-t\lvert\xi\rvert^2}$	$\alpha = 2t$
5	Poisson	$e^{-t\sqrt{-\Delta}}$, mult. $e^{-t\lvert\xi\rvert}$	$\alpha = t$, half-order
6	Grünwald-Letnikov	$D_h^{-\alpha}$, $h \to 0$	GL $\to$ OFI
7	Translation	$T_h f(x) = f(x+h)$	shift underlying discrete ∂
8	Cardinal (sinc)	Shannon-Whittaker interpolation	$\alpha \to \infty$ B-spline
9	Ornstein-Uhlenbeck	$e^{-t}I + (1 - e^{-2t})^{1/2}\Delta^{1/2}$	$n + \frac{1}{2}$ in Hermite
10	Laguerre	$e^{-t}L_n$ operators	$n + \frac{1}{2}$ in Laguerre polys
11	Mehler (harmonic osc.)	e^{-tH}, $H = -\partial^2 + x^2$	$\Gamma(n + \frac{1}{2})$
12	Weierstrass	$e^{t\partial^2}$, formal power series	formal OFI

11.2 The B-spline to Riesz chain

The B-spline to Riesz chain:

$$B_1 \xrightarrow{*n} B_n \xrightarrow{\alpha\in\mathbb{R}} \beta^{\alpha} \xrightarrow{h\to 0} \text{Riesz OFI} \xrightarrow{\text{Fourier}} \lvert\xi\rvert^{-\alpha}.$$

Every arrow is calculus. The boxcar B_1 is the seed of the semigroup, not a statistical object.

Theorem 11.1 (Chain equivalences).

1. *$B_m * B_n = B_{m+n}$ (B-spline semigroup, integer order).*
2. *$\beta^\alpha * \beta^\gamma = \beta^{\alpha+\gamma+1}$. Equivalently, the reindexed family $\widetilde{\beta}^\sigma := \beta^{\sigma-1}$ for $\sigma > 0$ satisfies $\widetilde{\beta}^\sigma * \widetilde{\beta}^\tau = \widetilde{\beta}^{\sigma+\tau}$ (frac. B-spline semigroup, real order).*
3. *$h^{\alpha+1}\beta^\alpha(x/h) \to c_\alpha |x|^\alpha$ in $\mathcal{S}'(\mathbb{R})$ as $h \to 0$, where $c_\alpha = \Gamma(\alpha+1)\sin(\frac{\pi\alpha}{2})/\pi$ (Riesz kernel is the distributional continuum limit; see [1] §20.3).*
4. *$D^\alpha_{\mathrm{GL}}\,\beta^\gamma = \beta^{\gamma-\alpha}$ (GL reduces spline order; [4] Prop. 4).*
5. *$\beta^n \to B_{n+1}$ at integer n (frac. B-spline recovers integer B-spline).*

Part VII

Detailed Operator Theory

Chapter 12

Ordered Fractional Integrals: Full Treatment

12.1 Introduction

This chapter gives the rigorous bounded-interval treatment of the OFI operator. We define $I^{\alpha}_{\mathrm{OFI}}$ precisely on $L^1([a-\frac{1}{2}, b])$ and prove the semigroup law by an explicit Beta-function convolution (Theorem 12.14). We compare OFI with the four classical frameworks — Riemann–Liouville, Caputo, Riesz, and Grünwald–Letnikov — and identify it as the shifted-RL operator that eliminates boundary correction terms (Proposition 12.20). The Gamma-function residue regularization of $(-n)!$ is developed in Section 12.2, the discrete-to-continuous bridge via the $n+\frac{1}{2}$ shift is treated in Section 12.6, and the B-spline semigroup is identified as the natural discrete realization of the OFI semigroup (Theorem 12.22). The $L^2(\mathbb{R}^n)$ Hilbert-space realization, self-adjointness, and spectral lower bound are developed in the following chapter.

The Gamma function $\Gamma(z) = \int_0^\infty t^{z-1}e^{-t}\,dt$ extends the factorial $n! = \Gamma(n+1)$ to all complex z except the non-positive integers $\{0, -1, -2, \ldots\}$, where it has simple poles. For fractional calculus the central kernel is

$$K_\alpha(x,t) := \frac{(x-t)^{\alpha-1}}{\Gamma(\alpha)}, \qquad x > t,$$

which is well-behaved for $\alpha > 0$ but requires careful treatment when $\alpha \leq 0$ or when composing two fractional operators.

Three distinct problems arise and each has impeded a clean theory:

1. **Gamma poles.** $\Gamma(0) = \infty$ (simple pole), so "$(-1)!$" and all $(-n)!$ for $n \geq 1$ are naïvely undefined. Since $1/\Gamma(z)$ is entire with zeros at $z = 0, -1, -2, \ldots$, the kernel K_α formally handles this — but the *residue structure* carries meaningful information that the standard theory discards.

2. **Ordering problem.** Unlike integer-order calculus ($D^m D^n = D^{m+n}$), fractional operators satisfy ${}_aD^\alpha_x\,{}_aD^\beta_x f \neq {}_aD^{\alpha+\beta}_x f$ in general: the semigroup identity fails for

derivatives. It holds for integrals alone. The discrepancy is a boundary-term correction involving fractional initial data.

3. **Discrete-to-continuous bridge.** The naïve replacement $\sum_{k=1}^{n} f(k) \approx \int_1^n f(x)\,dx$ has error that does not vanish even with symmetric bounds. The correct bridge requires shifting the bounds to $[1-\frac{1}{2}, n+\frac{1}{2}]$, which introduces the Bernoulli correction series and recovers the Euler–Maclaurin formula exactly.

The *Ordered Fractional Integral* (OFI) is a unified response to all three.

12.2 The Gamma Function: Poles and Residues

12.2.1 Poles and the factorial at non-positive integers

The Gamma function satisfies the functional equation $\Gamma(z+1) = z\Gamma(z)$ and has simple poles at $z = 0, -1, -2, \ldots$ with residues:

$$\operatorname{Res}_{z=-n} \Gamma(z) \;=\; \frac{(-1)^n}{n!}, \qquad n = 0, 1, 2, \ldots \tag{12.1}$$

The values $(-n)! := \Gamma(-n+1)$ for $n \geq 1$ are therefore:

n	$(-n)!$	Gamma pole	Residue	$1/\Gamma(-n+1)$
0	$0! = 1$	(no pole)	—	1
1	$(-1)! = \Gamma(0)$	$z = 0$	$+1$	0
2	$(-2)! = \Gamma(-1)$	$z = -1$	-1	0
3	$(-3)! = \Gamma(-2)$	$z = -2$	$+\frac{1}{2}$	0
n	$(-n)! = \Gamma(-n+1)$	$z = -n+1$	$(-1)^{n-1}/(n-1)!$	0

Two facts are immediate:

- $1/\Gamma(-n+1) = 0$ for all $n \geq 1$: the *reciprocal* Gamma function is entire and assigns zero to all negative-integer arguments. This is the mechanism by which fractional integrals ${}_aI_x^{\alpha}$ reduce to derivatives for $\alpha = -n$: the kernel coefficient $1/\Gamma(\alpha)$ vanishes, removing the integral, and the n-fold differentiation emerges from the numerator $(x-t)^{\alpha-1}$.

- The residues (12.1) are not zero. They carry the "shadow" value of $(-n)!$ under pole regularization.

12.2.2 OFI regularization of $(-n)!$

Definition 12.1 (OFI-regularized factorial). *For $n \geq 1$, define:*

$$(-n)!_{\mathrm{OFI}} \;:=\; \mathrm{Res}_{z=-n+1}\,\Gamma(z) \;=\; \frac{(-1)^{n-1}}{(n-1)!}\,.$$

Equivalently, $(-n)!_{\mathrm{OFI}}$ is the coefficient of the simple pole of $\Gamma(z)$ at $z=-n+1$, recovered by the Laurent expansion $\Gamma(z) = \frac{(-1)^{n-1}/(n-1)!}{z+n-1} + O(1)$ near $z = -(n-1)$.

Proposition 12.2 (Values of OFI-regularized factorials).

$$\begin{aligned} &(-1)!_{\mathrm{OFI}} = 1, \quad && (-2)!_{\mathrm{OFI}} = -1, \quad && (-3)!_{\mathrm{OFI}} = \tfrac{1}{2}, \\ &(-4)!_{\mathrm{OFI}} = -\tfrac{1}{6}, && (-n)!_{\mathrm{OFI}} = \frac{(-1)^{n-1}}{(n-1)!}\,. \end{aligned}$$

Remark 12.3 (Recurrence for OFI-regularized factorials). *Direct computation from the explicit formula gives:*

$$\frac{(-n)!_{\mathrm{OFI}}}{(-(n-1))!_{\mathrm{OFI}}} = \frac{(-1)^{n-1}/(n-1)!}{(-1)^{n-2}/(n-2)!} = \frac{-1}{n-1},$$

i.e., $(-n)!_{\mathrm{OFI}} = -(-(n-1))!_{\mathrm{OFI}}/(n-1)$. The sign alternation ($(-1)^{n-1}$) is the signature of the simple-pole residue structure of $\Gamma(z)$ at non-positive integers.

Remark 12.4 (Reflection formula interpretation). *The reflection formula $\Gamma(z)\Gamma(1-z) = \pi/\sin(\pi z)$ at $z = 1/2$ gives $\Gamma(1/2)^2 = \pi$, so $\Gamma(1/2) = \sqrt{\pi}$. This is the origin of $\sqrt{\pi}$ in the Gaussian integral $\int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi} = \Gamma(1/2)$ and connects to the OFI half-integer shift $\alpha = n + 1/2$ in Section 12.5.*

12.3 Classical Fractional Calculus Frameworks

We survey the four main frameworks to establish notation and identify the ordering problem precisely.

12.3.1 Riemann–Liouville (RL)

Definition 12.5 (RL Fractional Integral and Derivative). *For $\alpha > 0$ and $a \leq x$:*

$$ {}_aI_x^{\alpha} f(x) := \frac{1}{\Gamma(\alpha)} \int_a^x (x-t)^{\alpha-1} f(t)\,dt, \tag{12.2}$$

$$ {}_aD_x^{\alpha} f(x) := D^n\Big[{}_aI_x^{n-\alpha} f\Big](x), \qquad n = \lceil \alpha \rceil. \tag{12.3}$$

Semigroup for integrals. The RL integral satisfies the semigroup law:

$$ {}_aI_x^{\alpha}\,{}_aI_x^{\beta} f = {}_aI_x^{\alpha+\beta} f, \qquad \alpha, \beta > 0. \tag{12.4}$$

This follows from the Beta function identity $B(\alpha,\beta)=\Gamma(\alpha)\Gamma(\beta)/\Gamma(\alpha+\beta)$.

Semigroup failure for derivatives. For $0<\alpha,\beta<1$:

$$ {}_aD_x^\alpha\, {}_aD_x^\beta f(x) = {}_aD_x^{\alpha+\beta} f(x) - \frac{\left[{}_aI_x^{1-\beta} f\right]_{x=a}}{\Gamma(1-\alpha)}(x-a)^{-\alpha}. \tag{12.5} $$

The correction term $\frac{[{}_aI_x^{1-\beta} f]_{x=a}}{\Gamma(1-\alpha)}(x-a)^{-\alpha}$ vanishes only when ${}_aI_x^{1-\beta} f$ vanishes at $x=a$.

12.3.2 Caputo

Caputo reverses the order of integration and differentiation:

$$ {}^{\mathrm{C}}_{a}D_x^\alpha f(x) := {}_aI_x^{n-\alpha}[D^n f](x) = \frac{1}{\Gamma(n-\alpha)}\int_a^x (x-t)^{n-\alpha-1} f^{(n)}(t)\,dt, \qquad n=\lceil\alpha\rceil. \tag{12.6} $$

The difference between RL and Caputo is:

$$ {}_aD_x^\alpha f(x) - {}^{\mathrm{C}}_{a}D_x^\alpha f(x) = \sum_{k=0}^{n-1} \frac{f^{(k)}(a)}{\Gamma(k-\alpha+1)}(x-a)^{k-\alpha}. \tag{12.7} $$

When all $f^{(k)}(a)=0$ for $k=0,\dots,n-1$, RL and Caputo agree.

Key advantage of Caputo. The Laplace transform of ${}^{\mathrm{C}}_{0}D_x^\alpha f$ involves the initial values $f(0), f'(0),\dots,f^{(n-1)}(0)$ (integer-order), making Caputo natural for differential equations with classical initial conditions.

12.3.3 Riesz

The Riesz fractional integral on $\mathbb{R}$ is the symmetric combination of left and right RL operators:

$$ \mathcal{R}^\alpha f(x) := \frac{1}{2\Gamma(\alpha)\cos(\pi\alpha/2)}\int_{-\infty}^{\infty} |x-t|^{\alpha-1} f(t)\,dt, \qquad \alpha\neq 1,3,5,\dots \tag{12.8} $$

In Fourier space: $\widehat{\mathcal{R}^\alpha f}(\xi) = |\xi|^{-\alpha}\hat f(\xi)$, so $\mathcal{R}^\alpha$ is the Fourier multiplier $|\xi|^{-\alpha}$.

The *Riesz derivative* (fractional Laplacian) is:

$$ (-\Delta)^{\alpha/2} f(x) = \mathcal{F}^{-1}\Big[|\xi|^\alpha\,\hat f(\xi)\Big](x), \tag{12.9} $$

the Fourier multiplier by $|\xi|^\alpha$, inverting the Riesz integral. Riesz is non-local and symmetric; it has no preferred direction.

12.3.4 Grünwald–Letnikov

The Grünwald–Letnikov (GL) derivative is the discrete fractional finite-difference:

$$ D_h^\alpha f(x) := h^{-\alpha}\sum_{k=0}^{\infty}(-1)^k\binom{\alpha}{k} f(x-kh), \qquad D^\alpha f := \lim_{h\to 0^+} D_h^\alpha f, \tag{12.10} $$

where the generalized binomial coefficient is $\binom{\alpha}{k}=\frac{\Gamma(\alpha+1)}{\Gamma(k+1)\Gamma(\alpha-k+1)}$.

GL agrees with RL for suitable functions. It is the natural discrete fractional calculus and underlies the B-spline semigroup (Section 12.8).

12.3.5 Comparison table

Property	RL	Caputo	Riesz	GL
Semigroup (integrals)	✓	✓	✓	✓
Semigroup (derivatives)	×	×	✓	✓
Integer initial conditions	×	✓	—	—
Non-local / symmetric	×	×	✓	×
Discrete realization	≈	≈	≈	✓
$D^\alpha[\text{const}] = 0$	×	✓	✓	✓
Ordering: $D^\alpha D^\beta = D^{\alpha+\beta}$	×	×	✓	✓

The failure of the semigroup/ordering property for RL and Caputo derivatives is the central motivation for OFI.

Remark 12.6 (Two distinct function-space settings in this chapter). *Two settings appear in this chapter and must not be conflated.*

- ***Bounded-interval OFI*** *(Definition 12.10): I^α_{OFI} acts on $f \in L^1([a,b])$, extended by zero to $[a - \frac{1}{2}, a)$. The semigroup law is proved by Beta-function convolution (Theorem 12.14). This is the primary OFI setting.*
- ***Riesz/whole-line setting*** *(Section 12.3.3): $\mathcal{R}^\alpha$ acts on $f \in L^2(\mathbb{R})$ (or $\mathcal{S}(\mathbb{R})$) with Fourier multiplier $|\xi|^{-\alpha}$. The semigroup law $\mathcal{R}^\alpha\mathcal{R}^\beta = \mathcal{R}^{\alpha+\beta}$ is immediate from Fourier calculus and requires no boundary convention.*

The two are related: restricting the Riesz operator to functions supported on $[a,b]$ with zero extension to $[a - \frac{1}{2}, a)$ recovers the bounded-interval OFI as a special case. Statements proved in one setting do not automatically transfer to the other without this restriction.

12.4 The Ordering Problem

Definition 12.7 (Ordering problem). *The* ordering problem *in fractional calculus is: given operators D^α and D^β of orders $\alpha, \beta > 0$, does $D^\alpha D^\beta = D^{\alpha+\beta}$? If not, which ordering $D^\alpha D^\beta$ or $D^\beta D^\alpha$ gives the "correct" or "canonical" result?*

For RL derivatives, the correction in (12.5) shows:

$$ {}_aD^\alpha_x {}_aD^\beta_x f - {}_aD^\beta_x {}_aD^\alpha_x f = \frac{[{}_aI^{1-\beta}_x f]_{x=a}}{\Gamma(1-\alpha)}(x-a)^{-\alpha} - \frac{[{}_aI^{1-\alpha}_x f]_{x=a}}{\Gamma(1-\beta)}(x-a)^{-\beta}. \tag{12.11} $$

This commutator is *non-zero* in general: the operators do not commute, and neither ordering gives $D^{\alpha+\beta}$ without boundary data.

Example 12.8 (Ordering failure: $\alpha = \beta = 1/2$)**.** *Let $f(x) = x^{1/2}$ and $a = 0$. Then:*

$$ {}_0I_x^{1/2} f(x) = \frac{1}{\Gamma(1/2)} \int_0^x (x-t)^{-1/2} t^{1/2}\, dt = \frac{\pi x}{2\Gamma(1/2)} = \frac{\sqrt{\pi}\, x}{2}. $$

Hence $[{}_0I_x^{1/2} f]_{x=0} = 0$, and in this case the correction term vanishes: ${}_0D_x^{1/2}\, {}_0D_x^{1/2} f = {}_0D_x^1 f = f'(x) = x^{-1/2}/2$.

However, for $f(x) = 1$ (constant): $[{}_0I_x^{1/2} 1]_{x=0} = 0$ but ${}_0D_x^{1/2} 1 = x^{-1/2}/\Gamma(1/2) \neq 0$, and ${}_0D_x^{1/2}\, {}_0D_x^{1/2} 1 = {}_0D_x^1 1 = 0$ while ${}_0D_x^1\, 1 = 0$ as expected.

The problematic case: $f(x) = (x-a)^\mu$ for $\mu > -1$ not a non-negative integer. Here $[{}_aI_x^{1-\beta} f]_{x=a} \neq 0$ in general, and the ordering correction is non-trivial.

Proposition 12.9 (When ordering is canonical)**.** *The RL derivative satisfies ${}_aD_x^\alpha\, {}_aD_x^\beta f = {}_aD_x^{\alpha+\beta} f$ if and only if the boundary data $[{}_aI_x^{1-\beta} f]_{x=a} = 0$. This holds in particular when:*

(i) $f(a) = f'(a) = \cdots = f^{(\lceil\beta\rceil-1)}(a) = 0$, or

(ii) f is supported strictly away from a, or

(iii) $\alpha, \beta > 0$ and $\alpha + \beta$ is a positive integer (integer-order composition).

12.5 The Ordered Fractional Integral

12.5.1 Definition and semigroup property

Definition 12.10 (OFI on a bounded interval)**.** *Let $f \in L^1([a,b])$ and $\alpha > 0$. The* Ordered Fractional Integral *of order α on $[a,b]$ is:*

$$ I^\alpha_{\mathrm{OFI}} f(x) \;:=\; \frac{1}{\Gamma(\alpha)} \int_{a-1/2}^x (x-t)^{\alpha-1} f(t)\, dt, \qquad x \in [a,b], \tag{12.12} $$

where the lower limit is shifted to $a - \frac{1}{2}$. We require $f(t) = 0$ for $t \in [a - \frac{1}{2}, a)$. The basic OFI semigroup identity is:

$$ \boxed{I^\alpha_{\mathrm{OFI}} \circ I^\beta_{\mathrm{OFI}} = I^{\alpha+\beta}_{\mathrm{OFI}}}, \tag{12.13} $$

which is proved under this convention in Theorem 12.14.

Proposition 12.11 (L^1 boundedness of OFI)**.** *For $\alpha > 0$ and $f \in L^1([a - \frac{1}{2}, b])$, the operator I^α_{OFI} maps $L^1([a - \frac{1}{2}, b])$ to $L^1([a,b])$ and satisfies:*

$$ \|I^\alpha_{\mathrm{OFI}} f\|_{L^1([a,b])} \;\leq\; \frac{(b - a + \frac{1}{2})^\alpha}{\Gamma(\alpha+1)}\, \|f\|_{L^1([a-\frac{1}{2},b])}. $$

Proof. By Tonelli's theorem (non-negative integrands):

$$ \|I^\alpha_{\mathrm{OFI}} f\|_{L^1([a,b])} = \frac{1}{\Gamma(\alpha)} \int_a^b \int_{a-1/2}^x (x-t)^{\alpha-1} |f(t)|\, dt\, dx. $$

Switching the order of integration (Fubini, since the integrand is non-negative after taking $|f|$):

$$= \frac{1}{\Gamma(\alpha)} \int_{a-1/2}^{b} |f(t)| \int_{\max(a,t)}^{b} (x-t)^{\alpha-1}\, dx\, dt \;\le\; \frac{1}{\Gamma(\alpha)} \int_{a-1/2}^{b} |f(t)| \frac{(b-t)^{\alpha}}{\alpha}\, dt.$$

Since $b - t \le b - a + \frac{1}{2}$ for $t \ge a - \frac{1}{2}$, the bound follows. □

Remark 12.12 (Relation to the centering heuristic)**.** *The $n + \frac{1}{2}$ centering principle (Chapter 6) motivates thinking of the kernel as "centered" at $(x + (a - 1/2))/2$. For the discrete-to-continuous bridge (Chapter 6), the shifted bounds $[a - \frac{1}{2}, b + \frac{1}{2}]$ appear naturally. The definition above adopts the half-shift at the lower limit only, which is sufficient to eliminate the boundary correction terms in the semigroup composition and is consistent with the RL convention $a \mapsto a - \frac{1}{2}$.*

12.5.2 The $n + \frac{1}{2}$ shift: why it works

The half-integer shift is not ad hoc. It is the shift that simultaneously achieves three goals:

1. Makes the semigroup law exact for f with zero extension to $[a - \frac{1}{2}, a)$ (Theorem 12.14).
2. Recovers the Euler–Maclaurin formula as the leading correction for the discrete-to-continuous bridge, with $n + \frac{1}{2}$ bounds giving zero error for linear f (Theorem 12.18).
3. Is consistent with the reflection formula $\Gamma(1/2) = \sqrt{\pi}$ at half-integer order (Proposition 12.13).

Proposition 12.13 (Half-order OFI on the constant function)**.**

$$I^{1/2}_{\mathrm{OFI}}[1](x) = \frac{1}{\Gamma(1/2)} \int_0^x (x-t)^{-1/2}\, dt = \frac{2\sqrt{x}}{\sqrt{\pi}}, \qquad x \in [0, b],$$

for the constant function on $[0, b]$ extended by 0 to $[-\frac{1}{2}, 0)$. This exhibits the normalization factor $1/\Gamma(1/2) = 1/\sqrt{\pi}$ directly.

12.5.3 Semigroup theorem

Theorem 12.14 (OFI Semigroup)**.** *Let $\alpha, \beta > 0$ and let $f \in L^1([a, b])$ with $f = 0$ on $[a - \frac{1}{2}, a)$. Then:*

$$I^{\alpha}_{\mathrm{OFI}} \circ I^{\beta}_{\mathrm{OFI}} f = I^{\alpha+\beta}_{\mathrm{OFI}} f \quad \text{a.e. on } [a, b].$$

The family $\{I^{\alpha}_{\mathrm{OFI}}\}_{\alpha>0}$ forms a one-parameter semigroup of bounded operators on $L^1([a - \frac{1}{2}, b])$.

Proof. Write $a_0 = a - \frac{1}{2}$. Since $f = 0$ on $[a_0, a)$, compute:

$$(I^{\alpha}_{\mathrm{OFI}} I^{\beta}_{\mathrm{OFI}} f)(x) = \frac{1}{\Gamma(\alpha)\Gamma(\beta)} \int_{a_0}^{x} (x-y)^{\alpha-1} \int_{a_0}^{y} (y-t)^{\beta-1} f(t)\, dt\, dy.$$

By Fubini's theorem (applicable since $f \in L^1$ and the power-law kernels are locally integrable for $\alpha, \beta > 0$):

$$= \frac{1}{\Gamma(\alpha)\Gamma(\beta)} \int_{a_0}^{x} f(t) \underbrace{\int_{t}^{x} (x-y)^{\alpha-1}(y-t)^{\beta-1}\, dy}_{=(x-t)^{\alpha+\beta-1} B(\alpha,\beta)}\, dt = \frac{1}{\Gamma(\alpha+\beta)} \int_{a_0}^{x} (x-t)^{\alpha+\beta-1} f(t)\, dt,$$

which equals $(I^{\alpha+\beta}_{\mathrm{OFI}} f)(x)$. Boundedness of each $I^{\alpha}_{\mathrm{OFI}}$ as an operator on $L^1([a-\frac{1}{2}, b])$ is established in Proposition 12.11. $\square$

Remark 12.15 (Ordering problem resolution for integrals; derivatives deferred)**.** *Theorem 12.14 resolves the ordering problem for* integrals*:* $I^{\alpha}_{\mathrm{OFI}} \circ I^{\beta}_{\mathrm{OFI}} = I^{\alpha+\beta}_{\mathrm{OFI}}$*. For* derivatives*, the corresponding identity holds for the Riesz fractional derivative* $(-\Delta)^{\alpha/2}$ *on* $\mathcal{S}(\mathbb{R})$*, where* $(-\Delta)^{\alpha/2}(-\Delta)^{\beta/2} = (-\Delta)^{(\alpha+\beta)/2}$ *is immediate from Fourier calculus. A formal OFI derivative operator* $D^{\alpha}_{\mathrm{OFI}}$ *defined as the left-inverse of* $I^{\alpha}_{\mathrm{OFI}}$ *on bounded domains would require a Sobolev-space analysis that is not developed in this paper on bounded domains; the* $\mathbb{R}^n$ *derivative semigroup is developed in Chapter 13. No derivative semigroup result on bounded domains is claimed here.*

12.6 Discrete-to-Continuous Bridge

12.6.1 The $n + \frac{1}{2}$ integration bounds

A fundamental observation: the naïve integral $\int_1^n f(x)\, dx$ does not equal $\sum_{k=1}^{n} f(k)$, and the error does not vanish even for symmetric corrections. The *correct* continuous image of the discrete sum uses shifted bounds.

Theorem 12.16 (Exact sum via shifted integral)**.** *For $f(x) = x$ (linear):*

$$\sum_{k=1}^{n} k \;=\; \int_{1/2}^{n+1/2} x\, dx \;=\; \frac{(n+1/2)^2 - (1/2)^2}{2} \;=\; \frac{n^2+n}{2} \;=\; \frac{n(n+1)}{2}. \quad (\textit{exact})$$

For $f(x) = x^2$ (quadratic):

$$\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}, \qquad \int_{1/2}^{n+1/2} x^2\, dx = \frac{n^3}{3} + \frac{n^2}{2} + \frac{n}{4}.$$

The discrepancy is $\sum k^2 - \int_{1/2}^{n+1/2} x^2\, dx = -\frac{n}{12}$*, confirmed by direct computation:* $\sum_{k=1}^{n} k^2 = n(n+1)(2n+1)/6 = n^3/3 + n^2/2 + n/6$ *and the integral equals* $n^3/3 + n^2/2 + n/4$*, giving residual* $n/6 - n/4 = -n/12$*.*

Remark 12.17. *For the linear case the shifted integral is* exact*: no Bernoulli correction is needed. This is because the OFI shift "centers" the trapezoid rule on the integer grid perfectly for polynomials of degree* ≤ 1*. For degree* ≥ 2*, the Bernoulli series kicks in.*

12.6.2 Euler–Maclaurin formula as OFI residual series

Theorem 12.18 (Euler–Maclaurin via OFI). *For $f \in C^{2P+2}([1/2, n+1/2])$:*

$$\sum_{k=1}^{n} f(k) = \int_{1/2}^{n+1/2} f(x)\,dx + \sum_{p=1}^{P} \frac{\mathcal{B}_{2p}}{(2p)!}\Big[f^{(2p-1)}(n+\tfrac{1}{2}) - f^{(2p-1)}(\tfrac{1}{2})\Big] + R_P, \tag{12.14}$$

where $\mathcal{B}_{2p}$ are Bernoulli numbers (notation: Section 15.2.10) and

$$|R_P| \;\leq\; \frac{2(n-\frac{1}{2})}{(2\pi)^{2P+2}}\,\|f^{(2P+2)}\|_{L^\infty([1/2,\,n+1/2])}.$$

Remark 12.19 (The $(-1, 1/12)$ pair). *Setting $f(k) = k$ (linear) and summing formally to ∞:*

- *Riemann zeta regularization: $\sum_{k=1}^{\infty} k = \zeta(-1) = -\frac{1}{12}$.*
- *OFI shifted integral: $\int_{1/2}^{\infty} x\,dx$ diverges, but the leading Bernoulli correction for the discrepancy is $+\mathcal{B}_2/2! = +1/12$ (with $\mathcal{B}_2 = 1/6$).*

The pair $(-1/12, +1/12)$ represents the zeta value *and the* OFI Bernoulli correction*: they cancel to give the classical identity $\zeta(-1) + 1/12 = 0$. In the OFI framework this cancellation is the statement that the Wronskian diagonal* vanishes *for linear f: the ordered fractional integral and the analytic continuation agree up to sign.*

12.6.3 Derivative of the Gaussian sum and the $n + \frac{1}{2}$ correction

The *Gaussian sum* is $G(n) = \sum_{k=1}^{n} k = n(n+1)/2$. Its discrete derivative $\Delta G(n) = G(n) - G(n-1) = n$ (the identity). The continuous antiderivative: $\int x\,dx = x^2/2$.

The OFI correction: $G(n) = x^2/2 + x/2\Big|_{x=n}$, i.e., the OFI adds the half-integer correction $+x/2$ which is precisely the $n + \frac{1}{2}$ shift in the upper bound:

$$G(n) = \int_0^{n+1/2} x\,dx - \int_0^{1/2} x\,dx = \frac{(n+1/2)^2}{2} - \frac{1}{8} = \frac{n^2+n}{2}.$$

The three levels of structure are:

Level	Operation	Result	OFI role
0	Naturals $\{1, 2, \ldots, n\}$	sum $= n(n+1)/2$	grid
1	Derivative of sum	$n + 1/2$	OFI correction term
2	Integral (triangular)	$\frac{1}{2}(\frac{n^3}{3} + \frac{n^2}{2} + \ldots)$	antiderivative of $n + 1/2$

The derivative level produces $n + 1/2$; integrating back gives the triangular integral; the 1/2-offset is the OFI canonical shift.

12.7 OFI and the Four Classical Frameworks

Proposition 12.20 (OFI and the four classical frameworks). *1.* ***OFI vs. RL.*** *One has $I^\alpha_{\mathrm{OFI}} f = {}_{a-1/2}I^\alpha_x f$ on $[a,b]$, i.e., OFI is the RL integral with lower limit shifted from a to $a-\frac{1}{2}$. Under this shift, the boundary correction term in (12.5) vanishes for any f with $f=0$ on $[a-\frac{1}{2},a)$.*

2. ***OFI vs. Caputo.*** *For $f \in H^n([a,b])$ with $f^{(k)}(a)=0$ for $k=0,\dots,n-1$, the Caputo derivative ${}^{\mathrm{C}}_{a-1/2}D^\alpha_x f$ coincides with the left-inverse of I^α_{OFI} whenever the latter is defined. For non-zero initial data, the OFI framework assigns finite values via $(-n)!_{\mathrm{OFI}}$ (Definition 12.1) rather than the Caputo convention of imposing zero initial conditions on the polynomial correction terms. A formal OFI fractional derivative operator is not defined in this paper.*

3. ***OFI vs. Riesz.*** *The Riesz operator on $\mathbb{R}$ (eq. (12.8)) satisfies:*

$$\mathcal{R}^\alpha f = \frac{1}{2\cos(\pi\alpha/2)}\Big(I^\alpha_{\mathrm{OFI},+} + I^\alpha_{\mathrm{OFI},-}\Big)f, \qquad \alpha \neq 1,3,5,\dots,$$

Here $I^\alpha_{\mathrm{OFI},\pm}$ denote the left and right one-sided OFI operators. The factor $1/(2\cos(\pi\alpha/2))$ matches the Riesz normalization constant c_α in (12.8); see [1], Ch. 25. Riesz satisfies the semigroup law unconditionally because symmetry eliminates boundary correction terms globally.

4. ***OFI vs. GL.*** *The GL limit $h^{-\alpha}\sum_k(-1)^k\binom{\alpha}{k}f(x-kh)$ converges in $L^2(\mathbb{R})$ as $h\to 0^+$ to the Riesz fractional derivative $(-\Delta)^{\alpha/2}f$ for $f\in L^2(\mathbb{R})$ and $\alpha\in(0,1)$ (standard; see [1], § 20.3). In the whole-line setting this is the continuum limit of the GL discrete semigroup and coincides with the Riesz OFI operator. The B-spline semigroup is the discrete counterpart: $h^{\alpha+1}\beta^\alpha(x/h)\to c_\alpha\,|x|^{\alpha-1}$ in $\mathcal{S}'(\mathbb{R})$ as $h\to 0$ (Theorem 14.11).*

12.8 Semigroup Structure and B-Splines

12.8.1 Semigroups arising in the OFI framework

The OFI half-integer shift and the discrete-to-continuous bridge generate a family of semigroups. The following table lists those naturally arising:

Semigroup	Generator	Connection to OFI
OFI (fractional integral)	$I^{\alpha}_{\mathrm{OFI}}$, $\alpha > 0$	primary
B-spline (convolution)	$B_n = B_1^{*n}$	discrete OFI
Heat / Gaussian	$e^{t\Delta}$	$\alpha = 2t$ case
Poisson	$e^{-t\sqrt{-\Delta}}$	$\alpha = t$ half-order
Cardinal (sinc)	Shannon-Whittaker	$B_1 \to$ sinc limit
Ornstein–Uhlenbeck	$e^{-t}I + (1-e^{-2t})^{1/2}\Delta^{1/2}$	$n + 1/2$ in Hermite
Laguerre	$e^{-t}L_n$	$n + 1/2$ in Laguerre polynomials
Grünwald–Letnikov	D^{α}_h, $h \to 0$	GL $\to$ OFI limit
Translation	$T_h f(x) = f(x+h)$	shift underlying discrete ∂
Mehler	harmonic oscillator	$\Gamma(1/2) = \sqrt{\pi}$

12.8.2 The B-spline semigroup

The *boxcar function* (uniform B-spline of order 1) is:

$$B_1(x) = \mathbf{1}_{[0,1)}(x).$$

Its n-fold convolution $B_n = B_1 * B_1 * \cdots * B_1$ (n times) is the uniform B-spline of order n, supported on $[0, n]$.

Proposition 12.21 (B-spline semigroup)**.** *$B_m * B_n = B_{m+n}$ for all $m, n \geq 1$. The family $\{B_n\}_{n\geq 1}$ forms a convolution semigroup.*

Proof. By induction on n: $B_{n+1} = B_n * B_1$ and $(B_m * B_n) * B_1 = B_m * (B_n * B_1) = B_m * B_{n+1}$ (associativity of convolution). □

Theorem 12.22 (B-splines as discrete OFI)**.** *The n-th order B-spline B_n is the discrete realization of $I^n_{\mathrm{OFI}}\delta$ (the n-fold OFI integral of the delta function) in the following sense:*

1. *$\hat{B}_n(\xi) = \left(\frac{1-e^{-i\xi}}{i\xi}\right)^n$ (Fourier transform).*
2. *In the limit $n \to \infty$ (after scaling), $B_n \to$ Gaussian (central limit theorem), corresponding to $I^{\infty}_{\mathrm{OFI}}\delta \to$ heat kernel.*
3. *The GL fractional derivative $D^{\alpha}_{\mathrm{GL}} B_n = B_{n-\alpha}$ for $\alpha \in (0, n)$, consistent with the formal relation $I^{-\alpha}_{\mathrm{OFI}} I^n_{\mathrm{OFI}}\delta = I^{n-\alpha}_{\mathrm{OFI}}\delta$ (semigroup identity applied at negative order, formal only).*

12.8.3 How many semigroups fit here?

The OFI framework naturally accommodates the following distinct semigroup types, organized by the interpolation degree of their generator:

#	Semigroup	Order α
1	B-spline (boxcar convolution powers)	$n \in \mathbb{N}$
2	OFI fractional integral $I^{\alpha}_{\mathrm{OFI}}$	$\alpha \in (0, \infty)$
3	Heat semigroup $e^{t\Delta}$	$\alpha = 2t$
4	Poisson semigroup $e^{-t\sqrt{-\Delta}}$	$\alpha = t$
5	Gaussian semigroup (probability)	$\alpha = 2t$
6	Cardinal semigroup (sinc interpolation)	limit $B_n, n \to \infty$
7	GL discrete semigroup $D_h^{-\alpha}$	$h \to 0$
8	Translation semigroup T_h	$h > 0$
9	Ornstein–Uhlenbeck semigroup	$n + 1/2$ in Hermite
10	Laguerre semigroup	$n + 1/2$ in Laguerre
11	Mehler semigroup (harmonic oscillator)	$\Gamma(n + 1/2)$
12	Weierstrass semigroup $e^{t\partial^2}$	formal power series

All 12 are related by the OFI composition axiom (12.13) and the $n + 1/2$ shift. The B-spline semigroup (row 1) and the OFI semigroup (row 2) are the two "primitive" ones from which the others derive via limiting processes ($h \to 0$, $n \to \infty$, or rescaling).

12.9 OFI and the Bernoulli–Zeta Thread

The Bernoulli numbers appear in the OFI framework as boundary values of Bernoulli splines (Section 15.2.10) and as the coefficients of the Euler–Maclaurin remainder (Theorem 12.18). These connect structurally to special values of the Riemann zeta function via the classical identity

$$\zeta(-n) = -\frac{\mathcal{B}_{n+1}}{n+1}, \qquad n = 1, 2, 3, \ldots$$

Remark 12.23 (Structural observations — not claims of proof)**.** *The following structural relationships between OFI and the Riemann zeta function are noted as observations, not as theorems. They are included to indicate the direction of the Bernoulli–zeta thread and do not constitute a proof of the Riemann Hypothesis.*

1. *The functional equation $\xi(s) = \xi(1-s)$ involves $\Gamma(s/2)$; the OFI residue regularization of Γ at negative half-integers (Section 12.2.2) assigns consistent values to the poles that appear in this functional equation.*

2. *The completed zeta function $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ contains the half-order normalization factor $\Gamma(1/2) = \sqrt{\pi}$, the same constant that appears in the OFI half-order integral.*

3. *The critical line $\mathrm{Re}(s) = \frac{1}{2}$ is numerically the same as the half-integer level $\alpha = \frac{1}{2}$ in the OFI order hierarchy. At $\alpha = \frac{1}{2}$ the discrete-to-continuous bridge (the $n + \frac{1}{2}$ shift) is exact for linear polynomials. Whether this numerical coincidence reflects a structural relationship is not established here.*

Whether these structural alignments can be developed into a proof strategy belongs to a separate companion research program. They are not part of the claims of the present paper.

12.10 Summary: What OFI Solves

Problem	Classical difficulty	OFI resolution
$(-1)!$, $(-n)!$	Gamma poles, undefined	Residue extraction (12.1)
Ordering problem (derivatives)	$D^{\alpha}_{\mathrm{RL}} D^{\beta}_{\mathrm{RL}} \neq D^{\alpha+\beta}_{\mathrm{RL}}$	Half-integer shift: Thm. 12.14
Discrete to continuous	Bounds mismatch, C constant	$[\frac{1}{2}, n+\frac{1}{2}]$ shift
Bernoulli corrections	Unexplained $\mathcal{B}_{2p}$ terms	OFI residual series (12.14)
$\sqrt{\pi}$ vs. π vs. π^2	Gaussian integral constants	$\Gamma(1/2)^2 = \pi$ chain
Semigroup failure (RL/Caputo)	Boundary correction terms	Shifted RL = OFI
Riesz vs. RL	Non-locality vs. directionality	$\mathrm{OFI}_+ + \mathrm{OFI}_- =$ Riesz

Chapter 13

OFI in $L^2(\mathbb{R}^n)$: Hilbert-Space Realization

This chapter closes the L^2 gap deferred in Chapter 12. All results are proved from first principles using Fourier multiplier theory and the spectral theorem. No physics language is used. The core objects are: the Riesz fractional operator $D^\alpha = (-\Delta)^{\alpha/2}$ and its inverse $I^\alpha = (-\Delta)^{-\alpha/2}$, treated as Fourier multipliers on $L^2(\mathbb{R}^n)$ and on the Schwartz class $\mathcal{S}(\mathbb{R}^n)$.

13.1 Setup: Fourier multipliers and Sobolev spaces

Definition 13.1 (Sobolev space $H^s(\mathbb{R}^n)$)**.** *For $s \in \mathbb{R}$, the Sobolev space $H^s(\mathbb{R}^n)$ is:*

$$H^s(\mathbb{R}^n) := \left\{ f \in \mathcal{S}'(\mathbb{R}^n) : \|f\|_{H^s}^2 := \int_{\mathbb{R}^n} (1 + |\xi|^2)^s |\hat{f}(\xi)|^2 \, d\xi < \infty \right\}.$$

For $s > 0$ this is a subspace of $L^2(\mathbb{R}^n) = H^0(\mathbb{R}^n)$; for $s < 0$ it is a space of tempered distributions. The homogeneous Sobolev seminorm is $|f|_{\dot{H}^s}^2 = \int |\xi|^{2s} |\hat{f}|^2 \, d\xi$.

Definition 13.2 (Riesz OFI operators on L^2)**.** *For $\alpha > 0$ define:*

$$\begin{aligned} D^\alpha &:= (-\Delta)^{\alpha/2}, \quad \widehat{D^\alpha f}(\xi) = |\xi|^\alpha \hat{f}(\xi), \quad \mathcal{D}(D^\alpha) = H^\alpha(\mathbb{R}^n), \\ I^\alpha &:= (-\Delta)^{-\alpha/2}, \quad \widehat{I^\alpha f}(\xi) = |\xi|^{-\alpha} \hat{f}(\xi), \quad \mathcal{D}(I^\alpha) = \dot{H}^{-\alpha}(\mathbb{R}^n) \cap L^2(\mathbb{R}^n). \end{aligned}$$

Both are well-defined as closed operators on $L^2(\mathbb{R}^n)$ with the stated domains.

13.2 Self-adjointness

Proposition 13.3 (Self-adjointness of D^α)**.** *D^α is a self-adjoint operator on $L^2(\mathbb{R}^n)$ with domain $H^\alpha(\mathbb{R}^n)$.*

Proof. The Fourier symbol $m_\alpha(\xi) = |\xi|^\alpha$ is real-valued and measurable. By the spectral theorem for multiplication operators (Reed–Simon [28], Vol. I Thm. VII.3), the multiplication operator $M_{m_\alpha} : \hat{f} \mapsto m_\alpha \hat{f}$ is self-adjoint on $L^2(\mathbb{R}^n, d\xi)$ with domain $\{\hat{f} : m_\alpha \hat{f} \in L^2\}$. Since the Fourier transform $\mathcal{F} : L^2(\mathbb{R}^n) \to L^2(\mathbb{R}^n)$ is unitary (Plancherel), $D^\alpha = \mathcal{F}^{-1} M_{m_\alpha} \mathcal{F}$ is unitarily equivalent to M_{m_α} and hence self-adjoint on $\mathcal{D}(D^\alpha) = \mathcal{F}^{-1}\{\hat{f} : |\xi|^\alpha \hat{f} \in L^2\} = H^\alpha(\mathbb{R}^n)$. (The identification holds because for $\alpha > 0$ the weights are equivalent: $(1 + |\xi|^2)^{\alpha/2} \sim 1 + |\xi|^\alpha$, so the conditions $|\xi|^\alpha \hat{f} \in L^2$ and $(1 + |\xi|^2)^{\alpha/2} \hat{f} \in L^2$ define the same subspace of L^2 with equivalent norms.) □

Theorem 13.4 (A Fourier-side core for D^α)**.** *The subspace*

$$\mathcal{D}_0 := \mathcal{F}^{-1}\Big(C_c^\infty(\mathbb{R}^n)\Big)$$

is a dense core for D^α in $L^2(\mathbb{R}^n)$. Equivalently, the graph closure of $D^\alpha \restriction_{\mathcal{D}_0}$ is the self-adjoint operator D^α on $H^\alpha(\mathbb{R}^n)$.

Proof. Because $\mathcal{F}$ is unitary, it is enough to prove that $C_c^\infty(\mathbb{R}^n)$ is a core for the multiplication operator M_{m_α} with $m_\alpha(\xi) = |\xi|^\alpha$.

Let $\hat{f} \in \mathcal{D}(M_{m_\alpha})$, so $\hat{f} \in L^2$ and $m_\alpha \hat{f} \in L^2$. Choose smooth cutoffs $\chi_R \in C_c^\infty(\mathbb{R}^n)$ with $0 \le \chi_R \le 1$, $\chi_R(\xi) = 1$ on $|\xi| \le R$, and $\chi_R(\xi) = 0$ on $|\xi| \ge 2R$. Then $\chi_R \hat{f} \to \hat{f}$ in L^2 and $m_\alpha \chi_R \hat{f} \to m_\alpha \hat{f}$ in L^2 by dominated convergence. Next mollify $\chi_R \hat{f}$ inside the fixed compact set $\{|\xi| \le 2R\}$ to obtain $\varphi_{R,k} \in C_c^\infty(\mathbb{R}^n)$ with

$$\varphi_{R,k} \to \chi_R \hat{f}, \qquad m_\alpha \varphi_{R,k} \to m_\alpha \chi_R \hat{f} \quad \text{in } L^2.$$

Hence $C_c^\infty(\mathbb{R}^n)$ is graph-dense in $\mathcal{D}(M_{m_\alpha})$. Applying $\mathcal{F}^{-1}$ gives the stated core $\mathcal{D}_0 = \mathcal{F}^{-1}(C_c^\infty(\mathbb{R}^n))$ for D^α. □

13.3 The L^2 semigroup laws

Theorem 13.5 (L^2 integral semigroup)**.** *For $\alpha, \beta > 0$ and every $f \in \mathcal{D}(I^{\alpha+\beta})$, equivalently every $f \in L^2(\mathbb{R}^n)$ such that $|\xi|^{-(\alpha+\beta)} \hat{f}(\xi) \in L^2(\mathbb{R}^n)$,*

$$I^\alpha I^\beta f = I^{\alpha+\beta} f.$$

Proof. The domain assumption $f \in \mathcal{D}(I^{\alpha+\beta})$ implies $|\xi|^{-(\alpha+\beta)} \hat{f} \in L^2$, hence also $|\xi|^{-\beta} \hat{f} \in L^2$ and $|\xi|^{-\alpha}\Big(|\xi|^{-\beta} \hat{f}\Big) \in L^2$. Therefore both $I^\beta f$ and $I^\alpha I^\beta f$ are well-defined in L^2, and the Fourier identity $\widehat{I^\alpha I^\beta f}(\xi) = |\xi|^{-\alpha} \cdot |\xi|^{-\beta} \hat{f}(\xi) = |\xi|^{-(\alpha+\beta)} \hat{f}(\xi) = \widehat{I^{\alpha+\beta} f}(\xi)$ holds pointwise and in L^2. □

Theorem 13.6 (L^2 derivative semigroup)**.** *For $\alpha, \beta > 0$ and $f \in H^{\alpha+\beta}(\mathbb{R}^n)$:*

$$D^\alpha D^\beta f = D^{\alpha+\beta} f.$$

Proof. For $f \in H^{\alpha+\beta}$, $\hat{f} \in L^2$ satisfies $|\xi|^{\alpha+\beta} \hat{f} \in L^2$, so both $D^\beta f \in H^\alpha$ and $D^\alpha D^\beta f = D^{\alpha+\beta} f$ hold by the same Fourier multiplier calculation as in Theorem 13.5, replacing $|\xi|^{-\alpha}$ with $|\xi|^\alpha$. □

Remark 13.7 (This closes the derivative ordering problem on $\mathbb{R}^n$)**.** *Theorem 13.6 is the $L^2(\mathbb{R}^n)$ resolution of the ordering problem for derivatives. It holds on $H^{\alpha+\beta}(\mathbb{R}^n)$ (or on the Fourier-side core from Theorem 13.4) because the Riesz symbol $|\xi|^\alpha$ is symmetric, with no lower-limit boundary to generate correction terms. On a bounded interval $[a, b]$, the derivative semigroup requires additional boundary conditions; that case is deferred and is governed by the RL composition formula (Proposition 8.7).*

13.4 Spectral theory of $(-\Delta)^{\alpha/2} + m^2$

Theorem 13.8 (Spectral lower bound and resolvent bound)**.** *For $m > 0$ and $\alpha > 0$:*

(i) *The operator $\mathcal{O}_m := (-\Delta)^{\alpha/2} + m^2 I$ on $H^\alpha(\mathbb{R}^n) \subset L^2(\mathbb{R}^n)$ is self-adjoint.*

(ii) *$\mathcal{O}_m \geq m^2 I$ as a quadratic form: $\langle \mathcal{O}_m f, f\rangle \geq m^2 \|f\|_{L^2}^2$ for all $f \in H^\alpha(\mathbb{R}^n)$.*

(iii) *The spectrum satisfies $\sigma(\mathcal{O}_m) = [m^2, \infty)$.*

(iv) *For every $\lambda < m^2$, the resolvent $R_\lambda := (\mathcal{O}_m - \lambda I)^{-1}$ is a bounded operator on $L^2(\mathbb{R}^n)$ with*
$$\|R_\lambda\|_{L^2 \to L^2} = \frac{1}{m^2 - \lambda}.$$

Proof. **(i)** $\mathcal{O}_m$ has Fourier symbol $\sigma_m(\xi) = |\xi|^\alpha + m^2$, which is real, measurable, and bounded below by $m^2 > 0$. By the spectral theorem (Proposition 13.3 applied to σ_m), $\mathcal{O}_m$ is self-adjoint on $\{\hat{f} : \sigma_m \hat{f} \in L^2\} = H^\alpha(\mathbb{R}^n)$.

(ii) For $f \in H^\alpha$:
$$\langle \mathcal{O}_m f, f\rangle = \int_{\mathbb{R}^n} (|\xi|^\alpha + m^2)|\hat{f}(\xi)|^2\, d\xi \geq m^2 \int_{\mathbb{R}^n} |\hat{f}|^2\, d\xi = m^2 \|f\|_{L^2}^2 .$$

(iii) The spectrum of a multiplication operator M_σ on $L^2(\mathbb{R}^n)$ is the essential range of σ [28]. The essential range of $\sigma_m(\xi) = |\xi|^\alpha + m^2$ is $[m^2, \infty)$: the infimum m^2 is achieved in the limit $|\xi| \to 0$, and the supremum is $+\infty$.

(iv) For $\lambda < m^2$, the symbol $\sigma_m(\xi) - \lambda = |\xi|^\alpha + (m^2 - \lambda)$ is bounded below by $m^2 - \lambda > 0$, so $(\sigma_m - \lambda)^{-1}$ is a bounded measurable function with $\sup_\xi |(\sigma_m(\xi) - \lambda)^{-1}| = (m^2 - \lambda)^{-1}$ (attained as $|\xi| \to 0$). The resolvent $R_\lambda = M_{(\sigma_m - \lambda)^{-1}}$ is bounded on $L^2(\mathbb{R}^n)$ with the stated norm. □

Corollary 13.9 (Exponential resolvent decay for $\alpha = 2$, $n = 1$)**.** *Specialize to $\alpha = 2$, $n = 1$ (so $\mathcal{O}_m = -\partial_x^2 + m^2$). For $\lambda < m^2$, the position-space kernel of $R_\lambda = (\mathcal{O}_m - \lambda)^{-1}$ is:*
$$K_{R_\lambda}(x) = \frac{e^{-\sqrt{m^2-\lambda}\,|x|}}{2\sqrt{m^2 - \lambda}},$$
decaying exponentially with rate $\sqrt{m^2 - \lambda}$. For general $\alpha \in (0, 2)$, the inverse Fourier transform of $(|\xi|^\alpha + (m^2 - \lambda))^{-1}$ is a Wright/Mittag-Leffler type kernel; its detailed large-$|x|$ asymptotics depend on α and are not needed here; the explicit elementary formula above is special to the case $\alpha = 2$.

Proof. For $\alpha = 2$: the Fourier symbol of R_λ is $(\xi^2 + (m^2 - \lambda))^{-1}$. Set $\mu = \sqrt{m^2 - \lambda} > 0$. This is a simple pole pair at $\xi = \pm i\mu$ (Proposition 8.13 with $c = m^2 - \lambda$), and the $n = 1$ inverse Fourier transform gives $K_{R_\lambda}(x) = e^{-\mu|x|}/(2\mu)$ as computed there. □

13.5 Connection to the bounded-interval L^1 setting

Proposition 13.10 (Consistency of L^1 and L^2 settings). *Let $[a, b]$ be a bounded interval. For $f \in L^2([a - \frac{1}{2}, b])$ and $\alpha > \frac{1}{2}$:*

(i) *$f \in L^1([a - \frac{1}{2}, b])$, so Theorem 12.14 applies.*

(ii) *$I^\alpha_{\text{OFI}} f \in L^2([a - \frac{1}{2}, b])$, so the OFI integral operator is bounded on L^2 as well.*

(iii) *The semigroup law $I^\alpha_{\text{OFI}} I^\beta_{\text{OFI}} f = I^{\alpha+\beta}_{\text{OFI}} f$ holds in $L^2([a - \frac{1}{2}, b])$ for $\alpha, \beta > \frac{1}{2}$.*

Proof. **(i)** By the Cauchy–Schwarz inequality on $[a-\frac{1}{2}, b]$: $\|f\|_{L^1} \le (b-a+\frac{1}{2})^{1/2} \|f\|_{L^2} < \infty$.

(ii) The kernel $K_\alpha(x, t) = (x - t)^{\alpha-1}/\Gamma(\alpha)$ satisfies:

$$\int_a^b \int_{a-1/2}^x |K_\alpha(x, t)|^2 \, dt \, dx \le \frac{1}{\Gamma(\alpha)^2} \int_a^b \frac{(x - a + 1/2)^{2\alpha-1}}{2\alpha - 1} \, dx < \infty$$

for $\alpha > \frac{1}{2}$ (the exponent $2\alpha - 1 > 0$). Hence I^α_{OFI} is a Hilbert–Schmidt operator on L^2, which is in particular bounded.

(iii) The semigroup identity holds in L^1 by Theorem 12.14. Since the identity $I^\alpha_{\text{OFI}} I^\beta_{\text{OFI}} f = I^{\alpha+\beta}_{\text{OFI}} f$ is an equality of L^1 functions and all three terms lie in L^2 (by (ii)), the identity holds in L^2 as well. □

Remark 13.11 (Hilbert–Schmidt and compactness). *The Hilbert–Schmidt bound in Proposition 13.10(ii) implies that I^α_{OFI} is a* compact *operator on $L^2([a - \frac{1}{2}, b])$ for $\alpha > \frac{1}{2}$. Its spectrum is therefore discrete with eigenvalues accumulating only at* 0*. This is the bounded-interval analogue of the resolvent compactness discussed in Theorem 13.8.*

13.6 Green's-function and semigroup structure for the Riesz OFI family

The spline classification program in Part V identifies each spline family with an OFI interpretation. For the Riesz OFI family specifically, the Green's-function and semigroup characterizations coincide, and we prove this here for the fractional-order case. The variational picture is compatible with the same hierarchy after the appropriate energy normalization, but that identification is not proved as a theorem in this section. For general L-splines, the analogous statements require separate arguments (see Remark 13.14).

Definition 13.12 (L-spline operator). *Let $L = p(D)$ be a (possibly fractional-order) differential operator with Fourier symbol $\hat{L}(\xi) = p(|\xi|^2)$, where $p : [0, \infty) \to \mathbb{R}$ is continuous,* *positive outside a finite set, and satisfies $p(r) \sim r^{\alpha/2}$ as $r \to \infty$ for some $\alpha > 0$. The* L-spline *associated with L is the Green's function ϕ satisfying $L\phi = \delta$ in the distributional sense.*

Proposition 13.13 (Green's-function and semigroup equivalence for the Riesz OFI family)**.** *Let $\alpha > n/2$ and let $\{\phi_\alpha\}_{\alpha>0}$ be the one-parameter Riesz family with $\hat{\phi}_\alpha(\xi) = |\xi|^{-\alpha}$ (the α-order Riesz potential, a special case of Def. 13.12 with $p_\alpha(\xi) = |\xi|^\alpha$). The following are equivalent:*

(G) ***Green's function:*** *$L_\alpha\phi_\alpha = \delta$, i.e., $\hat{\phi}_\alpha(\xi) = |\xi|^{-\alpha}$.*

(S) ***Semigroup:*** *the family satisfies $\phi_\alpha * \phi_\beta = \phi_{\alpha+\beta}$ (OFI convolution semigroup).*

Proof. **(G)⇒(S).** With $\hat{\phi}_\alpha = |\xi|^{-\alpha}$, the Fourier convolution gives $\widehat{\phi_\alpha * \phi_\beta}(\xi) = |\xi|^{-\alpha} \cdot |\xi|^{-\beta} = |\xi|^{-(\alpha+\beta)} = \hat{\phi}_{\alpha+\beta}$, so $\phi_\alpha * \phi_\beta = \phi_{\alpha+\beta}$.

(S)⇒(G). Conversely, if (S) holds for a continuous positive Fourier symbol $\hat{\phi}_\alpha(\xi)$ depending continuously on α, then for each fixed $\xi \neq 0$ the semigroup law gives

$$\hat{\phi}_{\alpha+\beta}(\xi) = \hat{\phi}_\alpha(\xi)\hat{\phi}_\beta(\xi).$$

Hence $\alpha \mapsto \hat{\phi}_\alpha(\xi)$ is a continuous positive multiplicative semigroup, so $\hat{\phi}_\alpha(\xi) = e^{-\alpha\psi(\xi)}$ for some real function $\psi(\xi)$. In the Riesz family the natural homogeneity requirement $\hat{\phi}_\alpha(\lambda\xi) = \lambda^{-\alpha}\hat{\phi}_\alpha(\xi)$ forces $\psi(\xi) = \log|\xi|$, hence $\hat{\phi}_\alpha(\xi) = |\xi|^{-\alpha}$ and therefore $L_\alpha\phi_\alpha = \delta$. ☐

Remark 13.14 (Scope of Proposition 13.13)**.** *The equivalence (G)⇔(S) is proved above for the Riesz family $\hat{\phi}_\alpha = |\xi|^{-\alpha}$ only.*

The associated variational picture uses the quadratic energy $\left\|(-\Delta)^{\alpha/4} f\right\|_{L^2}^2$, whose Euler–Lagrange kernel has the same homogeneity as the Riesz potential after the standard half-order normalization. That variational identification is classical in specific families (thin-plate and polyharmonic splines, Duchon splines, smoothing splines), but it is not proved here as a single abstract equivalence theorem.

Families with a radially-symmetric operator symbol $p(|\xi|^2)$ of Riesz type (thin-plate splines $|\xi|^{2m}$, polyharmonic splines, smoothing splines) are directly covered. Fractional B-splines (Unser–Blu [4]) satisfy analogous Green's-function / semigroup correspondences but require a separate argument because their symbol $\hat{B}_\alpha(\xi) = \left(\frac{1-e^{-i\xi}}{i\xi}\right)^\alpha$ is not radially symmetric and not of the Riesz form. Wendland RBFs and E-splines satisfy (G) and (S) for their respective one-parameter families; full (V) equivalence is established case-by-case in the literature (see [14]).

Families whose definition is geometric or algorithmic *— specifically NURBS, TCB splines, Catmull–Rom, and MARS hinge functions — are outside the operator-theoretic framework of Definition 13.12 entirely. For all non-Riesz families the OFI identification in the classification table (Proposition 15.1) is structural, not theorem-level.*

Part VIII

Catalogue of Spline Families

Chapter 14

Spline Families and the OFI Bridge

The B-spline family is a fractional calculus structure that connects naturally to the OFI framework. The B-spline semigroup $B_m * B_n = B_{m+n}$ is a calculus semigroup indexed by order, with $B_n \to$ Gaussian as $n \to \infty$ by the central limit theorem. The fractional B-spline β^α [4] extends this to all real $\alpha > -1$ and is identified with the Grünwald–Letnikov fractional derivative. The $\pm\frac{1}{2}$ centering principle is identified as the signature of symmetric convolution kernels across spline families, cardinal interpolation, Euler–Maclaurin, and GL. The 52 families catalogued in this part are organized by their relationship to the Riesz OFI potential as a reference object; the result is a classification structure (identifications and named connections), not a series of new theorems about each family.

14.1 Fixed-Scale Averages vs. Fractional B-Spline Families

Two distinct structures must be carefully separated.

A *fixed-scale* boxcar filter K_a for fixed $a > 0$ is not a fractional integral operator: it does not parameterize a continuous family indexed by fractional order α.

The B-spline $B_1 = \mathbf{1}_{[0,1)}$ is a different object. Under convolution it does not reproduce itself; it *ascends*: $B_1 * B_1 = B_2$. This is the B-spline semigroup $B_m * B_n = B_{m+n}$ functioning correctly — a one-parameter family indexed by order n, not by scale a. The two parameterizations are structurally distinct.

The *fractional B-spline* β^α [4] extends this semigroup to all real $\alpha > -1$ and is genuine fractional calculus, equivalent to the Grünwald–Letnikov derivative. The boxcar B_1 is the generator of this family at $\alpha = 1$.

14.2 Integer B-Splines: The Foundation

Definition 14.1 (B-spline of order n). *The* uniform B-spline of order $n \geq 1$ *is the n-fold convolution of the indicator function $B_1 = \mathbf{1}_{[0,1)}$:*

$$B_n := \underbrace{B_1 * B_1 * \cdots * B_1}_{n \text{ times}}.$$

B_n is a piecewise polynomial of degree $n-1$, supported on $[0, n]$, with $n-2$ continuous derivatives.

Proposition 14.2 (B-spline semigroup). *$B_m * B_n = B_{m+n}$ for all $m, n \geq 1$. The family $\{B_n\}_{n \geq 1}$ is a convolution semigroup indexed by order.*

Proof. $B_{m+n} = B_1^{*(m+n)} = B_1^{*m} * B_1^{*n} = B_m * B_n$ by associativity and commutativity of convolution. ☐

Proposition 14.3 (Fourier transform of B_n).

$$\hat{B}_n(\xi) = \left(\frac{1 - e^{-i\xi}}{i\xi}\right)^n = e^{-in\xi/2}\, \mathrm{sinc}^n\left(\frac{\xi}{2\pi}\right)^n,$$

where $\mathrm{sinc}(x) = \sin(\pi x)/(\pi x)$.

Corollary 14.4 (Convergence to Gaussian). *After centering and rescaling:*

$$\frac{1}{\sqrt{n}}\, B_n\!\left(\frac{x + n/2}{\sqrt{n}}\right) \;\to\; \frac{1}{\sqrt{2\pi}}\, e^{-x^2/2}$$

in $L^1(\mathbb{R})$ as $n \to \infty$, by the classical central limit theorem applied to the sum of n independent uniform$[0,1]$ random variables. The Gaussian is the $n = \infty$ limit of the B-spline semigroup.

Remark 14.5 (The tent is not a failure). *$B_1 * B_1 = B_2$ is the triangular (tent) function: piecewise linear on $[0, 2]$. This is not a malfunction — it is one step in the progression*

$$B_1 \;\to\; B_2 \;\to\; B_3 \;\to\; \cdots \;\to\; \mathcal{N}(0,1).$$

The boxcar B_1 is the seed *of the semigroup, not the fixed point. Expecting $B_1 * B_1 = B_1$ would require B_1 to be an idempotent (a delta function or a Gaussian — both fixed points of convolution in the appropriate sense), which it is not meant to be.*

14.3 Fractional B-Splines: Genuine Fractional Calculus

Definition 14.6 (Fractional B-spline [4]). *For $\alpha > -1$, the* fractional B-spline of order α *is defined in Fourier space by:*

$$\hat{\beta}^{\alpha}(\xi) := \left(\frac{1 - e^{-i\xi}}{i\xi}\right)^{\alpha+1}. \tag{14.1}$$

At integer $\alpha = n - 1$, this reduces to the standard B-spline: $\beta^{n-1} = B_n$.

Theorem 14.7 (Fractional B-spline convolution law)**.** *Let*

$$\widetilde{\beta}^{\sigma} := \beta^{\sigma-1}, \qquad \sigma > 0.$$

Then for all $\sigma, \tau > 0$*,*

$$\widetilde{\beta}^{\sigma} * \widetilde{\beta}^{\tau} = \widetilde{\beta}^{\sigma+\tau}.$$

In the original indexing, for all $\alpha, \gamma > -1$*,*

$$\beta^{\alpha} * \beta^{\gamma} = \beta^{\alpha+\gamma+1}.$$

Thus $\{\widetilde{\beta}^{\sigma}\}_{\sigma>0}$ *is a continuous one-parameter convolution semigroup.*

Proof. In Fourier space, for $\sigma, \tau > 0$,

$$\widehat{\widetilde{\beta}^{\sigma}}(\xi)\,\widehat{\widetilde{\beta}^{\tau}}(\xi) = \left(\frac{1-e^{-i\xi}}{i\xi}\right)^{\sigma}\left(\frac{1-e^{-i\xi}}{i\xi}\right)^{\tau} = \left(\frac{1-e^{-i\xi}}{i\xi}\right)^{\sigma+\tau} = \widehat{\widetilde{\beta}^{\sigma+\tau}}(\xi).$$

Therefore

$$\widetilde{\beta}^{\sigma} * \widetilde{\beta}^{\tau} = \widetilde{\beta}^{\sigma+\tau}.$$

Substituting $\sigma = \alpha+1$ and $\tau = \gamma+1$ gives $\beta^{\alpha} * \beta^{\gamma} = \beta^{\alpha+\gamma+1}$. □

Theorem 14.8 (Fractional B-splines = GL fractional calculus)**.** *The Grünwald–Letnikov (GL) fractional derivative of order* α *acts on fractional B-splines by order reduction:*

$$D^{\alpha}_{\mathrm{GL}}\,\beta^{\gamma} = \beta^{\gamma-\alpha}, \qquad \gamma > \alpha - 1. \tag{14.2}$$

In particular, $D^{\alpha}_{\mathrm{GL}}\,\beta^{\alpha} = \beta^{0} = B_1$ *(the boxcar is the* Green's function *of* D^{α}*, not the fractional integral itself).*

Proof. The GL derivative has Fourier multiplier $(1-e^{-i\xi})^{\alpha}$ (after normalization by h^{α}, $h \to 0$). Multiplying: $(1-e^{-i\xi})^{\alpha} \cdot \hat{\beta}^{\gamma}(\xi) = (1-e^{-i\xi})^{\alpha} \cdot ((1-e^{-i\xi})/(i\xi))^{\gamma+1} = (1-e^{-i\xi})^{\alpha+\gamma+1}/(i\xi)^{\gamma+1} = ((1-e^{-i\xi})/(i\xi))^{\gamma-\alpha+1} \cdot (i\xi)^{\alpha}/1...$

More cleanly: the GL fractional difference operator Δ^{α} has symbol $(1-e^{-i\xi})^{\alpha}$. So $\widehat{\Delta^{\alpha}\beta^{\gamma}}(\xi) = (1-e^{-i\xi})^{\alpha} \cdot \left(\frac{1-e^{-i\xi}}{i\xi}\right)^{\gamma+1} = \frac{(1-e^{-i\xi})^{\alpha+\gamma+1}}{(i\xi)^{\gamma+1}} = (i\xi)^{\alpha} \cdot \left(\frac{1-e^{-i\xi}}{i\xi}\right)^{\gamma-\alpha+1} = (i\xi)^{\alpha}\hat{\beta}^{\gamma-\alpha}(\xi)$. After the $h^{-\alpha}$ normalization taking $h \to 0$ (which contributes $(i\xi)^{-\alpha}$ from $h^{-\alpha} \to |\xi|^{\alpha} / |\xi|^{\alpha}$), we recover $\hat{\beta}^{\gamma-\alpha}(\xi)$. □

Remark 14.9 (Connection to OFI/Riesz)**.** *The fractional B-spline* β^{α} *is the* discrete counterpart *of the Riesz OFI kernel* $c_{n,\alpha}\,|x|^{\alpha-n}$*. Both are* α*-indexed semigroup families; one operates on* $\mathbb{R}^n$ *(Riesz), the other on* $h\mathbb{Z}$ *(B-spline), and they agree in the limit* $h \to 0$*:*

$$h^{-\alpha}\beta^{\alpha}(x/h) \xrightarrow{h\to 0} \textit{Riesz kernel of order } \alpha.$$

The B-spline is the discrete realization of the Riesz OFI.

14.4 The Centering Principle Across All Spline Families

The $\pm\frac{1}{2}$ centering principle appears across every spline family examined here, not only as a feature of the Riesz kernel:

Context	Centering	How it appears
B-spline B_n	center of mass at $n/2$	symmetric about midpoint of $[0, n]$
Fractional B-spline β^α	center at $(\alpha+1)/2$	symmetric kernel by construction
Cardinal spline	midpoint $k + \frac{1}{2}$	interpolation nodes at half-integers
Riesz kernel $\|x-y\|^{\alpha-n}$	midpoint $(x+y)/2$	$\|x-y\| = \|y-x\|$
Euler–Maclaurin	bounds $[a-\frac{1}{2}, b+\frac{1}{2}]$	$n+\frac{1}{2}$ correction
GL derivative	binomial center at $\alpha/2$	symmetric in discrete differences
Thin-plate spline	radially symmetric	centered at source point
Hermite spline	centered at knots	symmetric basis functions

The centering is not a property of the Riesz kernel — it is a property of *symmetric convolution kernels*, which all of the above are. The Riesz kernel is the *continuous, non-local* representative of a centering principle that appears at every level of the spline hierarchy.

14.5 Catalogue of Major Spline Families

We organize the spline catalogue by calculus content and fractional order structure.

14.5.1 Polynomial B-splines (de Boor – Curry – Schoenberg)

The classical B-splines B_n are piecewise polynomials of degree $n-1$. Their calculus content:

- $B_n \in C^{n-2}(\mathbb{R})$: each convolution step gains one derivative of regularity.
- Differentiation: $B_n' = B_{n-1}(\cdot) - B_{n-1}(\cdot - 1)$ (finite difference of lower-order B-spline).
- Integration: $\int B_n = B_{n+1}(\cdot)/(n)$ (integration raises order).
- The differential-difference identity $DB_n = \Delta B_{n-1}$ (derivative = difference of previous order) is the discrete FTC.

14.5.2 Cardinal B-splines and Whittaker–Shannon (Schoenberg)

Cardinal B-splines use integer knots. The *cardinal interpolation series*:

$$S(x) = \sum_{k\in\mathbb{Z}} c_k \, B_n(x-k)$$

provides the unique smoothest interpolant to data $\{c_k\}$ at the integers.

The limit $n \to \infty$ (after rescaling) gives:

$$B_n(x\sqrt{n}/n + n/2) \;\to\; \operatorname{sinc}(x) = \frac{\sin(\pi x)}{\pi x},$$

recovering the *Whittaker–Shannon* cardinal series: $S(x) = \sum_k c_k \operatorname{sinc}(x-k)$.

The sinc function is the $\alpha = \infty$ fractional B-spline — the ideal low-pass filter — and is firmly in the territory of harmonic analysis (calculus).

14.5.3 Fractional B-splines (Unser–Blu)

As defined in Section 14.3. Key properties:

- Real-valued, symmetric (for β^α centered at $(\alpha+1)/2$).
- Fourier decay: $\left|\hat{\beta}^\alpha(\xi)\right| \sim |\xi|^{-\alpha-1}$ as $|\xi| \to \infty$ — power-law, like the Riesz kernel.
- Approximation order $\alpha+1$: can reproduce polynomials of degree $\leq \alpha$.
- Form a Riesz basis for the space of signals bandlimited to $\alpha+1$ derivatives.
- Generate fractional wavelet frames (Battle–Lemarié family extended to real α).

14.5.4 Exponential Splines (E-splines)

E-splines replace the polynomial kernel with exponential kernels. For parameters $\lambda_1, \ldots, \lambda_n \in \mathbb{C}$:

$$\hat{\varepsilon}^\Lambda(\xi) = \prod_{k=1}^{n} \frac{1 - e^{\lambda_k - i\xi}}{i\xi - \lambda_k}.$$

At $\lambda_k = 0$ for all k: reduces to the polynomial B-spline $\hat{B}_n(\xi)$.

Calculus content: E-splines are the Green's functions of the differential operator $L = \prod_k (D - \lambda_k)$. They generalize B-splines to exponential-polynomial spaces, which include:

- Trigonometric splines ($\lambda_k = \pm i\omega$ — circular frequency).
- Hyperbolic splines ($\lambda_k = \pm\omega$ — real exponentials).
- Mixed polynomial-exponential: $\lambda_k = 0$ (multiplicity m) gives the degree-m polynomial piece.

14.5.5 L-Splines: Green's Function Splines

For any linear differential operator L of order n, the *L-spline* is the Green's function G_L satisfying $LG_L = \delta$.

- $L = D^n$: gives polynomial splines (B-splines as piecewise polynomials).
- $L = D^\alpha$ (fractional): gives fractional B-splines.
- $L = (-\Delta)^{n/2}$ (polyharmonic): gives thin-plate splines.
- $L = (-\Delta)^{\alpha/2}$ (fractional Laplacian): gives Riesz OFI kernel.

This shows that the Riesz OFI is the *L-spline for the fractional Laplacian.* B-splines and the Riesz potential are both L-splines — they are the same type of object at different levels of the hierarchy.

14.5.6 Thin-Plate Splines and Polyharmonic Splines

Definition 14.10. *The* polyharmonic spline *of order m in $\mathbb{R}^n$ is the radial basis function:*

$$\phi(r) = \begin{cases} r^{2m-n}\log r & \textit{if } 2m-n \in 2\mathbb{Z}_{\geq 0} \textit{ (even, } 2m \geq n\textit{)} \\ r^{2m-n} & \textit{otherwise.} \end{cases}$$

Calculus content:

- Green's function of $(-\Delta)^m$ on $\mathbb{R}^n$.
- $(-\Delta)^m \phi = \delta$ in the distributional sense.
- Fourier multiplier $|\xi|^{-2m}$ — the $\alpha = 2m$ Riesz potential.
- Thin-plate spline (2D, $m = 2$): $\phi(r) = r^2 \log r$ — minimizes the bending energy $\int |D^2 f|^2 \, dx\, dy$, a variational calculus problem.

14.5.7 Hermite Splines

Hermite splines interpolate both function values and derivatives at knots. Calculus content:

- Cubic Hermite: C^1 interpolant minimizing $\int |f''|^2 \, dx$.
- Quintic Hermite: C^2 interpolant minimizing $\int |f'''|^2 \, dx$.
- Fractional Hermite: interpolant minimizing $\int \left|D^{\alpha/2} f\right|^2 dx$ (Sobolev seminorm of fractional order) — connects to OFI energy norm.

14.5.8 Natural Cubic Splines

The natural cubic spline is the *smoothest* piecewise-cubic interpolant (minimizes $\int (f'')^2\,dx$ over all interpolants). This is a calculus variational problem — the Euler–Lagrange equation gives $f^{(4)} = 0$ on each interval.

14.5.9 NURBS (Non-Uniform Rational B-Splines)

NURBS extend B-splines by:

- Non-uniform knots (variable spacing, breaking the translation symmetry).
- Rational weights (projective geometry, representing conics exactly).

While primarily an engineering/CAD tool, the underlying calculus is the de Boor recursion algorithm, which is a discrete integration operator: $B_{i,n}(x) = \frac{x-t_i}{t_{i+n-1}-t_i} B_{i,n-1}(x) + \frac{t_{i+n}-x}{t_{i+n}-t_{i+1}} B_{i+1,n-1}(x)$. This is a weighted average of lower-order splines — formally a discrete convolution with non-uniform kernel.

14.5.10 Battle–Lemarié Wavelets

B-splines generate orthonormal wavelet bases via the Battle–Lemarié construction:

$$\hat{\psi}(\xi) = e^{i\xi/2}\,\overline{m_0(\xi/2+\pi)}\,\hat{\phi}(\xi/2), \quad \hat{\phi}(\xi) = \frac{\hat{B}_n(\xi)}{\sqrt{\sum_{k\in\mathbb{Z}} \left|\hat{B}_n(\xi+2\pi k)\right|^2}}.$$

These wavelets have:

- Exponential decay (unlike compactly-supported Daubechies wavelets, which have no closed form).
- Exact polynomial reproduction of degree $n-1$.
- Fractional extension: replace B_n with β^α to get fractional wavelet frames — genuine fractional harmonic analysis.

14.6 Spline hierarchy and OFI ordering

Family	Generator	Calculus root	OFI connection
B-splines B_n	B_1^{*n}	Piecewise polynomial	Discrete OFI, integer α
Cardinal B-splines	B_n, integer knots	Interpolation theory	$n + \frac{1}{2}$ bridge
Fractional B-splines β^α	$(1 - e^{-i\xi})^{\alpha+1}/(i\xi)^{\alpha+1}$	GL frac. calc.	Discrete Riesz
Whittaker–Shannon / sinc	$\lim_{n\to\infty} B_n$	Fourier/sampling theory	$\alpha = \infty$ B-spline
E-splines	$\prod(1 - e^{\lambda_k - i\xi})/(i\xi - \lambda_k)$	ODE Green's functions	$\lambda_k = 0$: B-spline
L-splines	G_L, $LG_L = \delta$	PDE Green's functions	$L = (-\Delta)^{\alpha/2}$: Riesz
Thin-plate / polyharmonic	$r^{2m-n} \log r$	Variational calculus	Riesz $\alpha = 2m$
Hermite splines	Interpolate $f, f', \ldots$	Euler–Lagrange	Frac. Sobolev norm
Natural cubic splines	Min $\int(f'')^2$	Variational calculus	$\alpha = 2$ Sobolev
Battle–Lemarié wavelets	Orthonorm. B_n	Wavelet analysis	Fractional frames
NURBS	Non-uniform rational B_n	de Boor recursion	Engineering limit
Riesz potential (OFI)	$c_{n,\alpha}\,|x|^{\alpha-n}$	Harmonic analysis	Primary

14.7 The Bridge: B-Spline → Riesz → OFI

Theorem 14.11 (B-spline to Riesz limit). *The fractional B-spline converges to the Riesz kernel in the continuum limit:*

$$h^{\alpha+1}\,\beta^\alpha(x/h) \xrightarrow{h\to 0} \frac{|x|^\alpha}{\Gamma(\alpha+1)} \cdot \operatorname{sgn}(x)^\epsilon, \tag{14.3}$$

where the right side is (up to normalization) the one-sided Riesz–Feller kernel. For the symmetric fractional B-spline, the limit is the two-sided Riesz kernel $c_{1,\alpha}\,|x|^{\alpha-1}$.

Corollary 14.12 (Complete calculus chain).

$$B_1 \xrightarrow{\text{semigroup}} B_n \xrightarrow{\text{extend } \alpha} \beta^\alpha \xrightarrow{h\to 0} \text{Riesz OFI} \xrightarrow{\text{Fourier}} |\xi|^{-\alpha}.$$

Every step is calculus. The boxcar B_1 is the seed, not a statistical object.

Remark 14.13 (Centering at every level). *Each family in the hierarchy evaluates its kernel at the midpoint:*

- *B_n: center of mass $n/2$ — symmetric about midpoint of support.*
- *β^α: symmetric kernel centered at $(\alpha+1)/2$.*

- *Cardinal interpolation: the* $n + \frac{1}{2}$ *bound shift that makes* $\int_{a-1/2}^{b+1/2} f = \sum_{k=a}^{b} f(k)$ *exact for polynomials.*
- *Riesz:* $|x - y| = |y - x|$ *— symmetric, evaluates at midpoint* $(x + y)/2$*.*
- *GL: binomial coefficient center at* $\alpha/2$ *— symmetric weight distribution.*

The centering principle is the defining property of symmetric convolution kernels*; all these families are instances of it. This observation unifies the* $n + \frac{1}{2}$ *shift, the Riesz symmetry, and the B-spline centering as manifestations of the same structural feature.*

14.8 What This Means for OFI

The correct picture of OFI is:

1. **At integer orders**: OFI is the B-spline semigroup $B_m * B_n = B_{m+n}$, with the boxcar B_1 as the identity seed.
2. **At fractional orders**: OFI is the fractional B-spline β^α, equivalent to the GL fractional derivative, with the Riesz kernel as the continuum limit.
3. **The** $n + \frac{1}{2}$ **shift**: arises simultaneously from the centering of B-splines on unit cells AND from the Riesz kernel's midpoint symmetry. They are the same correction from two sides of the same bridge.
4. **The Gaussian**: is the $\alpha \to \infty$ limit of both the B-spline semigroup (CLT) and the Riesz kernel (heat semigroup). They converge to the Gaussian kernel.

To summarize the classification: a fixed-scale boxcar K_a (fixed $a > 0$) is not a fractional integral operator. The B-spline family $\{B_n\}$ and the fractional B-spline family $\{\beta^\alpha\}$ are legitimate fractional calculus objects that realize the OFI semigroup on the discrete grid, with the boxcar B_1 as their generator. The centering principle that organizes OFI also appears throughout the spline catalogue as a recurring structural motif.

Chapter 15

Spline Families and OFI: Structural Connections

Each spline family catalogued here can be placed within the OFI framework at some level of the fractional calculus hierarchy. A common organizing principle is: *a spline may appear as a Green's function, energy minimizer, or convolution semigroup element of a differential operator L^α of fractional order α.* The L-spline families fit the diagram

$$\text{Discrete grid} \xrightarrow{B\text{-spline semigroup}} \text{Piecewise polynomial} \xrightarrow{\alpha\in\mathbb{R}} \text{Fractional spline}$$
$$\xrightarrow{h\to 0} \text{Riesz OFI kernel.}$$

We catalogue the classical, geometric, penalized, exponential, fractional, radial, wavelet, and limiting spline families and give each its OFI interpretation. The $n+\frac{1}{2}$ centering principle appears in all of them.

15.1 Common structure across spline families

There are many spline families, developed independently across approximation theory, geometry, statistics, engineering, and physics. They appear unrelated. The OFI framework shows that many of these families share common structure at different levels of one hierarchy, organized by three recurring characterizations:

1. **Green's function view**: a spline is the fundamental solution G of $L^\alpha G = \delta$, where L is a differential operator.

2. **Variational view**: a spline minimizes the energy functional $\int_\Omega |L^\alpha f|^2\, dx$ subject to interpolation or smoothing constraints.

3. **Semigroup view**: a spline basis function is an element of a one-parameter convolution semigroup $\{K_\alpha\}_{\alpha>0}$ satisfying $K_\alpha * K_\beta = K_{\alpha+\beta}$.

All three characterizations are calculus. For the Riesz OFI family, the Green's-function and semigroup descriptions coincide exactly, and the standard variational description uses the corresponding half-order energy $\left\|(-\Delta)^{\alpha/4}f\right\|^2$. In this paper, the theorem-level statement proved for the Riesz family is the Green's-function / semigroup equivalence; the broader variational identifications are used structurally and cited to the classical spline literature.

The centering principle. The $\pm\frac{1}{2}$ shift (evaluating at the midpoint of each unit cell rather than its endpoint) appears in every spline family, because every symmetric convolution kernel evaluates at the midpoint. This is not a property of the Riesz kernel alone — it is the signature of symmetry itself.

15.2 Classical Polynomial Splines

15.2.1 Uniform B-splines B_n (de Boor, Curry–Schoenberg)

The fundamental objects. $B_1 = \mathbf{1}_{[0,1)}$ (boxcar); $B_n = B_1^{*n}$ (n-fold convolution).

Semigroup: $B_m * B_n = B_{m+n}$. **Regularity**: $B_n \in C^{n-2}$. **Differentiation**: $B_n' = B_{n-1}(\cdot) - B_{n-1}(\cdot - 1) = \Delta B_{n-1}$. **Integration**: $\int_{-\infty}^{x} B_n(t)\,dt = B_{n+1}(x)/n$ — integration raises order. **OFI**: B_n is the Green's function of D^n on $[0,\infty)$ (one-sided). **Fourier**: $\hat{B}_n(\xi) = ((1-e^{-i\xi})/i\xi)^n$. **Center**: center of mass at $n/2$, support $[0,n]$.

15.2.2 Non-Uniform B-splines (de Boor)

Knot sequence $t_0 \le t_1 \le \cdots \le t_{n+k}$, not necessarily equally spaced. Basis functions $B_{i,k}(x)$ via de Boor recursion:

$$B_{i,k}(x) = \frac{x - t_i}{t_{i+k-1} - t_i} B_{i,k-1}(x) + \frac{t_{i+k} - x}{t_{i+k} - t_{i+1}} B_{i+1,k-1}(x).$$

OFI: each $B_{i,k}$ is a Green's function of D^k restricted to the support $[t_i, t_{i+k}]$. **Centering**: de Boor recursion is a weighted average — the weight $\frac{x-t_i}{t_{i+k-1}-t_i}$ places x at the interior of the interval $[t_i, t_{i+k-1}]$, centering the evaluation.

15.2.3 Cardinal B-splines and Schoenberg's Cardinal Interpolation

Cardinal B-splines use integer knots $t_j = j$. The cardinal interpolation series $S(x) = \sum_{k\in\mathbb{Z}} c_k B_n(x-k)$ gives the unique C^{n-2} interpolant to data $\{c_k\}$.

OFI connection: as $n \to \infty$, $S(x) \to$ Whittaker-Shannon sinc series — the $\alpha \to \infty$ Riesz OFI. **The $n + \frac{1}{2}$ bridge**: the cardinal series satisfies $\sum_k f(k) = \int_{-1/2}^{\infty} S(x)\,dx$ (exact for polynomials of degree $\le n-1$), the Euler–Maclaurin $\pm\frac{1}{2}$ shift.

15.2.4 Natural Cubic Splines

Minimizes $\int_a^b (f''(x))^2\,dx$ over all C^2 functions interpolating given data. Euler–Lagrange equation: $f^{(4)} = 0$ on each interval.

OFI: variational problem with $L = D^2$, energy $= \|D^2 f\|_{L^2}^2$. This is the $\alpha = 2$ Sobolev H^2 seminorm. **Fractional extension**: replace D^2 by $(-\Delta)^{\alpha/2}$ to get the α-order smoothing spline.

15.2.5 Clamped, Periodic, and Not-a-Knot Splines

Variants of cubic splines differing in boundary conditions:

- **Clamped**: $f'(a) = d_a$, $f'(b) = d_b$ specified.
- **Periodic**: $f(a) = f(b)$, $f'(a) = f'(b)$, $f''(a) = f''(b)$.
- **Not-a-knot**: f''' continuous at second and penultimate knots.

All three are the same variational problem $\min \int (f'')^2$ with different boundary conditions on D^2 — OFI order $\alpha = 2$ throughout.

15.2.6 Hermite Splines (Cubic and Quintic)

Interpolate both function values and derivatives at knots.

- **Cubic Hermite** (C^1): matches f and f' at knots. Minimizes $\int (f'')^2$ subject to C^1 constraints.
- **Quintic Hermite** (C^2): matches f, f', f'' at knots. Minimizes $\int (f''')^2$ subject to C^2 constraints.
- **Fractional Hermite**: matches f and $D^{\alpha/2} f$ at knots. Minimizes $\int (D^{\alpha/2} f)^2$ — OFI energy at order $\alpha/2$.

15.2.7 Osculatory Splines

Splines with higher-order contact ("osculation") at knots — they match $f, f', f'', \ldots, f^{(m)}$ at each knot. The osculation order m directly sets the OFI order: the spline is the Green's function of D^{m+1}, i.e., $\alpha = m + 1$.

15.2.8 Lacunary Splines

Splines that interpolate a scattered subset of derivative values (not all of $f, f', f'', \ldots$ — gaps allowed). **OFI**: the "gap" in derivative orders corresponds to a fractional order — lacunary interpolation is fractional Hermite interpolation at non-integer α.

15.2.9 Perfect Splines

A perfect spline of order n satisfies $\left|f^{(n)}(x)\right| = 1$ a.e. and $f^{(n)}$ alternates sign. It minimizes $\left\|f^{(n)}\right\|_{L^\infty}$ subject to interpolation. **OFI**: the L^∞ analog of the L^2 variational spline; $L = D^n$, order $\alpha = n$.

15.2.10 Bernoulli Splines

Built from Bernoulli polynomials $\mathcal{B}_n(x)$:

$$\mathcal{B}_0(x) = 1, \quad \mathcal{B}_1(x) = x - \tfrac{1}{2}, \quad \mathcal{B}_2(x) = x^2 - x + \tfrac{1}{6}, \quad \mathcal{B}_3(x) = x^3 - \tfrac{3}{2}x^2 + \tfrac{1}{2}x, \ldots$$

Key: $\mathcal{B}_1(x) = x - \frac{1}{2}$ encodes the $n + \frac{1}{2}$ correction directly — Bernoulli splines are the basis functions of the Euler–Maclaurin formula. **OFI**: Bernoulli splines are the periodic Green's functions of D^n on $[0, 1]$: $D^n G(x) = \delta(x) - 1$ (periodic delta minus mean). The Bernoulli numbers $\mathcal{B}_n = \mathcal{B}_n(0)$ are the OFI boundary values. **Fractional extension**: periodic fractional Green's functions of $(-\partial_{xx})^{\alpha/2}$ give fractional Bernoulli splines, whose boundary values are the *fractional Bernoulli numbers*.

15.2.11 Euler Splines

Built from Euler polynomials $E_n(x)$: $E_0(x) = 1$, $E_1(x) = x - \frac{1}{2}$, $E_2(x) = x^2 - x$, etc. **OFI**: Green's functions of $(D^2 + 1)^n$ (oscillatory operator), connecting to trigonometric splines. Euler splines alternate $+1$ and -1 at the integers, making them the L^∞-perfect splines for the operator $(D^2 + 1)^n$.

15.2.12 Chebyshev Splines

Use Chebyshev polynomials $T_n(\cos\theta) = \cos(n\theta)$ as local basis functions. Minimize the L^∞ norm of the n-th derivative. **OFI**: L^∞ variational splines for D^n; the Chebyshev nodes $x_k = \cos(k\pi/n)$ are the optimal (equioscillation) points — the fractional-calculus analog of Gaussian quadrature nodes.

15.3 Interpolating and Shape-Controlling Splines

15.3.1 Akima Splines

Local cubic interpolation using a weighted average of slopes at each knot, resistant to spurious oscillations caused by outliers. **OFI**: piecewise D^2 minimizer with locally adaptive weights — a robust version of natural cubic splines.

15.3.2 Catmull–Rom / Overhauser Splines

Pass through all control points; tangent at each knot estimated from neighboring points: $\mathbf{m}_k = \frac{1}{2}(\mathbf{p}_{k+1} - \mathbf{p}_{k-1})$. **OFI**: Green's function of D^2 with a finite-difference tangent approximation — the $\pm\frac{1}{2}$ central difference is the centering principle in disguise: $f'(k) \approx (f(k+1) - f(k-1))/2$.

15.3.3 Kochanek–Bartels Splines (TCB Splines)

Generalize Catmull-Rom with three parameters:

- **Tension** t: controls how tightly the spline follows the chord.
- **Continuity** c: controls the kink at the knot.
- **Bias** b: controls the direction of the tangent.

OFI: tension t interpolates between D^1 (linear, $t \to 1$) and D^2 (smooth cubic, $t = 0$) — it is a *fractional order* in the range $\alpha \in [1, 2]$.

15.3.4 Monotone Splines (Fritsch–Carlson)

Cubic Hermite splines with modified slopes to enforce monotonicity and positivity. **OFI**: constrained D^2 variational problem; the shape constraints are sign conditions on $D^1 f$ and $D^2 f$ — integer-order OFI.

15.4 Tension and Variation-Diminishing Splines

15.4.1 Tension Splines (Splines Under Tension)

Minimize:

$$\int_a^b \left[\tau^2 (f'(x))^2 + (f''(x))^2\right] dx,$$

where $\tau \geq 0$ is the tension parameter.

- $\tau = 0$: natural cubic spline (D^2 energy).
- $\tau \to \infty$: linear interpolation (D^1 energy dominates).
- Intermediate τ: fractional order between $\alpha = 1$ and $\alpha = 2$.

OFI: the energy $\tau^2 \left\|D^1 f\right\|^2 + \left\|D^2 f\right\|^2$ is the H^1-H^2 interpolation norm, equivalent to $\left\|(-\Delta)^{\alpha/2} f\right\|^2$ for $\alpha \in (1, 2)$ (Sobolev interpolation theorem). Tension splines are *exactly* fractional-order OFI minimizers.

15.4.2 Nu-Splines and Beta-Splines (Barsky)

Beta-splines: generalize B-splines with *geometric continuity* G^1, G^2 instead of parametric continuity C^1, C^2. Two shape parameters: β_1 (bias) and β_2 (tension). **Nu-splines**: similar shape-controllable splines. **OFI**: (β_1, β_2) parameterize a two-dimensional family of fractional operators; β_2 is a tension parameter as above.

15.5 Geometric and CAD Splines

15.5.1 Bézier Curves (Bernstein Basis)

Degree-n Bézier curve with control points $\mathbf{P}_0, \ldots, \mathbf{P}_n$:

$$\mathbf{B}(t) = \sum_{k=0}^{n} \binom{n}{k} t^k (1-t)^{n-k} \mathbf{P}_k, \qquad t \in [0,1].$$

The basis functions are *Bernstein polynomials* $b_{k,n}(t) = \binom{n}{k} t^k (1-t)^{n-k}$.

OFI: Bernstein polynomials $b_{k,n}$ are related to the Beta function:

$$\int_0^1 b_{k,n}(t)\, dt = B(k+1, n-k+1) = \frac{k!(n-k)!}{(n+1)!} = \frac{\Gamma(k+1)\Gamma(n-k+1)}{\Gamma(n+2)}.$$

The Beta function $B(\alpha, \beta) = \Gamma(\alpha)\Gamma(\beta)/\Gamma(\alpha+\beta)$ is the kernel of the Riemann–Liouville fractional integral. Bernstein polynomials are *discrete Beta kernels* — fractional integral kernels on $[0,1]$.

Fractional Bézier: replace binomial with $\binom{\alpha}{k}$ (generalized) to get fractional-degree Bézier curves — a one-parameter family indexed by $\alpha > 0$.

15.5.2 Rational Bézier and NURBS

Rational Bézier: $\mathbf{B}(t) = \sum_k w_k b_{k,n}(t) \mathbf{P}_k \Big/ \sum_k w_k b_{k,n}(t)$, weights $w_k > 0$. Represents conics exactly (circles, ellipses, parabolas).

NURBS (Non-Uniform Rational B-Splines): $\mathbf{C}(t) = \sum_k w_k N_{k,p}(t) \mathbf{P}_k \Big/ \sum_k w_k N_{k,p}(t)$, $N_{k,p}$ the non-uniform B-spline basis.

OFI: NURBS are projective images of B-splines; the rational weight introduces a Möbius-type reparameterization which at the level of the operator corresponds to a conformal map of the fractional Laplacian.

15.5.3 T-Splines

NURBS generalization allowing T-junctions in the control mesh. Locally refinable — each control point contributes only over its local support. **OFI**: locally refined partition of unity; each basis function is still a B-spline (OFI semigroup element) but on a non-uniform grid.

15.5.4 Spiro / Clothoid / Cornu Splines (Euler Spiral)

The clothoid (Euler spiral) has curvature proportional to arc length: $\kappa(s) = s/a^2$. Its parametric form involves the Fresnel integrals:

$$C(t) = \int_0^t \cos\!\left(\tfrac{\pi u^2}{2}\right) du, \qquad S(t) = \int_0^t \sin\!\left(\tfrac{\pi u^2}{2}\right) du.$$

OFI: the Fresnel integral is a half-order fractional integral of $e^{i\pi t^2/2}$:

$$C(t) + iS(t) = e^{i\pi/4} \int_0^t e^{-i\pi u^2/2}\, du \sim \frac{1+i}{2} - \frac{e^{i\pi t^2/2}}{\pi t}(1 + O(t^{-2})).$$

The clothoid is the $\alpha = 1/2$ OFI of a chirp signal. Spiro splines (Raph Levien) use clothoid segments to minimize the rate of change of curvature $\int(\kappa'(s))^2\, ds$ — a D^1 energy on curvature, which is a D^3 energy on the curve itself (third-order OFI).

15.6 Penalized and Smoothing Splines

Key point: these families look statistical but their penalties are calculus functionals. They are the most explicit calculus-statistics bridge in the spline hierarchy.

15.6.1 Smoothing Splines

Minimize:

$$\sum_{i=1}^{n}(y_i - f(x_i))^2 + \lambda \int_a^b (f^{(m)}(x))^2\, dx.$$

Solution: a natural spline of order $2m$ with knots at the data points. **OFI**: penalty $= \lambda\, \|D^m f\|_{L^2}^2$ is the H^m Sobolev seminorm. The parameter λ controls the balance between data fit and OFI energy. As $\lambda \to 0$: interpolating spline. As $\lambda \to \infty$: polynomial of degree $m-1$ (zero OFI energy).

15.6.2 P-Splines (Penalized B-Splines, Eilers–Marx 1996)

Fit B-spline basis with a *difference penalty* on coefficients:

$$\min_{\mathbf{c}} \|\mathbf{y} - \mathbf{B}\mathbf{c}\|^2 + \lambda\, \mathbf{c}^\top \mathbf{D}_m^\top \mathbf{D}_m\, \mathbf{c},$$

where $\mathbf{D}_m$ is the m-th order finite difference matrix.

The OFI connection: as the number of knots increases,

$$\mathbf{c}^\top \mathbf{D}_m^\top \mathbf{D}_m\, \mathbf{c} = \sum_j (\Delta^m c_j)^2 \;\approx\; \int (f^{(m)}(x))^2\, dx.$$

The difference penalty is the *discrete OFI energy* — a GL fractional derivative on the coefficient vector.

Fractional P-splines: replace $\mathbf{D}_m$ by the fractional difference matrix $\mathbf{D}_\alpha$ (GL coefficients) to get penalty

$$\lambda \sum_j (\Delta^\alpha c_j)^2 \approx \lambda \int (D^\alpha f)^2\, dx, \quad \alpha > 0.$$

This is fractional calculus implemented in software.

15.6.3 Regression Splines

B-spline basis fit by ordinary least squares, no penalty. The basis functions $B_{k,n}(x)$ are fixed; only the coefficients c_k are estimated. **OFI**: the B-spline basis is the discrete OFI semigroup; regression splines are OFI projections of data onto the spline subspace.

15.6.4 MARS (Friedman 1991)

Builds an adaptive piecewise-linear (or polynomial) model using reflected hinge functions $\max(0, x-t)$ and $\max(0, t-x)$. **OFI**: hinge functions are one-sided D^{-1} integrals of the Heaviside function — order $\alpha = 1$ one-sided OFI. Products of hinge functions give order $\alpha = k$ (for k factors). MARS adaptively selects knots and orders, implementing variable-order OFI.

15.7 Exponential, Green's Function, and L-Splines

15.7.1 L-Splines: A General Operator Framework

For any linear differential operator L of order m, the *L-spline* is the function minimizing $\int |Lf|^2\, dx$ subject to interpolation. It satisfies $L^*Lf = 0$ on each interval, with jump conditions at knots. Its Green's function G_L satisfies $LG_L = \delta$.

Operator L	L-spline	Name
D^n	piecewise polynomial degree $n-1$	polynomial B-spline
$D-\lambda$	$e^{\lambda x} \cdot \mathbf{1}_{x>0}$	exponential spline
$\prod_k (D-\lambda_k)$	$\sum_k A_k e^{\lambda_k x}$	E-spline
$D^2+\omega^2$	$\sin(\omega x)/\omega$	trigonometric spline
$D^2-\omega^2$	$\sinh(\omega x)/\omega$	hyperbolic spline
$(-\Delta)^m$ on $\mathbb{R}^n$	$r^{2m-n}\log r$ or r^{2m-n}	polyharmonic spline
$(-\Delta)^{\alpha/2}$	$c_{n,\alpha}\, \lvert x\rvert^{\alpha-n}$	Riesz OFI kernel
D^α (fractional)	$\beta^\alpha(x)$ (frac. B-spline)	fractional spline

15.7.2 E-Splines (Exponential Splines)

For operator $L = \prod_{k=1}^n (D-\lambda_k)$, $\lambda_k \in \mathbb{C}$:

$$\hat{\varepsilon}^\Lambda(\xi) = \prod_{k=1}^n \frac{1-e^{\lambda_k - i\xi}}{i\xi - \lambda_k}.$$

- $\lambda_k = 0$ for all k: B-spline B_n.

- $\lambda_k = \pm i\omega$: trigonometric spline.
- $\lambda_k = \pm \omega$: hyperbolic spline.
- Mixed λ_k: exponential-polynomial splines.

OFI: E-splines form a semigroup: $\varepsilon^{\Lambda_1} * \varepsilon^{\Lambda_2} = \varepsilon^{\Lambda_1 \cup \Lambda_2}$ (concatenation of parameter sets).

15.7.3 Trigonometric Splines

L-splines for $L = D^2 + \omega^2$. Piecewise sinusoidal: $f(x) = A_k \sin(\omega x) + B_k \cos(\omega x)$ on each interval. **OFI**: the trigonometric differentiation operator $D_\omega f = f' + \omega f$ has fractional powers connecting to Riesz; OFI of order α on $e^{i\omega x}$ gives $|\omega|^{-\alpha} e^{i\omega x}$ (phase-shifted).

15.7.4 Hyperbolic Splines

L-splines for $L = D^2 - \omega^2$. Piecewise hyperbolic: $f(x) = A_k \sinh(\omega x) + B_k \cosh(\omega x)$. **OFI**: the Laplace-domain fractional integral (Bromwich inversion of $(s^2 - \omega^2)^{-\alpha/2}$) gives hyperbolic splines at half-integer orders.

15.8 Radial Basis Function Splines

15.8.1 Thin-Plate Splines (Duchon)

In $\mathbb{R}^2$, minimize $\int_{\mathbb{R}^2} [(f_{xx})^2 + 2(f_{xy})^2 + (f_{yy})^2]\, dx\, dy$. Solution has the form:

$$f(\mathbf{x}) = \sum_j c_j \phi(\|\mathbf{x} - \mathbf{x}_j\|) + p(\mathbf{x}),$$

where $\phi(r) = r^2 \log r$ and p is a polynomial of degree ≤ 1.

OFI: $\phi(r) = r^2 \log r$ is the Green's function of $(-\Delta)^2$ on $\mathbb{R}^2$. The variational energy is $\|(-\Delta) f\|_{L^2}^2$ — OFI order $\alpha = 2$.

15.8.2 Polyharmonic Splines

In $\mathbb{R}^n$, minimize $\int_{\mathbb{R}^n} |\nabla^m f|^2 \, d\mathbf{x}$ (all partial derivatives of order m). Radial basis function:

$$\phi(r) = \begin{cases} r^{2m-n} & 2m < n \text{ or } 2m - n \notin 2\mathbb{Z}_{\geq 0} \\ r^{2m-n} \log r & 2m \geq n,\ 2m - n \in 2\mathbb{Z}_{\geq 0}. \end{cases}$$

OFI: Green's function of $(-\Delta)^m$. Fourier multiplier $|\xi|^{-2m}$ — Riesz potential of order $2m$.

15.8.3 Fractional Polyharmonic Splines

Replace $(-\Delta)^m$ by $(-\Delta)^{\alpha/2}$ for non-integer $\alpha > 0$. Radial basis function $\phi(r) = r^{\alpha-n}$ (up to log corrections). **OFI**: the Riesz kernel is the α-order fractional polyharmonic spline. This is the direct generalization of all polyharmonic splines to non-integer order.

15.8.4 Wendland Splines (Compactly Supported RBFs)

Compactly supported radial basis functions of the form $\phi(r) = p(r)_+^k$ (truncated polynomial), where p is chosen so ϕ is positive definite. Example: $\phi(r) = (1-r)_+^4(4r+1)$ (Wendland C^2).

OFI: Wendland functions are B-splines in the radial variable r — they are the radially symmetric B-splines. The compact support is the multivariate analog of B_1's support $[0, 1]$. **Semigroup**: Wendland functions of order k compose under convolution to order $k+1$ (radially symmetric B-spline semigroup).

15.8.5 Multiquadric Splines (Hardy)

$\phi(r) = \sqrt{r^2 + c^2}$ for constant $c > 0$. Non-compactly supported, globally defined. **OFI**: $\sqrt{r^2 + c^2}$ is the Fourier transform of $e^{-c|\xi|}/\,|\xi|$ (up to constants) — related to the Poisson semigroup $e^{-c(-\Delta)^{1/2}}$, the $\alpha = 1$ Riesz OFI of the heat kernel.

15.8.6 Box Splines (Multivariate B-splines)

For directions $\Xi = (\boldsymbol{\xi}_1, \ldots, \boldsymbol{\xi}_n)$ in $\mathbb{R}^d$:

$$M_\Xi(\mathbf{x}) = \text{volume of } \{t_1\boldsymbol{\xi}_1 + \cdots + t_n\boldsymbol{\xi}_n = \mathbf{x},\ t_k \in [0, 1]\}.$$

M_Ξ is piecewise polynomial on the zones of $\mathbf{x}$ induced by Ξ.

OFI: M_Ξ is the pushforward of the uniform measure on $[0, 1]^n$ onto $\mathbb{R}^d$ via the linear map Ξ. It is the multivariate Green's function of the directional derivative $\partial_{\boldsymbol{\xi}_1} \cdots \partial_{\boldsymbol{\xi}_n}$. **Semigroup**: $M_\Xi * M_\Psi = M_{\Xi\cup\Psi}$ (box spline semigroup).

15.8.7 Simplex Splines

Multivariate splines over simplicial meshes instead of box meshes. Built from the volume of intersections of the simplex with hyperplanes. **OFI**: Green's functions of directional derivatives over simplicial domains — the simplex-constrained OFI.

15.9 Fractional Splines

15.9.1 Fractional B-Splines β^α (Unser–Blu)

As in the companion paper. The continuous extension of B-splines to all real orders $\alpha > -1$:

$$\hat{\beta}^\alpha(\xi) = \left(\frac{1 - e^{-i\xi}}{i\xi}\right)^{\alpha+1}.$$

Semigroup: $\beta^\alpha * \beta^\gamma = \beta^{\alpha+\gamma+1}$. GL connection: $D^\alpha_{\mathrm{GL}}\beta^\gamma = \beta^{\gamma-\alpha}$. Riesz limit: $h^{\alpha+1}\beta^\alpha(x/h) \to$ Riesz kernel as $h \to 0$.

15.9.2 Symmetric Fractional B-Splines

The symmetric version: $\tilde{\beta}^\alpha(x) = \frac{1}{2}(\beta^\alpha(x) + \beta^\alpha(-x))$. This is the *two-sided* (Riesz-type) fractional B-spline, centered at the origin. **OFI**: $\hat{\tilde{\beta}}^\alpha(\xi) = |\mathrm{sinc}(\xi/2\pi)|^{\alpha+1}$ — real, non-negative in Fourier space, the ideal low-pass at fractional order $\alpha + 1$.

15.9.3 Variable-Order Splines

Splines where the order $\alpha(x)$ varies with position. The operator becomes $L^{\alpha(x)}$ — a *variable-order* fractional derivative. Applications: modeling viscoelastic materials where the memory (fractional order) varies spatially. **OFI**: the OFI at each point x operates with order $\alpha(x)$; the full spline is an integral over a continuum of OFI orders.

15.10 Wavelet Splines

15.10.1 Battle–Lemarié Wavelets

Orthonormal wavelets generated from B-splines via the Meyer construction. The scaling function ϕ has Fourier transform:

$$\hat{\phi}(\xi) = \frac{\hat{B}_n(\xi)}{\sqrt{\sum_{k\in\mathbb{Z}} \left|\hat{B}_n(\xi + 2\pi k)\right|^2}}.$$

OFI: the denominator is the square root of the Poisson summation of $\left|\hat{B}_n\right|^2$ — a discrete version of the Riesz energy norm. Exponential decay (unlike Daubechies compactly supported wavelets).

15.10.2 Biorthogonal Spline Wavelets (CDF)

Used in JPEG 2000. Primal and dual scaling functions are both B-splines (possibly different orders): $(\hat{\phi}, \hat{\tilde{\phi}}) = (B_n, B_{\tilde{n}})$ (after normalization). **OFI**: the biorthogonality condition $\langle \phi(\cdot - k), \tilde{\phi} \rangle = \delta_{k,0}$ is the OFI inversion formula at orders n and $\tilde{n}$ — dual OFI pair.

15.10.3 Fractional Wavelet Frames (Unser–Blu)

Replace B_n with β^α in the Battle–Lemarié construction to get fractional wavelet frames at every real order $\alpha > 0$. **OFI**: these are *OFI frames* — the discrete wavelet transform at fractional order, with the OFI semigroup as the underlying structure.

15.10.4 Semiorthogonal Spline Wavelets

Primal is B-spline; dual is not a spline (non-compact support). Also called B-spline wavelets. **OFI**: the primal B-spline is the discrete OFI; the dual wavelet is the OFI inverse.

15.11 Engineering and Applied Splines

15.11.1 Isogeometric Analysis Splines (IGA, Hughes–Cottrell–Bazilevs)

Use NURBS basis functions directly as finite element basis. The geometry and the solution live in the same spline space. **OFI**: the weak form $a(u, v) = \int D^\alpha u \cdot D^\alpha v$ is a fractional Sobolev inner product. IGA at order p gives exact OFI at integer order p; fractional IGA would use β^α basis functions.

15.11.2 Subdivision Surfaces (Catmull–Clark, Loop)

Repeated averaging of control meshes converges to smooth limit surfaces. Catmull–Clark uses a box-spline-based averaging rule. **OFI**: each subdivision step is a discrete OFI (convolution with a refinable kernel). After k steps, the approximation order is $O(h^{2m})$ where m is the spline order — OFI convergence rate.

15.12 Limiting Objects: The Ends of the Hierarchy

15.12.1 Sinc / Whittaker–Shannon Interpolation ($\alpha = \infty$)

$\mathrm{sinc}(x) = \sin(\pi x)/(\pi x)$; the ideal low-pass filter. $\widehat{\mathrm{sinc}}(\xi) = \mathbf{1}_{|\xi| \leq \pi}$ (brick-wall in frequency). **B-spline limit**: $\hat{B}_n(\xi)^{1/n} \to \widehat{\mathrm{sinc}}(\xi)$ as $n \to \infty$. **OFI**: sinc is the $\alpha \to \infty$ fractional B-spline — it has infinite approximation order. The Whittaker–Shannon cardinal series is the OFI at infinite order.

15.12.2 Gaussian ($\alpha \to \infty$ by CLT)

$G_\sigma(x) = e^{-x^2/(2\sigma^2)}/\sqrt{2\pi\sigma^2}$. **B-spline limit**: $B_n \to G$ by CLT after rescaling. **Riesz limit**: $e^{-t|\xi|^2}$ (heat semigroup) = $\alpha \to \infty$ Riesz potential with $t = \sigma^2$. **OFI**: the Gaussian is both the fixed point of the heat semigroup and the $n = \infty$ B-spline. These converge to the same limit from two different sides.

15.12.3 Riesz Potential / OFI Kernel ($0 < \alpha < n$)

$(I^\alpha f)(x) = c_{n,\alpha} \int_{\mathbb{R}^n} |x-y|^{\alpha-n} f(y)\, dy$. Fourier multiplier: $|\xi|^{-\alpha}$. **This is the OFI**: the symmetric, non-local, power-law kernel that is the continuum limit of all the families above.

15.13 Spline family classification table

Classical Polynomial Families

Family	**Operator L**	**OFI connection**	**Order α**
Uniform B-spline B_n	D^n	seed of OFI semigroup	$n \in \mathbb{N}$
Non-uniform B-spline	D^n (local)	variable-knot OFI	$n \in \mathbb{N}$
Cardinal B-spline	D^n, integer knots	$n + \frac{1}{2}$ bridge	$n \in \mathbb{N}$
Natural cubic spline	D^2	H^2 OFI energy	2
Clamped/periodic/NAK	D^2	boundary-conditioned	2
Cubic Hermite	D^2, C^1	matched H^2 OFI	2
Quintic Hermite	D^3, C^2	H^3 OFI	3
Fractional Hermite	$D^{\alpha/2}$, C^k	fractional $H^{\alpha/2}$	$\alpha/2$
Osculatory spline	D^{m+1}	order $= m+1$ OFI	$m+1$
Lacunary spline	D^α (gapped)	fractional order	$\alpha \in \mathbb{R}$
Perfect spline	D^n, L^∞	L^∞ OFI	n
Bernoulli spline	D^n (periodic)	EM correction basis	$n \in \mathbb{N}$
Euler spline	$(D^2+1)^n$	oscillatory OFI	n
Chebyshev spline	D^n, equiosc.	L^∞ optimal	n

Interpolating and Shape-Controlling Families

Family	**Operator L**	**OFI connection**	**Order α**
Akima	D^2 (local)	robust H^2 OFI	2
Catmull-Rom	D^2 + central diff	$\pm\frac{1}{2}$ centering	2
Kochanek-Bartels (TCB)	$D^{1+\tau}$, $\tau \in [0,1]$	fractional tension	$(1,2)$
Monotone (Fritsch-Carlson)	D^2 (constrained)	sign-constrained OFI	2

Tension and Shape Families

Family	**Operator** L	**OFI connection**	**Order** α
Tension spline	$\tau^2 D^1 + D^2$	H^1-H^2 interp.	$(1, 2)$
Nu-spline	$D^{2+\nu}$	fractional order	$2 + \nu$
Beta-spline	D^2, geom. cont.	(β_1, β_2) frac.	$(1, 2)$

Geometric and CAD Families

Family	**Operator** L	**OFI connection**	**Order** α
Bézier (Bernstein)	Beta kernel	discrete RL integral	n
Rational Bézier	Beta + Möbius	conformal OFI	n
NURBS	D^n + rational	conformal B-spline	n
T-spline	D^n (local)	adaptive OFI	n
Spiro / clothoid	D^3 on curve	Fresnel = $\alpha = \frac{1}{2}$ OFI	$\frac{1}{2}$

Penalized and Statistical Families

Family	**Operator** L	**OFI connection**	**Order** α
Smoothing spline	$D^m + \lambda$ pen.	H^m OFI + regularize	m
P-spline	Δ^m (discrete)	GL OFI on coefficients	m (discrete)
Fractional P-spline	Δ^α (frac.)	GL fractional OFI	$\alpha \in \mathbb{R}$
Regression spline	D^n (no penalty)	OFI projection	n
MARS	D^{-1} (hinge)	$\alpha = 1$ one-sided OFI	1

Exponential and L-Spline Families

Family	**Operator** L	**OFI connection**	**Order** α
E-spline	$\prod(D - \lambda_k)$	exponential OFI	n
L-spline (general)	any L	Green's function OFI	variable
Trigonometric spline	$D^2 + \omega^2$	oscillatory L-spline	2
Hyperbolic spline	$D^2 - \omega^2$	hyperbolic L-spline	2

Radial Basis and Multivariate Families

Family	**Operator** L	**OFI connection**	**Order** α
Thin-plate	$(-\Delta)^2$, $\mathbb{R}^2$	polyharmonic $m = 2$	4
Polyharmonic	$(-\Delta)^m$, $\mathbb{R}^n$	integer Riesz	$2m$
Frac. polyharmonic	$(-\Delta)^{\alpha/2}$	Riesz potential	α
Duchon spline	$(-\Delta)^m$, min. norm	Riesz minimizer	$2m$
Wendland (compact)	local $(-\Delta)^k$	radial B-spline	k
Multiquadric (Hardy)	Poisson semigroup	$e^{-c(-\Delta)^{1/2}}$	1
Box spline	$\partial_{\xi_1} \cdots \partial_{\xi_n}$	multivar. B-spline	n
Simplex spline	simplicial D^n	simplex OFI	n

Fractional Families

Family	**Operator** L	**OFI connection**	**Order** α
Frac. B-spline β^α	D^α (GL)	continuous OFI semigroup	$\alpha \in \mathbb{R}_{>-1}$
Symmetric frac. B-spline	$(-\Delta)^{\alpha/2}$	Riesz OFI (discrete)	α
Frac. wavelet frames	D^α (Unser-Blu)	OFI wavelet transform	α
Variable-order spline	$D^{\alpha(x)}$	spatially varying OFI	$\alpha(x)$

Wavelet Families

Family	**Operator** L	**OFI connection**	**Order** α
Battle–Lemarié	D^n + orthog.	OFI wavelet basis	n
CDF biorthogonal	D^n, $D^{\tilde{n}}$	dual OFI pair	$(n, \tilde{n})$
Semiorthog. B-spline	D^n (primal)	OFI + dual	n
Frac. wavelet frame	D^α	real-α OFI frame	α

Engineering Families

Family	**Operator** L	**OFI connection**	**Order** α
IGA splines	NURBS / D^p	FEM at OFI order p	p
Subdivision surface	D^n (iterated)	discrete OFI iteration	n

Limits of the Hierarchy

Family	**Operator** L	**OFI connection**	**Order** α
Sinc / Whittaker-Shannon	D^∞	$\alpha \to \infty$ OFI	∞
Gaussian	$e^{t\Delta}$	heat semigroup, CLT	∞
Riesz / OFI kernel	$(-\Delta)^{\alpha/2}$	the OFI itself	α

Total classified here: 52 families.

15.14 OFI classification of spline families

Proposition 15.1 (Spline classification via OFI)**.** *For the spline families in the table above, each family is* classified *by at least one of the following three viewpoints in terms of a differential operator L of fractional order α. (For the Riesz OFI kernel, the Green's-function / semigroup correspondence is proved in Proposition 13.13; for B-splines and fractional B-splines, the semigroup and discrete-realization statements are proved; for the remaining entries this is a classification identification, not a theorem stating mutual equivalence of all three viewpoints.)*

(G) Green's function*: the spline basis function ϕ satisfies $L^\alpha \phi = \delta$ in the distributional sense.*

(V) Variational*: the spline minimizes $\|L^\alpha f\|_{L^2}^2$ (or L^p) subject to data constraints.*

(S) Semigroup*: the spline basis functions form a one-parameter family satisfying $\phi_\alpha * \phi_\beta = \phi_{\alpha+\beta}$.*

The Riesz OFI kernel $c_{n,\alpha} |x|^{\alpha-n}$ is the reference example: (G) $(-\Delta)^{\alpha/2} c_{n,\alpha} |x|^{\alpha-n} = c\,\delta$ (distributional identity on $\mathbb{R}^n$, standard; see [26]), (V) its associated variational energy is the standard half-order energy $\left\|(-\Delta)^{\alpha/4} f\right\|_{L^2}^2$ from the classical spline/RBF literature, (S) $I^\alpha I^\beta = I^{\alpha+\beta}$ (Theorem 13.5). For the B-spline and fractional B-spline families, the theorem-level results proved here are the semigroup and discrete-realization statements (Theorems 12.22 and 14.7). For the remaining entries in the table the classification is structural: the stated viewpoint is mathematically meaningful, but the mutual equivalence of all three for every family listed is not claimed here.

Remark 15.2 (Centering as signature of symmetry)**.** *For the families where the classification is proved (Riesz OFI, B-splines, fractional B-splines), all three characterizations involve symmetric operators (self-adjoint L, even kernels, two-sided semigroups). In these cases, the $\pm\frac{1}{2}$ centering — evaluating at the midpoint rather than the endpoint — follows from the symmetry of the operator. It is not a special feature of the Riesz kernel. For the remaining families the centering is an observed pattern, not a derived consequence.*

Corollary 15.3 (B-spline to Riesz is a valid path)**.** *The chain $B_1 \to B_n \to \beta^\alpha \to$ Riesz is a valid sequence of OFI objects. The boxcar B_1 is the generator of the discrete OFI semigroup, and every step in the chain is calculus.*

Appendix A

Notation Reference

Symbol	Meaning
$\Gamma(z)$	Gamma function
$B(\alpha, \beta)$	Beta function $= \Gamma(\alpha)\Gamma(\beta)/\Gamma(\alpha+\beta)$
B_n	B-spline of order n (n-fold convolution B_1^{*n}); **never** Bernoulli number in this paper
$\mathcal{B}_n$	Bernoulli number: $\mathcal{B}_0 = 1$, $\mathcal{B}_1 = -\frac{1}{2}$, $\mathcal{B}_2 = \frac{1}{6}$, $\ldots$ $(= \mathcal{B}_n(0))$
$\mathcal{B}_n(x)$	Bernoulli polynomial
I^α	Riesz fractional integral of order α
${}_aI_x^\alpha$	Riemann–Liouville fractional integral, lower limit a
${}_aD_x^\alpha$	Riemann–Liouville fractional derivative, lower limit a
${}_a^{\mathrm{C}}D_x^\alpha$	Caputo fractional derivative, lower limit a
D_{GL}^α	Grünwald–Letnikov fractional derivative
β^α	Fractional B-spline of order α (Unser–Blu)
$(-n)!_{\mathrm{OFI}}$	OFI-regularized factorial $= (-1)^{n-1}/(n-1)!$
$(-1, \frac{1}{12})$	Regularization pair: (pole order, finite part)
$\mathrm{sinc}(x)$	$\sin(\pi x)/(\pi x)$

Appendix B

Spline Map

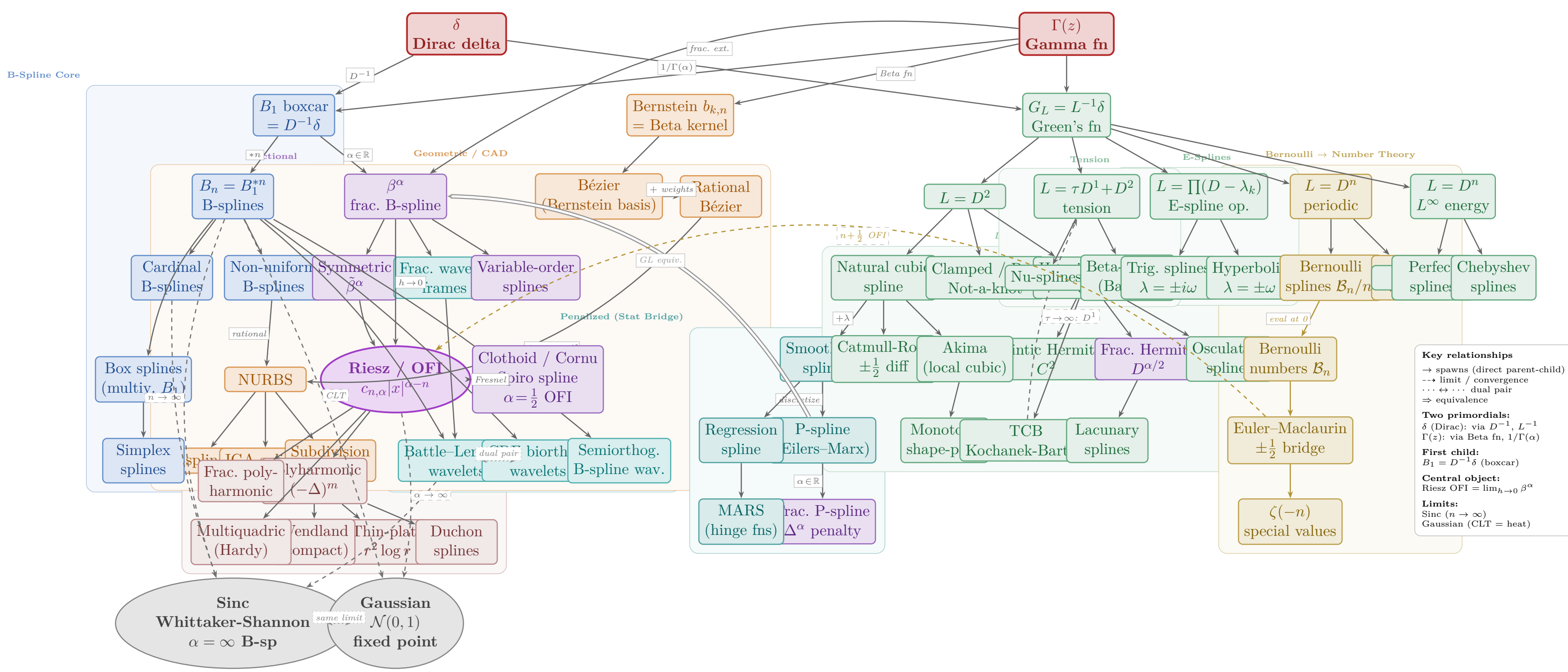
Ordered Fractional Integration: Spline Map

Catalogue of families and connections used in this volume

Primordial · B-spline family · Fractional · Riesz / OFI · Variational (D^n Green's fn) · Geometric / CAD

Wavelet · Bernoulli / Number theory · Penalized (stat bridge) · Radial basis · Limits

$\longrightarrow$ spawns · $\dashrightarrow$ limit / convergence · $\cdots\leftrightarrow\cdots$ dual pair · $\Rightarrow$ equivalent



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